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Statistical Mechanics

NOTES BY

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INTRODUCTION. Statistical mechanics is a branch of physics aiming at understanding the laws of the macroscopic behaviour of systems composed of many microscopic components. Critical phenomena, such as phase transitions, involve a drastic change in the macroscopic state by tuning model parameters. Critical phenomena are extremely universal far beyond physics, for instance in chemistry, biology or complex systems. In this course, we aim to give a mathematical foundation for the study of many component systems on the lattice and in the continuum space. Moreover, we would like to motivate the theory of Gibbs measures starting from basic principles in classical mechanics.

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Introduction

SECTION 1.1

| From Micro to Macro: Statistical Mechanics

The object of statistical mechanics is the study of *macroscopic systems* and of their behaviour. Due to the huge number of degrees of freedom (e.g., of order 10^{23}), it is practically impossible to solve the equations of motion for such a many-body system. However, we know from thermodynamics that, at equilibrium, one can characterise observations of a macroscopic system with only a handful of variables (e.g., volume, pressure, temperature, external magnetic field).

We abandon the idea of a deterministic description of the system and search instead for a probability distribution over the set of all microstates. The *probabilistic distribution* is indeed such that:

- (i) it describes the *observed macrostates* at equilibrium;
- (ii) the distribution of the *observable macrostates* (e.g., total energy) concentrates on the most probable value;
- (iii) it can describe *phase transitions* and *critical phenomena*;
- (iv) it allows us to analyse *fluctuations* of macroscopic quantities in large *finite* systems.

We denote with Ω the set of all *microstates* and we assume that $|\Omega| < \infty$. Hence, we assume to have a *discrete space*. We call *Hamiltonian* the interaction between microscopic constituents (e.g., domains) of the form $\mathcal{H}(\omega)$, associated with each microstate $\omega \in \Omega$:

$$\mathcal{H} : \Omega \rightarrow \mathbb{R}.$$

We denote with $\mathcal{M}_1(\Omega)$ the set of all *probability measures* on Ω . A measure $\mu \in \mathcal{M}_1(\Omega)$ is *fully characterised* by the collection of the probabilities associated to each *microstate* $\omega \in \Omega$ (i.e., by $(\mu(\omega))_{\omega \in \Omega}$). We call *observable* the result of a measurement of a system. In our formalism it corresponds to a *random variable* $f : \Omega \rightarrow \mathbb{R}$. We will denote the *expected value* of an observable f under $\mu \in \mathcal{M}_1(\Omega)$ by

$$\langle f \rangle_\mu := \sum_{\omega \in \Omega} f(\omega) \mu(\omega).$$

The Microcanonical Ensemble

LEMMA 1.1 (Ergodic Hypothesis).

The action of the evolution transformation S as a cell permutation of the phase space cells on the

surface of constant energy is a one-cycle permutation of the N phase space cells with the given energy

$$S\Delta_k = \Delta_{k+1}, \quad k \in \{1, \dots, N\},$$

if the cells are suitably enumerated.

Remark. In other words, as time evolves every cell evolves visiting successively all other cells with equal energy.

The *microcanonical ensemble* is the probability space of an *isolated system* at equilibrium. An isolated system has a set of *conserved quantities* (e.g., total energy, number of particles).

Consider a system in a volume Λ of *finite cardinality* $|\Lambda| = V < \infty$ (e.g., a lattice with V sites). We denote with Ω_Λ the set of all these microstates. Let us restrict to a system where the total energy is *fixed* at a value U . We consider the *energy shell*:

$$\Omega_{\Lambda;U} := \{\omega \in \Omega_\Lambda : H(\omega) = U\}.$$

DEFINITION 1.2 (Microcanonical Ensemble).

The *microcanonical ensemble* is the uniform measure on $\Omega_{\Lambda;U}$:

$$\mu_{\Lambda;U}^{\text{Mic}}(\omega) := \begin{cases} \frac{1}{|\Omega_{\Lambda;U}|}, & \omega \in \Omega_{\Lambda;U} \\ 0, & \text{otherwise} \end{cases}.$$

Remark. Any microstate in the microcanonical ensemble is equally likely.

The Canonical Ensemble

DEFINITION 1.3 (Shannon Entropy).

Given $\mu \in \mathcal{M}_1$, we define the (*Shannon*) *entropy* as:

$$S_{\text{Sh}}(\mu) := - \sum_{\omega \in \Omega} \mu(\omega) \log \mu(\omega).$$

Remark. Shannon entropy quantifies the unpredictability of the outcome of a probability distribution μ on Ω .

EXERCISE 1.1. Show that the Shannon entropy S_{Sh} is concave as a function $\mathcal{M}_1(\Omega) \rightarrow \mathbb{R}$, i.e., for all $\mu, \nu \in \mathcal{M}_1(\Omega)$ and $\alpha \in [0, 1]$, we have:

$$S_{\text{Sh}}(\alpha\mu + (1 - \alpha)\nu) \geq \alpha S_{\text{Sh}}(\mu) + (1 - \alpha)S_{\text{Sh}}(\nu).$$

SOLUTION. Writing

$$S_{\text{Sh}}(\mu) = \sum_{\omega \in \Omega} \psi(\mu(\omega)),$$

where $\psi(x) := -x \log x$, we see that S_{Sh} is concave. $\square$

The Shannon entropy can be used for deriving a *variational principle*. In fact, we have a characterisation of the uniform distribution.

LEMMA 1.4.

The uniform distribution on Ω (i.e., $\nu^{\text{Unif}}(\omega) := \frac{1}{|\Omega|}$), is the unique probability distribution at which S_{Sh} attains the maximum, i.e.:

$$S_{\text{Sh}}(\nu^{\text{Unif}}) = \sup_{\mu \in \mathcal{M}_1} S_{\text{Sh}}(\mu).$$

Proof. Consider $\psi(x) := -x \log x$, which is concave. Using *Jensen's Inequality* gives

$$S_{\text{Sh}}(\mu) = |\Omega| \sum_{\omega \in \Omega} \frac{1}{|\Omega|} \psi(\mu(\omega)) \leq |\Omega| \psi\left(\sum_{\omega \in \Omega} \frac{\mu(\omega)}{|\Omega|}\right) = |\Omega| \psi\left(\frac{1}{|\Omega|}\right) = \log |\Omega| = S_{\text{Sh}}(\nu^{\text{Unif}}),$$

with equality if and only if $\mu(\cdot)$ is constant, that is, if $\mu = \nu^{\text{Unif}}$. $\square$

EXAMPLE 1.5. The microcanonical distribution has the maximal entropy among *all* the distribution concentrated on the energy shell of energy U :

$$S_{\text{Sh}}(\mu_{\Lambda;U}^{\text{Mic}}) = \sup_{\mu \in \mathcal{M}_1(\Omega_{\Lambda}); \mu(\Omega_{\Lambda;U})=1} S_{\text{Sh}}(\mu).$$

We would like to derive now the probability distribution to describe a macroscopic system at equilibrium at a finite temperature: such a system is not isolated anymore, but in contact with a *thermal reservoir*. The total energy is not fixed anymore, but we can anyway compute the average energy and call it U .

We want to find the distribution μ that maximises the entropy S_{Sh} under the constraint that $\langle \mathcal{H} \rangle_{\mu} = U$. Therefore,

$$\begin{aligned} & \text{maximise} \quad - \sum_{\omega \in \Omega_{\Lambda}} \mu(\omega) \log \mu(\omega) \\ & \text{when} \quad \begin{cases} \sum_{\omega \in \Omega_{\Lambda}} \mu(\omega) = 1, & \text{probability normalisation} \\ \sum_{\omega \in \Omega_{\Lambda}} \mu(\omega) \mathcal{H}(\omega) = U. & \text{fixed average energy} \end{cases} \end{aligned}$$

Use the method of *Lagrange multipliers* to find the minimum. Since we have two constraints, we define the *Lagrange function*:

$$L(\mu) := \sum_{\omega \in \Omega_{\Lambda}} \mu(\omega) \log \mu(\omega) + \lambda \left(\sum_{\omega \in \Omega_{\Lambda}} \mu(\omega) - 1 \right) + \beta \left(\sum_{\omega \in \Omega_{\Lambda}} \mu(\omega) \mathcal{H}(\omega) - U \right).$$

We have to solve the following system of $|\Omega_{\Lambda}| + 2$ variables:

$$\begin{cases} \nabla L = 0, \\ \sum_{\omega \in \Omega_{\Lambda}} \mu(\omega) = 1, \\ \sum_{\omega \in \Omega_{\Lambda}} \mu(\omega) \mathcal{H}(\omega) = U, \end{cases}$$

where the first condition corresponds to the set of equations:

$$\frac{\partial L(\mu)}{\partial \mu(\omega)} = \log \mu(\omega) + 1 + \lambda + \beta \mathcal{H}(\omega) = 0, \quad \forall \omega \in \Omega_{\Lambda},$$

whose solution is $\mu(\omega) = e^{-\beta \mathcal{H}(\omega) - 1 - \lambda}$. The first constraint $\sum_{\omega \in \Omega_{\Lambda}} \mu(\omega) = 1$ implies that: $e^{1+\lambda} = \sum_{\omega \in \Omega_{\Lambda}} e^{-\beta \mathcal{H}(\omega)}$. Therefore:

$$\mu_{\beta}(\omega) := \frac{e^{-\beta \mathcal{H}(\omega)}}{\sum_{\omega \in \Omega_{\Lambda}} e^{-\beta \mathcal{H}(\omega)}}$$

where the Lagrange multiplier β has to be chosen such that

$$\sum_{\omega \in \Omega_\Lambda} \mu_\beta(\omega) \mathcal{H}(\omega) = U.$$

This solution possesses exactly one solution $\beta = \beta(U)$, when $U \in (U_{\min}, U_{\max})$. Hence β can always be chosen in such a way that the energy takes a given value: it can be used as a *natural parameter* of the canonical distribution.

EXERCISE 1.2. Given the canonical measure μ_β , prove the existence and the uniqueness of the solution $\beta(U)$ of the equation:

$$\sum_{\omega \in \Omega_\Lambda} \mu_\beta(\omega) \mathcal{H}(\omega) = U.$$

EXERCISE 1.3. Show that $\beta \mapsto \langle \mathcal{H} \rangle_{\mu_\beta}$ is continuously differentiable, decreasing and such that:

$$\lim_{\beta \rightarrow -\infty} \langle \mathcal{H} \rangle_{\mu_\beta} = \max_{\omega} \mathcal{H}(\omega), \quad \lim_{\beta \rightarrow +\infty} \langle \mathcal{H} \rangle_{\mu_\beta} = \min_{\omega} \mathcal{H}(\omega).$$

DEFINITION 1.6 (Canonical Gibbs Distribution).

The *canonical Gibbs distribution* at parameter β associated with a system in a domain Λ is the probability distribution on Ω_Λ defined by:

$$\mu_\beta(\omega) := \frac{e^{-\beta \mathcal{H}(\omega)}}{\mathbf{Z}_{\Lambda; \beta}},$$

where $e^{\beta \mathcal{H}(\omega)}$ is called the *Boltzmann weight*, and the normalisation

$$\mathbf{Z}_{\Lambda; \beta} := \sum_{\omega \in \Omega_\Lambda} e^{-\beta \mathcal{H}(\omega)}$$

is called the *canonical partition function*.

EXERCISE 1.4. By using the *Maximum Entropy Principle*, determine the probability of maximal entropy measure $\nu = \{\mu(1), \dots, \mu(6)\}$ for the outcomes of a dice whose expected value is 4.

SOLUTION. The desired probabilities are given by

$$\mu(i) = \frac{e^{-\beta i}}{\mathbf{Z}_\beta},$$

where $\mathbf{Z}_\beta = \sum_{i=1}^6 e^{-\beta i}$ and β must be chosen such that $\sum_i i \mu(i) = 4$. Numerically, one finds that

$$\mu(1) \approx 0.10, \quad \mu(2) \approx 0.12, \quad \mu(3) \approx 0.15, \quad \mu(4) \approx 0.17, \quad \mu(5) \approx 0.21, \quad \mu(6) \approx 0.25.$$

□

SECTION 1.2

| Linking Statistical Mechanics and Thermodynamics

Boltzmann's Principle and the Thermodynamic Limit

DEFINITION 1.7 (Boltzmann Entropy).

Given the energy shell

$$\Omega_{\Lambda;U} := \{\omega \in \Omega_{\Lambda} : \mathcal{H}(\omega) = U\},$$

we define the *Boltzmann entropy* S_{Boltz} of the system as:¹

$$S_{\text{Boltz}}(\Lambda; U) := \log |\Omega_{\Lambda;U}|.$$

Clearly $S_{\text{Sh}}(\mu_{\Lambda;U}^{\text{Mic}}) = S_{\text{Boltz}}(\Lambda; U)$. Therefore, all the thermodynamic quantities can be derived by the microcanonical ensemble, once a *thermodynamical limit* is performed. In particular, the *macroscopic temperature* T of the system is related to the entropy by the relation:

$$\frac{1}{T} = \left(\frac{\partial S_{\text{Boltz}}}{\partial U} \right)_V,$$

where $\frac{1}{T}$ is called the *inverse (absolute) temperature* and usually denoted by β . Identifying the Shannon entropy S_{Sh} of the canonical ensemble with the Boltzmann entropy, the parameter β can be interpreted as the *inverse temperature*.

LEMMA 1.8.

We have in fact that

$$\frac{\partial S_{\text{Sh}}(\mu_{\beta})}{\partial U} = \beta(U),$$

with $\beta(U)$ such that $\sum_{\omega \in \Omega_{\Lambda}} \mu_{\beta} \mathcal{H}(\omega) = U$.

Proof. By the definition of Shannon entropy and Gibbs measures we have:

$$S_{\text{Sh}}(\mu_{\beta}) = \beta \langle \mathcal{H} \rangle_{\mu_{\beta}} + \log \mathbf{Z}_{\Lambda;\beta} = \beta U + \log \mathbf{Z}_{\Lambda;\beta}.$$

By differentiability of the map $U \mapsto \beta$, it follows:

$$\frac{\partial S_{\text{Sh}}(\mu_{\beta})}{\partial U} = \frac{\partial \beta}{\partial U} U + \beta + \frac{\partial \log \mathbf{Z}_{\Lambda;\beta}}{\partial \beta} \frac{\partial \beta}{\partial U} = \frac{\partial \beta}{\partial U} U + \beta - U \frac{\partial \beta}{\partial U} = \beta.$$

□

Remark. Notice that:

$$F = U - TS_{\text{Sh}}(\mu_{\beta}) = -\frac{1}{\beta} \log \mathbf{Z}_{\Lambda;\beta}.$$

So that $-\frac{1}{\beta} \log \mathbf{Z}_{\Lambda;\beta}$ will be called *free energy of the canonical ensemble*. It contains all the thermodynamical properties of the system at equilibrium.

¹ We will assume that the units are chosen so that the Boltzmann's constant $k_B = 1$.

SECTION 1.3

| Magnetic Systems

Phenomenology: Paramagnets and Ferromagnets

A *ferromagnet* exhibits *spontaneous magnetisation*: global ordering of the *spins* (better: magnetic dipole of the Curie-Weiss domain) even in the absence of the external magnetic field.

If $T < T_c$, where T_c is called the *Curie temperature*, as the field goes to zero, the magnetisation has a discontinuity. This corresponds to a *first-order phase transition*. We will introduce a very simplified microscopic model (i.e., the *Ising model*) in order to explain this macro ordering of the spins.

A Simple Model for a Magnet: the Ising Model

The *Ising model* has been the first system of *locally interacting particles* for which it was possible to prove the existence of a *phase transition*. It models the magnetic atoms in a *crystalline structure*.

We consider a *non-oriented graph* $G = (\Lambda, \mathcal{E})$, whose set of vertices Λ is a subset of $\mathbb{Z}^d$ (i.e., a lattice). The edges of the graph will be between *nearest neighbors* (*n.n.*): i, j such that $\|i - j\|_1 = 1$. We write $i \sim j$ to indicate that the sites i, j are *n.n.*

We will consider lattices Λ consisting of *boxes of radius n* :

$$B(n) := \{-n, \dots, n\}^d.$$

At each vertex i we assign a *spin variable* ω_i attaining two possible values: $+1$ (spin up) or -1 (spin down). Hence, $\omega_i \in \{-1, +1\}$. A *microstate* ω of the system is a configuration $\omega \in \Omega_\Lambda$, where $\Omega_\Lambda := \{-1, +1\}^\Lambda$.

The *Hamiltonian* of the Ising model is given by:

$$\mathcal{H}_\Lambda(\omega) := - \sum_{\substack{i, j \in \Lambda \\ i \sim j}} \omega_i \omega_j - h \sum_{i \in \Lambda} \omega_i, \quad \omega \in \Omega_\Lambda, \quad (1.1)$$

where the previous term is the pair interaction that favours alignment of *n.n.* spins, and the latter term is the interaction with an external magnetic field $h \in \mathbb{R}$, which favours alignment with this field.

The *Gibbs distribution* is:

$$\mu_{\Lambda; \beta}(\omega) := \frac{e^{-\beta \mathcal{H}_\Lambda(\omega)}}{\mathbf{Z}_{\Lambda; \beta}}.$$

In case $h = 0$, it is *invariant* under global spin-flip, i.e., $\mu_{\Lambda; \beta}(\omega) = \mu_{\Lambda; \beta}(-\omega)$.

EXAMPLE 1.9.

Thermodynamic Behaviour

We study the Ising model in a *large region* Λ and take the *thermodynamic limit* (e.g., if $\Lambda = B(n)$, taking the limit $n \rightarrow \infty$). We consider now the simpler case $h = 0$. In order to quantify the

notion of *global order*, we introduce the *total magnetisation*:

$$M_\Lambda(\omega) := \sum_{i \in \Lambda} \omega_i.$$

Then, the magnetisation density:

$$m_\Lambda(\omega) := \frac{M_\Lambda(\omega)}{|\Lambda|} \in [-1, 1].$$

LEMMA 1.10.

The average magnetisation is zero at all temperatures:

$$\langle M_\Lambda \rangle_{\mu_{\Lambda, \beta}} = 0.$$

Proof. Since $M_\Lambda(-\omega) = -M_\Lambda(\omega)$,

$$\langle M_\Lambda \rangle_{\mu_{\Lambda, \beta}} = \sum_{\omega \in \Omega_\Lambda} M_\Lambda(\omega) \mu_{\Lambda, \beta}(\omega) = \frac{1}{2} \sum_{\omega \in \Omega_\Lambda} M_\Lambda(\omega) \underbrace{\{\mu_{\Lambda, \beta}(\omega) - \mu_{\Lambda, \beta}(-\omega)\}}_{=0} = 0.$$

□

This means that in average the densities of + and − are equal. We want to study now the *fluctuations* of M_Λ around its typical value. To give the heuristics, we take Λ and study two trivial regimes: *infinite temperature* $\beta = \frac{1}{T} \rightarrow 0$ and *zero temperature* $\beta = \frac{1}{T} \rightarrow \infty$.

Now we are interested at the *high temperature limit* of the measure. We have that for the Gibbs canonical measures:

$$\lim_{\beta \rightarrow 0} \mu_{\Lambda, \beta}(\omega) = \frac{1}{|\Omega_\Lambda|},$$

which converges to the uniform distribution on $\Omega_{B(n)}$ ($\omega \sim \mathcal{U}(\Omega_\Lambda)$). In this regime M_Λ is a sum of *independent and identically distributed (i.i.d.)* random variables.

Now we are interested at the *low temperature limit* of the measure. The Gibbs measure concentrates on configurations that minimise the Hamiltonian (*ground states*). In the Ising model when $h = 0$, there are two ground states: the configuration with all +1 spins (i.e., $+1_i = 1, \forall i \in \Lambda$); the configuration with all −1 spins (i.e., $+1_i = -1, \forall i \in \Lambda$). For any configuration $\omega \notin \{-1, +1\}$, there is at least a *n.n.* pair $\{i, j\}$ such that $\omega_i \neq \omega_j$, so that:

$$\mathcal{H}_\Lambda(\omega) - \mathcal{H}_\Lambda(\pm 1) \geq 2.$$

Consequently,

$$\frac{\mu_{\Lambda, \beta}(\omega)}{\mu_{\Lambda, \beta}(\pm 1)} = \frac{e^{-\beta \mathcal{H}_\Lambda(\omega)}}{e^{-\beta \mathcal{H}_\Lambda(\pm 1)}} \leq e^{-2\beta} \rightarrow 0 \quad \text{as } \beta \rightarrow \infty.$$

Since $\mu_{\Lambda, \beta}(+1) = \mu_{\Lambda, \beta}(-1)$, we have:

$$\lim_{\beta \rightarrow \infty} \mu_{\Lambda, \beta}(\omega) = \begin{cases} \frac{1}{2}, & \omega \in \{-1, +1\} \\ 0, & \text{otherwise} \end{cases}.$$

The two limiting regimes suggest two possible scenarios for high and low temperature behaviour of the Ising model in a large box:

- when β is small, the magnetisation density is close to zero, i.e., with high probability:

$$\frac{M_\Lambda}{|\Lambda|} \approx 0$$

- in a typical configuration the fractions of $+$ and of -1 are almost equal;
- when β is large, with high probability:

$$\frac{M_\Lambda}{|\Lambda|} \approx +1 \quad \text{or} \quad \frac{M_\Lambda}{|\Lambda|} \approx -1.$$

In this scenario, *spontaneous magnetisation (global order)* is observed: a majority of spin has the same sign. The *Law of Large Numbers* does not hold in this regime (dependence). The global spin-flip symmetry is *spontaneously broken*.

The main problem is to determine which of these behaviours (if any) gives the correct description of the system for intermediate values $0 < \beta < \infty$.

Consider a *deformation* of the *ground states (excitations)* at exactly k vertices (in the other it coincides with the ground state). Denote with $\mathcal{A}_k$ the set of all the configurations coinciding with either $+1$ or -1 , except at exactly k vertices. For each $\omega \in \mathcal{A}_k$:

$$\mathcal{H}_\Lambda(\omega) - \mathcal{H}_\Lambda(\pm 1) \leq 8k.$$

Observe that

$$|\mathcal{A}_k| = \binom{|\Lambda|}{k}.$$

For any $k \geq 1$, since $k < \frac{|\Lambda|}{2}$, we have:

$$\frac{\mu_{\Lambda;\beta}(\mathcal{A}_k)}{\mu_{\Lambda;\beta}(\pm 1)} = \sum_{\omega \in \mathcal{A}_k} e^{-\beta(\mathcal{H}_\Lambda(\omega) - \mathcal{H}_\Lambda(\pm 1))} \geq \binom{|\Lambda|}{k} e^{-8\beta k} \geq \frac{C}{k!} \left(\frac{1}{2} |\Lambda| e^{-8\beta} \right)^k \gg 1$$

for $|\Lambda|$ large enough at a fixed β . Even at very low temperatures, excitations are much more probable than the ground states.

We have the *usual competition between energy and entropy*: configurations at low energy are favoured by the Gibbs weight, while the number of configurations with a given number of excitations grows fast with the system size.

The Ising Model

SECTION 2.1

Finite-Volume Gibbs Distributions

Finite Volumes with Free Boundary Condition

The configurations of the Ising model in a finite volume $\Lambda \Subset \mathbb{Z}^d$ with free boundary condition are the elements of the set

$$\Omega_\Lambda := \{-1, 1\}^\Lambda.$$

A configuration $\omega \in \Omega_\Lambda$ is thus of the form $\omega = (\omega_i)_{i \in \Lambda}$. The basic random variable associated to the model is the *spin* at a vertex $i \in \mathbb{Z}^d$, which is the random variable $\sigma_i : \Omega_\Lambda \rightarrow \{-1, 1\}$ defined by $\sigma_i(\omega) := \omega_i$.

We will often identify a finite set Λ with the graph that contains all edges formed by *n.n.* pairs of vertices of Λ . We denote the latter set of edges by

$$\mathcal{E}_\Lambda := \{\{i, j\} \subset \Lambda : i \sim j\}.$$

To each configuration $\omega \in \Omega_\Lambda$, we associate its *energy*, given by the Hamiltonian

$$\mathcal{H}_{\Lambda, \beta, h}^\emptyset(\omega) := -\beta \sum_{\{i, j\} \in \mathcal{E}_\Lambda} \sigma_i(\omega) \sigma_j(\omega) - h \sum_{i \in \Lambda} \sigma_i(\omega),$$

where $\beta \in \mathbb{R}^*$ is the inverse temperature and $h \in \mathbb{R}$ is the magnetic field. The superscript $\emptyset$ indicates that this model has *free boundary condition*: spins in Λ do not interact with other spins located outside of Λ .

DEFINITION 2.1.

The Gibbs distribution of the Ising model in Λ with *free boundary condition*, at parameters β and h , is the distribution on Ω_Λ defined by

$$\mu_{\Lambda, \beta, h}^\emptyset(\omega) := \frac{1}{Z_{\Lambda, \beta, h}^\emptyset} \exp\left(-\mathcal{H}_{\Lambda, \beta, h}^\emptyset(\omega)\right).$$

The normalisation constant

$$Z_{\Lambda, \beta, h}^\emptyset := \sum_{\omega \in \Omega_\Lambda} \exp\left(-\mathcal{H}_{\Lambda, \beta, h}^\emptyset(\omega)\right)$$

is called the *partition function* in Λ with free boundary condition.

Finite Volumes with Periodic Boundary Condition

We now consider the Ising model on the torus $\mathbb{T}_n$. Its set of vertices is given by

$$V_n := \{0, \dots, n-1\}^d,$$

and there is an edge between each pair of vertices $i = (i_1, \dots, i_d)$, $j = (j_1, \dots, j_d)$ such that $i - j$ has only one nonzero component and the latter is equal to $\pm 1 \pmod n$. We denote by $\mathcal{E}_{V_n}^{\text{per}}$ the set of edges of $\mathbb{T}_n$.

Configurations of the model are now the elements of $\{-1, 1\}^{V_n}$ and have an energy given by

$$\mathcal{H}_{V_n; \beta, h}^{\text{per}}(\omega) := -\beta \sum_{\{i, j\} \in \mathcal{E}_{V_n}^{\text{per}}} \sigma_i(\omega) \sigma_j(\omega) - h \sum_{i \in V_n} \sigma_i(\omega).$$

DEFINITION 2.2.

The Gibbs distribution of the Ising model in V_n with *periodic boundary condition*, at parameters β and h , is the probability distribution on $\{-1, 1\}^{V_n}$ defined by

$$\mu_{V_n; \beta, h}^{\text{per}}(\omega) := \frac{1}{Z_{V_n; \beta, h}^{\text{per}}} \exp\left(-\mathcal{H}_{V_n; \beta, h}^{\text{per}}(\omega)\right).$$

The normalisation constant

$$Z_{V_n; \beta, h}^{\text{per}} := \sum_{\omega \in \Omega_{V_n}} \exp\left(-\mathcal{H}_{V_n; \beta, h}^{\text{per}}(\omega)\right)$$

is called the *partition function* in V_n with periodic boundary condition.

Finite Volumes with Configurations as Boundary Condition

¹Consider configurations of the Ising model on the infinite lattice $\mathbb{Z}^d$, that is, elements of

$$\Omega := \{-1, 1\}^{\mathbb{Z}^d}.$$

Fixing a finite set $\Lambda \subset \mathbb{Z}^d$ and a configuration $\eta \in \Omega$, we define a *configuration of the Ising model in Λ with boundary condition η* as an element of the finite set

$$\Omega_{\Lambda}^{\eta} := \{\omega \in \Omega : \omega_i = \eta_i, \forall i \notin \Lambda\}.$$

The *energy* of a configuration $\omega \in \Omega_{\Lambda}^{\eta}$ is defined by

$$\mathcal{H}_{\Lambda; \beta, h}^{\eta}(\omega) := -\beta \sum_{\{i, j\} \in \mathcal{E}_{\Lambda}^b} \sigma_i(\omega) \sigma_j(\omega) - h \sum_{i \in \Lambda} \sigma_i(\omega),$$

where we have introduced

$$\mathcal{E}_{\Lambda}^b := \{(i, j) \subset \mathbb{Z}^d : i, j \cap \Lambda \neq \emptyset, i \sim j\}.$$

Note that $\mathcal{E}_{\Lambda}^b$ differs from $\mathcal{E}_{\Lambda}$ by the addition of all the edges connecting vertices inside Λ to their neighbours outside Λ .

¹ It will turn out to be useful to consider the Ising model on the full lattice $\mathbb{Z}^d$, but with configurations which are frozen outside a finite set.

DEFINITION 2.3.

The Gibbs distribution of the Ising model in Λ with boundary condition η , at parameters β and h , is the probability distribution on Ω_Λ^η defined by

$$\mu_{\Lambda;\beta,h}^\eta(\omega) := \frac{1}{Z_{\Lambda;\beta,h}^\eta} \exp\left(-\mathcal{H}_{\Lambda,\beta,h}^\eta(\omega)\right).$$

The normalisation constant

$$Z_{\Lambda;\beta,h}^\eta := \sum_{\omega \in \Omega_\Lambda^\eta} \exp\left(-\mathcal{H}_{\Lambda,\beta,h}^\eta(\omega)\right)$$

is called the *partition function* with η -boundary condition.

Two boundary conditions play a particularly important role in the analysis of the Ising model: the $+$ boundary condition η^+ , for which $\eta_i^+ := +1$ for all i , and the $-$ boundary condition η^- , similarly defined by $\eta_i^- := -1$ for all i . The corresponding Gibbs distributions will be simply denoted by $\mu_{\Lambda;\beta,h}^+$ and $\mu_{\Lambda;\beta,h}^-$; similarly, we will write Ω_Λ^+ , Ω_Λ^- for the corresponding sets of configurations.

In the following, we will use the symbol $\#$ to denote a generic type of boundary condition. For instance, $Z_{\Lambda;\beta,h}^\#$ can denote $Z_{\Lambda;\beta,h}^\emptyset$, $Z_{V_n;\beta,h}^{\text{per}}$, or $Z_{\Lambda;\beta,h}^\eta$. In the case of periodic boundary condition, Λ will always implicitly be assumed to be a cube.

Following the custom in statistical physics, expectation of a function of ω with respect to a probability distribution μ will be denoted by a bracket: $\langle f \rangle_\mu$. When the distribution is identified by indices, we will apply the same indices to the bracket. For example, expectation of a function f under $\mu_{\Lambda;\beta,h}^\#$ will be denoted by

$$\langle f \rangle_{\Lambda;\beta,h}^\# := \sum_{\omega \in \Omega_\Lambda^\#} f(\omega) \mu_{\Lambda;\beta,h}^\#(\omega).$$

We will often use $\langle \cdot \rangle_{\Lambda;\beta,h}^\#$ and $\mu_{\Lambda;\beta,h}^\#(\cdot)$ interchangeably.

SECTION 2.2

| Thermodynamic Limit, Pressure and Magnetisation

Convergence of Subsets

To define the Ising model on the whole lattice $\mathbb{Z}^d$ (one often says “in infinite volume”), the *thermodynamic limit* will be considered along sequences of finite subsets $\Lambda_n \Subset \mathbb{Z}^d$ which converge to $\mathbb{Z}^d$, denoted by $\Lambda_n \uparrow \mathbb{Z}^d$, in the sense that

- Λ_n is *increasing*: $\Lambda_n \subset \Lambda_{n+1}$;
- Λ_n *invades* $\mathbb{Z}^d$: $\cup_{n \geq 1} \Lambda_n = \mathbb{Z}^d$.

Sometimes, in order to control the influence of the boundary condition and of the shape of the box on thermodynamic quantities, it will be necessary to impose a further regularity property on the sequence Λ_n . We say that a sequence $\Lambda_n \uparrow \mathbb{Z}^d$ *converges to $\mathbb{Z}^d$ in the sense of van Hove*, which we denote by $\Lambda_n \uparrow\!\!\uparrow \mathbb{Z}^d$, if and only if

$$\lim_{n \rightarrow \infty} \frac{|\partial^{\text{in}} \Lambda_n|}{|\Lambda_n|} = 0, \quad (2.1)$$

where $\partial^{\text{in}} \Lambda := \{i \in \Lambda : \exists j \notin \Lambda, j \sim i\}$. The simplest sequence to satisfy this condition is the sequence

$$B(n) := \{-n, \dots, n\}^d.$$

EXERCISE 2.1. Let $B(n) = \{-n, \dots, n\}^d$ for all $n \in \mathbb{N}$. Show that $B(n) \uparrow\!\!\uparrow \mathbb{Z}^d$ as $n \rightarrow \infty$ in the sense of van Hove. Give an example of an increasing sequence $(\Lambda_n)_n$ which converges to $\mathbb{Z}^d$, but not in the sense of van Hove.

SOLUTION. Notice that $|B(n)| = (2n+1)^d$ and that

$$|\partial^{\text{in}} B(n)| = |B(n) \setminus B(n-1)| = (2n+1)^d - (2n-1)^d \leq d(2n+1)^{d-1},$$

which shows that

$$\frac{|\partial^{\text{in}} B(n)|}{|B(n)|} \rightarrow 0.$$

Any sequence $\Lambda_n \uparrow \mathbb{Z}^d$ whose boundary grows as fast as its volume, such as

$$\Lambda_n = B(n) \cup \{(i, 0, \dots, 0) \in \mathbb{Z}^d : 0 \leq i \leq e^n\},$$

will not converge in the sense of van Hove. □

Pressure

DEFINITION 2.4 (Pressure).

The *pressure in $\Lambda \subset \mathbb{Z}^d$* , with boundary condition of the type $\#$, is defined by

$$\psi_\Lambda^\#(\beta, h) := \frac{1}{|\Lambda|} \log Z_{\Lambda; \beta, h}^\#.$$

EXERCISE 2.2. Show that for all $\Lambda \subset \mathbb{Z}^d$, $\beta \geq 0$ and $h \in \mathbb{R}$ we have that

$$\Psi_\Lambda^\varnothing(\beta, h) = \Psi_\Lambda^\varnothing(\beta, -h), \quad \Psi_\Lambda^{\text{per}}(\beta, h) = \Psi_\Lambda^{\text{per}}(\beta, -h), \quad \Psi_\Lambda^+(\beta, h) = \Psi_\Lambda^+(\beta, -h).$$

LEMMA 2.5.

For each type of boundary condition $\#$, $(\beta, h) \mapsto \psi_\Lambda^\#(\beta, h)$ is convex.

Proof. We consider $\psi_\Lambda^\eta(\beta, h)$, but the other cases are similar. Let $\alpha \in [0, 1]$. Since $\mathcal{H}_{\Lambda; \beta, h}$ is an affine function of the pair (β, h) , Hölder's inequality yields

$$\begin{aligned} Z_{\Lambda; \alpha\beta_1 + (1-\alpha)\beta_2, \alpha h_1 + (1-\alpha)h_2}^\eta &= \sum_{\omega \in \Omega_\Lambda^\eta} e^{-\alpha \mathcal{H}_{\Lambda; \beta_1, h_1}(\omega) - (1-\alpha) \mathcal{H}_{\Lambda; \beta_2, h_2}(\omega)} \\ &\leq \left(\sum_{\omega \in \Omega_\Lambda^\eta} e^{-\mathcal{H}_{\Lambda; \beta_1, h_1}(\omega)} \right)^\alpha \left(\sum_{\omega \in \Omega_\Lambda^\eta} e^{-\mathcal{H}_{\Lambda; \beta_2, h_2}(\omega)} \right)^{1-\alpha}. \end{aligned}$$

Therefore, ψ_Λ^η is convex:

$$\psi_\Lambda^\eta(\alpha\beta_1 + (1-\alpha)\beta_2, \alpha h_1 + (1-\alpha)h_2) \leq \alpha \psi_\Lambda^\eta(\beta_1, h_1) + (1-\alpha) \psi_\Lambda^\eta(\beta_2, h_2).$$

□

Of course, the finite-volume pressure $\psi_\Lambda^\#$ depends on Λ and on the boundary condition used. However, as the following theorem shows, when Λ is so large that $|\Lambda| \gg |\partial\Lambda|$, the boundary condition and the shape of Λ only provide negligible corrections: there exists a function $\psi(\beta, h)$ such that

$$\psi_\Lambda^\#(\beta, h) = \psi(\beta, h) + \mathcal{O}\left(\frac{|\partial\Lambda|}{|\Lambda|}\right).$$

$\psi(\beta, h)$ then provides a better candidate for the corresponding thermodynamic potential, since the latter does not depend on the “details” of the observed system, such as its shape.

THEOREM 2.6.

In the thermodynamic limit, the pressure

$$\psi(\beta, h) := \lim_{\Lambda \uparrow \mathbb{Z}^d} \psi_\Lambda^\#(\beta, h)$$

is well defined and independent of the sequence $\Lambda \uparrow \mathbb{Z}^d$ and of the type of boundary condition. Moreover, ψ is convex (as a function on $\mathbb{R}^* \times \mathbb{R}$) and is even as a function of h .

Proof. Existence of the limit. We start by proving convergence in the case of free boundary condition. The proof is done in two steps. We will first show existence of the limit $\lim_{n \rightarrow \infty} \psi_{D_n}^\varnothing(\beta, h)$ where $D_n := \{1, 2, \dots, 2^n\}^d$. After that, we extend the convergence to any sequence $\Lambda_n \uparrow \mathbb{Z}^d$. Since the pair (β, h) is fixed, we will omit it from the notations most of the time, until the end of the proof.

The pressure associated to the box D_{n+1} will be shown to be close to the one associated to the box D_n . Indeed, let us decompose D_{n+1} into 2^d disjoint translates of D_n , denoted by $D_n^{(1)}, \dots, D_n^{(2^d)}$.

The energy of ω in D_{n+1} can be written as

$$\mathcal{H}_{D_{n+1}}^\varnothing = \sum_{i=1}^{2^d} \mathcal{H}_{D_n^{(i)}}^\varnothing + R_n,$$

where R_n represents the energy of interaction between pairs of spins that belong to different sub-boxes. Since each face of D_{n+1} contains $(2^{n+1})^{d-1}$ points, we have $|R_n(\omega)| \leq \beta d (2^{n+1})^{d-1}$. To obtain an upper

bound on the partition function, we can write

$$\mathcal{H}_{D_{n+1}}^{\varnothing} \geq -\beta d (2^{n+1})^{d-1} + \sum_{i=1}^{2^d} \mathcal{H}_{D_n^{(i)}}^{\varnothing},$$

which yields

$$\mathbf{Z}_{D_{n+1}}^{\varnothing} \leq e^{\beta d 2^{(n+1)(d-1)}} \sum_{\omega \in \Omega_{D_{n+1}}} \prod_{i=1}^{2^d} \exp\left(-\mathcal{H}_{D_n^{(i)}}^{\varnothing}(\omega)\right).$$

Splitting the sum over $\omega \in D_{n+1}$ into 2^d sums over $\omega^{(i)} \in D_n^{(i)}$,

$$\sum_{\omega \in \Omega_{D_{n+1}}} \prod_{i=1}^{2^d} \exp\left(-\mathcal{H}_{D_n^{(i)}}^{\varnothing}(\omega)\right) = \prod_{i=1}^{2^d} \sum_{\omega^{(i)} \in \Omega_{D_n^{(i)}}} \exp\left(-\mathcal{H}_{D_n^{(i)}}^{\varnothing}(\omega)\right) = \left(\mathbf{Z}_{D_n}^{\varnothing}\right)^{2^d},$$

where we have used the fact that $\mathbf{Z}_{D_n^{(i)}}^{\varnothing} = \mathbf{Z}_{D_n}^{\varnothing}$ for all i . A lower bound can be obtained in a similar fashion, leading to

$$e^{-\beta d 2^{(n+1)(d-1)}} \left(\mathbf{Z}_{D_n}^{\varnothing}\right)^{2^d} \leq \mathbf{Z}_{D_{n+1}}^{\varnothing} \leq e^{\beta d 2^{(n+1)(d-1)}} \left(\mathbf{Z}_{D_n}^{\varnothing}\right)^{2^d}.$$

After taking the logarithm, dividing by $|D_{n+1}| = 2^{d(n+1)}$ and taking n large enough,

$$\left|\psi_{D_{n+1}}^{\varnothing} - \psi_{D_n}^{\varnothing}\right| \leq \beta d 2^{-(n+1)}.$$

This implies that ψ_{D_n} is a Cauchy sequence: for all $n \leq m$,

$$\left|\psi_{D_m}^{\varnothing} - \psi_{D_n}^{\varnothing}\right| \leq \beta d \sum_{k=n+1}^m 2^{-k} = \beta d (2^{-n} - 2^{-m}).$$

Therefore, $\lim_{n \rightarrow \infty} \psi_{D_n}^{\varnothing}$ exists; we denote it by ψ .

Let us now consider an arbitrary sequence $\Lambda_n \uparrow \mathbb{Z}^d$. We fix some integer k and consider a partition of $\mathbb{Z}^d$ into adjacent disjoint translates of D_k . For each n , consider a minimal covering of Λ_n by elements $D_k^{(j)}$ of the partition, and let $[\Lambda_n] := \cup_j D_k^{(j)}$. We use the estimate

$$\left|\psi_{\Lambda_n}^{\varnothing} - \psi\right| \leq \left|\psi_{\Lambda_n}^{\varnothing} - \psi_{[\Lambda_n]}^{\varnothing}\right| + \left|\psi_{[\Lambda_n]}^{\varnothing} - \psi_{D_k}^{\varnothing}\right| + \left|\psi_{D_k}^{\varnothing} - \psi\right|. \quad (2.2)$$

Fix $\epsilon > 0$. Since $\psi_{D_k}^{\varnothing} \rightarrow \psi$ when $k \rightarrow \infty$, there exists k_0 , depending on β and ϵ , such that $\left|\psi_{D_k}^{\varnothing} - \psi\right| \leq \frac{\epsilon}{3}$ for all $k \geq k_0$. We then compute $\psi_{\Lambda_n}^{\varnothing}$ by writing

$$\mathcal{H}_{[\Lambda_n]}^{\varnothing} = \sum_j \mathcal{H}_{D_k^{(j)}}^{\varnothing} + W_n,$$

where

$$|W_n| \leq \beta \frac{|\partial[\Lambda_n]|}{|D_k|} d (2^k)^{d-1} = \beta d 2^{-k} |[\Lambda_n]|.$$

Therefore, there exists k_1 (also depending on β and ϵ) such that

$$\left|\psi_{\Lambda_n}^{\varnothing} - \psi_{D_k}^{\varnothing}\right| \leq \beta d 2^{-k} < \frac{\epsilon}{3},$$

for all $k \geq k_1$. Let us then fix $k \geq \max\{k_0, k_1\}$. Let us write $\Delta_n := [\Lambda_n] \setminus \Lambda_n$. We observe that

$$\left|\mathcal{H}_{[\Lambda_n]}^{\varnothing} - \mathcal{H}_{\Lambda_n}^{\varnothing}\right| \leq (2d\beta + |h|)|\Delta_n|.$$

Therefore,

$$\mathbf{Z}_{\Lambda_n}^{\varnothing} = \sum_{\omega \in \Omega_{[\Lambda_n]}} e^{-\mathcal{H}_{[\Lambda_n]}^{\varnothing}(\omega)} \leq \sum_{\omega \in \Omega_{\Lambda_n}} e^{-\mathcal{H}_{\Lambda_n}^{\varnothing}(\omega)} \sum_{\omega' \in \Omega_{\Delta_n}} e^{(2d\beta + |h|)|\Delta_n|} = e^{(2d\beta + |h| + \log 2)|\Delta_n|} \mathbf{Z}_{\Lambda_n}^{\varnothing}.$$

Proceeding similarly to get a lower bound and observing that Δ_n contains at most $|\partial[\Lambda_n]|D_k$ vertices, this yields

$$\left| \log \mathbf{Z}_{\Lambda_n}^\varnothing - \log \mathbf{Z}_{[\Lambda_n]}^\varnothing \right| \leq |\partial^{\text{in}}[\Lambda_n]|D_k(2d\beta + |h| + \log 2). \quad (2.3)$$

Since

$$1 \leq \frac{|[\Lambda_n]|}{|\Lambda_n|} \leq 1 + \frac{|\partial^{\text{in}}[\Lambda_n]|D_k}{|\Lambda_n|}$$

and since ψ_Λ^η is uniformly bounded (for example, by $2d\beta + |h| + \log 2$), it follows from (2.1) and (2.3) that

$$\left| \psi_{\Lambda_n}^\varnothing - \psi_{[\Lambda_n]}^\varnothing \right| \leq \frac{\epsilon}{3}$$

for all n large enough. Combining all these estimates, we conclude from (2.2) that, when n is sufficiently large,

$$|\psi_\Lambda^\omega - \psi| \leq \epsilon.$$

Independence of boundary condition. Let $\Lambda \Subset \mathbb{Z}^d$, $\eta \in \Omega$ and $\omega \in \Omega_\Lambda$. Denote by ω' the configuration in Ω_Λ^η coinciding with ω inside Λ . Then,

$$|\mathcal{H}_\Lambda(\omega') - \mathcal{H}_\Lambda^\varnothing(\omega)| \leq 2d\beta|\partial^{\text{in}}\Lambda|.$$

This observation implies that

$$e^{-\beta 2d|\partial^{\text{in}}\Lambda|} \mathbf{Z}_\Lambda^\varnothing \leq \mathbf{Z}_\Lambda^\eta \leq e^{\beta 2d|\partial^{\text{in}}\Lambda|} \mathbf{Z}_\Lambda^\varnothing.$$

Applying this to each Λ_n and using (2.1) shows that $\lim_{\Lambda_n \uparrow \mathbb{Z}^d} \psi_{\Lambda_n}^\eta$ exists and coincides with ψ . A completely similar argument, comparing $\mathbf{Z}_{\Lambda_n}^\varnothing$ and $\mathbf{Z}_{\Lambda_n}^{\text{per}}$, shows that $\lim_{n \rightarrow \infty} \psi_{\Lambda_n}^{\text{per}} = \psi$.

Convexity. Since $(\beta, h) \mapsto \psi^\alpha(\beta, h)$ is convex (Lemma 2.5), its limit $\Lambda \uparrow \mathbb{Z}^d$ is also convex.

Symmetry. The fact that $h \mapsto \psi(\beta, h)$ is even is a direct consequence of the above. $\square$

Magnetisation

The *magnetisation density* in $\Lambda \Subset \mathbb{Z}^d$ is the random variable

$$m_\Lambda := \frac{M_\Lambda}{|\Lambda|},$$

where $M_\Lambda := \sum_{i \in \Lambda} \sigma_i$ is the *total magnetisation*. We also define, for any $\Lambda \Subset \mathbb{Z}^d$,

$$m_\Lambda^\#(\beta, h) := \langle m_\Lambda \rangle_{\Lambda; \beta, h}^\#.$$

As can be easily checked,

$$m_\Lambda^\#(\beta, h) = \frac{\partial \psi_\Lambda^\#}{\partial h}(\beta, h).$$

EXERCISE 2.3. Show that the cumulant generating function associated with M_Λ can be expressed as

$$\log \left(\langle e^{t\beta M_\Lambda} \rangle_{\Lambda; \beta, h}^\# \right) = |\Lambda| (\Psi_\Lambda^\#(\beta, h + t) - \Psi_\Lambda^\#(\beta, h)).$$

Deduce the r -th cumulant $c_r(M_\Lambda)$.

COROLLARY 2.7.

For all $h \notin \mathfrak{B}_\beta$, the average magnetisation density

$$m(\beta, h) := \lim_{\Lambda \uparrow \mathbb{Z}^d} m_\Lambda^\#(\beta, h)$$

is well defined, independent of the sequence $\Lambda \uparrow \mathbb{Z}^d$ and of the boundary condition and satisfies

$$m(\beta, h) = \frac{\partial \psi}{\partial h}(\beta, h). \quad (2.4)$$

Moreover, the function $h \mapsto m(\beta, h)$ is non-decreasing on $\mathbb{R} \setminus \mathfrak{B}_\beta$ and is continuous at every $h \notin \mathfrak{B}_\beta$. It is however discontinuous at each $h \in \mathfrak{B}_\beta$; for any $h \in \mathfrak{B}_\beta$,

$$\lim_{h' \uparrow h} m(\beta, h') = \frac{\partial \psi}{\partial h^-}(\beta, h), \quad \lim_{h' \downarrow h} m(\beta, h') = \frac{\partial \psi}{\partial h^+}(\beta, h).$$

In particular, the spontaneous magnetisation

$$m^*(\beta) := \lim_{h \downarrow 0} m(\beta, h)$$

is always well defined.

SECTION 2.3

| The One-Dimensional Ising model

THEOREM 2.8.

For all $\beta \geq 0$ and all $h \in \mathbb{R}$, the pressure $\psi(\beta, h)$ of the one-dimensional Ising model is given by

$$\psi(\beta, h) = \log \left\{ e^\beta \cosh(h) + \sqrt{e^{2\beta} \cosh^2(h) - 2 \sinh(2\beta)} \right\}. \quad (2.5)$$

Proof.

□

The explicit expression (2.5) shows that $h \mapsto \psi(\beta, h)$ is differentiable (real-analytic in fact) everywhere, for all $\beta \geq 0$, thus showing that $\mathfrak{B}_\beta = \emptyset$ when $d = 1$.

Consequently, as seen in [Corollary 2.7](#), the average magnetisation density $m(\beta, h)$ is given by

$$m(\beta, h) = \frac{\partial \psi}{\partial h}(\beta, h), \quad \forall h \in \mathbb{R}.$$

Since $h \mapsto \psi(\beta, h)$ is analytic, its derivative $h \mapsto m(\beta, h)$ is also analytic, in particular continuous. Therefore,

$$m^*(\beta) = \lim_{h \downarrow 0} m(\beta, h) = m(\beta, 0).$$

But, since $\psi(\beta, h) = \psi(\beta, -h)$, we get $\frac{\partial \psi}{\partial h}(\beta, 0) = 0$. This shows that the spontaneous magnetisation of the one-dimensional Ising model is zero at all temperatures:

$$m^*(\beta) = 0, \quad \forall \beta > 0.$$

EXERCISE 2.4. Using (2.5) compute $m(\beta, h)$. Check that

$$\lim_{h \rightarrow \infty} m(\beta, h) = \pm 1, \quad \forall \beta \geq 0,$$

and

$$\lim_{\beta \rightarrow \infty} m(\beta, h) = \begin{cases} +1 & h > 0 \\ 0 & h = 0 \\ -1 & h < 0 \end{cases}.$$

SOLUTION. By a straightforward computation,

$$m_\beta(h) = \frac{\sinh(h)}{\sqrt{\sinh^2(h) + e^{-4\beta}}}.$$

□

EXERCISE 2.5. Assume that $h = 0$.

- (i) Configurations can be characterised by the collection of edges $\{i, i+1\}$ such that $\omega_i \neq \omega_{i+1}$. What is the contribution of a configuration with k such edges? Use that to compute the pressure for the one-dimensional Ising model.
- (ii) Express the partition function in terms of the variables $(\omega_1, \tau_1, \dots, \tau_{n-1})$ where $\tau_i = \omega_{i-1} \omega_i$. Use this to compute the pressure.

SOLUTION. (i) The partition function with free boundary condition can be expressed as

$$Z_{B(n);\beta,h}^{\emptyset} = \sum_{\omega_i=\pm 1} \prod_{i \in B(n)} e^{\beta \omega_i \omega_{i+1}} = e^{2\beta n} \sum_{\omega_i=\pm 1} \prod_{i=-n}^{n-1} e^{\beta(\omega_i \omega_{i+1} - 1)}.$$

Each factor in the last product is either equal to 1 (if $\omega_i = \omega_{i+1}$) or to $e^{-2\beta}$. Therefore,

$$Z_{B(n);\beta,h}^{\emptyset} = 2e^{2\beta n} \sum_{k=0}^{2n} \binom{2n}{k} (e^{-2\beta})^k = 2e^{2\beta n} (1 + e^{-2\beta})^{2n}.$$

This yields $\psi(\beta) = \log \cosh(\beta) + \log 2$, which of course coincides with (2.5).

(ii) In terms of the variables τ_i ,

$$Z_{B(n);\beta,h}^{\emptyset} = \sum_{\omega_{-n}=\pm 1} \sum_{\tau_i=\pm 1} \prod_{i=-n+1}^n e^{\beta \tau_i} = 2(e^{\beta} + e^{-\beta})^{2n}.$$

□

EXERCISE 2.6. Consider the Ising model in $1d$ with periodic boundary conditions. The Hamiltonian is given by

$$-\mathcal{H}_{N;\beta,h}(\omega) = \sum_{1 \leq i \leq N} (\sigma_i(\omega) \sigma_{i+1}(\omega) + h \sigma_i(\omega))$$

where $\sigma_{N+1} = \sigma_1$.

(i) Find a 2×2 matrix T such that

$$\mathbf{Z}_{N;\beta,h}^{\text{per}} = \sum_{\omega \in \Sigma_N} e^{-\beta \mathcal{H}_{N;\beta,h}(\omega)} = \text{Tr}(T^N).$$

(ii) Compute the pressure

$$\Psi(\beta, h) = \lim_{N \rightarrow \infty} \Psi_N^{\text{per}}(\beta, h).$$

(iii) Show that

$$\langle \sigma_1 \rangle_{N;\beta,h}^{\text{per}} = \left\langle \frac{1}{N} \sum_{i=1}^N \sigma_i \right\rangle_{N;\beta,h}^{\text{per}} = \frac{\partial}{\partial h} \Psi_N^{\text{per}}(\beta, h).$$

(iv) Compute

$$\langle \sigma_1 \rangle_{\beta,h} = \lim_{N \rightarrow \infty} \langle \sigma_1 \rangle_{N;\beta,h}^{\text{per}}.$$

SECTION 2.4

| Infinite-Volume Gibbs States

The pressure only provides information about the thermodynamical behaviour of the system in large volumes. If one is interested in the statistical properties of general observables, such as the fluctuations of the magnetisation density in a finite region or the correlations between far apart spins, one needs to understand the behaviour of the Gibbs distribution $\mu_{\Lambda;\beta,h}^\#$ in large volumes. A *state* (in infinite volume) will be identified with an assignment of an average value to each *local function*, that is, to each observable whose value only depends on finitely many spins.

DEFINITION 2.9 (Local Functions).

A function $f : \Omega \rightarrow \mathbb{R}$ is *local* if there exists $\Lambda \in \mathbb{Z}^d$ such that $f(\omega) = f(\omega')$ as soon as ω and ω' coincide on Λ . The smallest such set Λ is called the *support* of f and denoted by $\text{supp}(f)$.

EXAMPLE 2.10. The value taken by the spin at the origin, σ_0 , or the magnetisation density in a set $\Lambda \in \mathbb{Z}^d$, $m_\Lambda = \frac{1}{|\Lambda|} \sum_{i \in \Lambda} \sigma_i$, are local functions with supports given respectively by $\{0\}$ and Λ .

Remark. In the sequel, we will occasionally make the following mild abuse of notation: if $f : \Omega \rightarrow \mathbb{R}$ is a local function and $\Delta \supset \text{supp}(f)$, then, for any $\omega' \in \Omega_\Delta$, $f(\omega')$ is defined as the value of f evaluated at any configuration $\omega \in \Omega$ such that $\omega_i = \omega'_i$ for all $i \in \Delta$. (Clearly, that value does not depend on the choice of ω .)

DEFINITION 2.11 (States).

An *infinite-volume state* (or simply a *state*) is a mapping associating to each local function f a real number $\langle f \rangle$ and satisfying:

- (i) *normalisation*: $\langle 1 \rangle = 1$;
- (ii) *positivity*: if $f \geq 0$, then $\langle f \rangle \geq 0$;
- (iii) *linearity*: for any $\lambda \in \mathbb{R}$, $\langle f + \lambda g \rangle = \langle f \rangle + \lambda \langle g \rangle$.

The number $\langle f \rangle$ is called the *average* of f in the state $\langle \cdot \rangle$.

DEFINITION 2.12.

Let $\Lambda_n \uparrow \mathbb{Z}^d$ and $(\#_n)_{n \geq 1}$ be a sequence of boundary conditions. The sequence of Gibbs distributions $\left(\mu_{\Lambda_n;\beta,h}^{\#_n} \right)_{n \geq 1}$ is said to *converge to the state* $\langle \cdot \rangle$ if and only if

$$\lim_{n \rightarrow \infty} \langle f \rangle_{\Lambda_n;\beta,h}^{\#_n} = \langle f \rangle,$$

for every local function f . The state $\langle \cdot \rangle$ is then called a *Gibbs state* (at (β, h)).

Remark. We simply write, as a shorthand,

$$\langle \cdot \rangle = \lim_{n \rightarrow \infty} \langle \cdot \rangle_{\Lambda_n;\beta,h}^{\#_n}.$$

to indicate that $\langle \cdot \rangle_{\Lambda_n;\beta,h}^{\#_n}$ converges to $\langle \cdot \rangle$.

Since states are defined on the infinite lattice, it is natural to distinguish those that are *translation invariant*. The translation by $j \in \mathbb{Z}^d$, $\theta_j : \mathbb{Z}^d \rightarrow \mathbb{Z}^d$ is defined by $\theta_j i := i + j$. Translations can naturally be made to act on configurations: if $\omega \in \Omega$, then $\theta_j \omega$ is defined by $(\theta_j \omega)_i := \omega_{i-j}$.

DEFINITION 2.13 (Translation Invariant States).

A state $\langle \cdot \rangle$ is *translation invariant* if $\langle f \circ \theta_j \rangle = \langle f \rangle$ for every local function f and for all $j \in \mathbb{Z}^d$.

The following theorem shows that the constant-spin boundary conditions η^+ and η^- can be used to construct two states which will play a central role in the sequel.

THEOREM 2.14.

Let $\beta \geq 0$ and $h \in \mathbb{R}$. Along any sequence $\Lambda_n \uparrow \mathbb{Z}^d$, the finite-volume Gibbs distributions with + or – boundary condition converge to infinite-volume Gibbs states:

$$\langle \cdot \rangle_{\beta, h}^+ = \lim_{n \rightarrow \infty} \langle \cdot \rangle_{\Lambda_n; \beta, h}^+, \quad \langle \cdot \rangle_{\beta, h}^- = \lim_{n \rightarrow \infty} \langle \cdot \rangle_{\Lambda_n; \beta, h}^-.$$

The states $\langle \cdot \rangle_{\beta, h}^+, \langle \cdot \rangle_{\beta, h}^-$ do not depend on the sequence $(\Lambda_n)_{n \geq 1}$ and are both translation invariant.

Proof.

□

Remark. The theorem does not claim that $\langle \cdot \rangle_{\beta, h}^+$ and $\langle \cdot \rangle_{\beta, h}^-$ are distinct Gibbs states.

EXERCISE 2.7. Let $(\eta_n)_{n \geq 1}$ be a sequence of boundary conditions and $\Lambda_n \uparrow \mathbb{Z}^d$. Prove that there exists an increasing sequence $(n_k)_{k \geq 1}$ of integers and a Gibbs state $\langle \cdot \rangle$ such that

$$\langle \cdot \rangle = \lim_{k \rightarrow \infty} \langle \cdot \rangle_{\Lambda_{n_k}; \beta, h}^{\eta_{n_k}}$$

is well-defined.

SOLUTION. Notice that any local function can be expressed as a finite linear combination of *cylinder functions*, which are of the following form: $f(\omega) = 1$ if ω coincides, on a finite region Λ , with some configuration τ , and zero otherwise. Since each spin ω_i takes only two values, there are countably many cylinder functions, we denote them by $f_1, f_2, \dots$. Since, for each j , the sequence $\left(\langle f_j \rangle_{\Lambda_n; \beta, h}^{\eta_n} \right)_{n \geq 1}$ is bounded, a standard diagonalization argument allows one to extract a subsequence $(n_k)_{k \geq 1}$ such that

$$\lim_{k \rightarrow \infty} \langle f_j \rangle_{\Lambda_{n_k}; \beta, h}^{\eta_{n_k}}$$

exists for all j . The existence of $\langle f \rangle := \lim_{k \rightarrow \infty} \langle f \rangle_{\Lambda_{n_k}; \beta, h}^{\eta_{n_k}}$ for all local functions f follows by linearity and defines a Gibbs state. □

SECTION 2.5

Two Families of Local Functions

The construction of Gibbs states consists in proving the existence of the limit

$$\lim_{n \rightarrow \infty} \langle f \rangle_{\Lambda_n; \beta, h}^{\eta_n}$$

for each local function f . Ideally, one would like to test convergence only on a restricted family of functions. The following lemma provides two particularly convenient such families, which will be especially well suited for the use of the correlation inequalities introduced in the next section. Define, for all $A \subset \mathbb{Z}^d$,

$$\sigma_A := \prod_{j \in A} \sigma_j, \quad n_A := \prod_{j \in A} n_j,$$

where $n_j := \frac{1}{2}(1 + \sigma_j)$ is the occupation variable at j .

LEMMA 2.15.

Let f be local. There exist real coefficients $(\hat{f}_A)_{A \subset \text{supp}(f)}$ and $(\tilde{f}_A)_{A \subset \text{supp}(f)}$ such that both of the following representations hold:

$$f = \sum_{A \subset \text{supp}(f)} \hat{f}_A \sigma_A, \quad f = \sum_{A \subset \text{supp}(f)} \tilde{f}_A n_A.$$

Proof. The following orthogonality relation will be proved below: for all $B \subset \mathbb{Z}^d$ and all configurations ω, ω' ,

$$2^{-|B|} \sum_{A \subset B} \sigma_A(\tilde{\omega}) \sigma_A(\omega') = \mathbb{1}_{\{\omega_i = \omega'_i, \forall i \in B\}}. \quad (2.6)$$

Applying (2.6) with $B = \text{supp}(f)$,

$$\begin{aligned} f(\omega) &= \sum_{\omega' \in \Omega_{\text{supp}(f)}} f(\omega') \mathbb{1}_{\{\omega'_i = \omega_i, \forall i \in \text{supp}(f)\}} \\ &= \sum_{\omega' \in \Omega_{\text{supp}(f)}} f(\omega') 2^{-|\text{supp}(f)|} \sum_{A \subset \text{supp}(f)} \sigma_A(\omega) \sigma_A(\omega') \\ &= \sum_{A \subset \text{supp}(f)} \left\{ 2^{-|\text{supp}(f)|} \sum_{\omega' \in \Omega_{\text{supp}(f)}} f(\omega') \sigma_A(\omega') \right\} \sigma_A(\omega). \end{aligned}$$

This shows that the first identity holds with

$$\hat{f}_A = 2^{-|\text{supp}(f)|} \sum_{\omega' \in \Omega_{\text{supp}(f)}} f(\omega') \sigma_A(\omega').$$

Since $\sigma_A = \prod_{i \in A} (2n_i - 1)$, the second identity follows from the first one. $\square$

Remark. Checking convergence of $\left(\langle f \rangle_{\Lambda_n; \beta, h}^{\eta_n} \right)_{n \geq 1}$ for all local functions can now be reduced to showing convergence of $\left(\langle \sigma_A \rangle_{\Lambda_n; \beta, h}^{\eta_n} \right)_{n \geq 1}$ or $\left(\langle n_A \rangle_{\Lambda_n; \beta, h}^{\eta_n} \right)_{n \geq 1}$ for all finite $A \subseteq \mathbb{Z}^d$.

SECTION 2.6

Correlation Inequalities

Correlation inequalities are one of the major tools in the mathematical analysis of the Ising model. We will use them to construct $\langle \cdot \rangle_{\beta,h}^+$ and $\langle \cdot \rangle_{\beta,h}^-$ and to study many other properties.

The GKS Inequalities

As a motivation, consider the Ising model in a volume Λ , with + boundary condition. First, the ferromagnetic nature of the model makes it likely that the + boundary condition will favor a nonnegative magnetisation inside the box, at least when $h \geq 0$. Therefore, if i is any point of Λ , it seems reasonable to expect that $h \geq 0$ implies

$$\langle \sigma_i \rangle_{\Lambda;\beta,h}^+ \geq 0. \quad (2.7)$$

Similarly, knowing that the spin at some vertex j takes the value +1 should not decrease the probability of observing a + spin at another given vertex i , that is, one would expect that

$$\mu_{\Lambda;\beta,h}^+(\sigma_i = 1 \mid \sigma_j = 1) \geq \mu_{\Lambda;\beta,h}^+(\sigma_i = 1),$$

which can equivalently be written

$$\mu_{\Lambda;\beta,h}^+(\sigma_i = 1, \sigma_j = 1) \geq \mu_{\Lambda;\beta,h}^+(\sigma_i = 1) \mu_{\Lambda;\beta,h}^+(\sigma_j = 1).$$

Since $\mathbb{1}_{\{\sigma_i=1\}} = \frac{1}{2}(1 + \sigma_i)$, this can also be expressed as

$$\langle \sigma_i \sigma_j \rangle_{\Lambda;\beta,h}^+ \geq \langle \sigma_i \rangle_{\Lambda;\beta,h}^+ \langle \sigma_j \rangle_{\Lambda;\beta,h}^+. \quad (2.8)$$

This is equivalent to asking whether σ_i and σ_j are *positively correlated* under $\mu_{\Lambda;\beta,h}^+$.

Inequalities (2.7) and (2.8) are actually true, and will be particular instances of the GKS *inequalities* (named after Griffiths, Kelly and Sherman) which hold in a more general setting.

Namely, let $\mathbf{J} = (J_{ij})$ be a collection of *non-negative* real numbers J_{ij} indexed by pairs $\{i, j\} \in \mathcal{E}_\Lambda^b$. Let also $\mathbf{h} = (h_i)$ be a collection of real numbers indexed by vertices of Λ . We write $\mathbf{h} \geq 0$ as a shortcut for $h_i \geq 0$ for all $i \in \Lambda$. We then write, for $\omega \in \Omega_\Lambda^\eta$,

$$\mathcal{H}_{\Lambda;\mathbf{J},\mathbf{h}}(\omega) := - \sum_{\{i,j\} \in \mathcal{E}_\Lambda^b} J_{ij} \sigma_i(\omega) \sigma_j(\omega) - \sum_{i \in \Lambda} h_i \sigma_i(\omega).$$

We denote the corresponding finite-volume Gibbs distribution by $\mu_{\Lambda;\mathbf{J},\mathbf{h}}^\eta$. Of course, we recover $\mathcal{H}_{\Lambda;\beta,h}$ and $\mu_{\Lambda;\beta,h}^\eta$ by setting $J_{ij} = \beta$ for all $\{i, j\} \in \mathcal{E}_\Lambda^b$ and $h_i = h$ for all $i \in \Lambda$.

The GKS inequalities are mostly restricted to +, free and periodic boundary conditions and to non-negative magnetic fields. They deal with expectations and covariances of random variables of the type σ_Λ , which is precisely what is needed for the study of the thermodynamic limit.

THEOREM 2.16 (GKS Inequalities).

Let $\mathbf{J}, \mathbf{h}$ be as above and $\Lambda \Subset \mathbb{Z}^d$. Assume that $h \geq 0$. Then, for all $A, B \subset \Lambda$,

$$\begin{aligned} \langle \sigma_A \rangle_{\Lambda;\mathbf{J},\mathbf{h}}^+ &\geq 0, \\ \langle \sigma_A \sigma_B \rangle_{\Lambda;\mathbf{J},\mathbf{h}}^+ &\geq \langle \sigma_A \rangle_{\Lambda;\mathbf{J},\mathbf{h}}^+ \langle \sigma_B \rangle_{\Lambda;\mathbf{J},\mathbf{h}}^+. \end{aligned}$$

These inequalities remain valid for $\langle \cdot \rangle_{\Lambda; \mathbf{J}, h}^\emptyset$ and $\langle \cdot \rangle_{\Lambda; \mathbf{J}, h}^{\text{per}}$.

Let $\Lambda \in \mathbb{Z}^d$ and let $\mathbf{K} = (K_C)_{C \subset \Lambda}$ be a family of real numbers, called *coupling constants*. Consider the following probability distribution on Ω_Λ :

$$\nu_{\Lambda; \mathbf{K}}(\omega) := \frac{1}{Z_{\Lambda, \mathbf{K}}} \exp \left(\sum_{C \subset \Lambda} K_C \omega_C \right),$$

where $\omega_C := \prod_{i \in C} \omega_i$ and $Z_{\Lambda, \mathbf{K}}$ is the associated partition function. The Gibbs distributions $\mu_{\Lambda; \mathbf{J}, h}^+$, $\mu_{\Lambda; \mathbf{J}, h}^-$ and $\mu_{\Lambda; \mathbf{J}, h}^{\text{per}}$ can all be written in this form, with $K_C \geq 0$, for all $C \subset \Lambda$, if $h \geq 0$. For example, $\mu_{\Lambda; \mathbf{J}, h}^+ = \nu_{\Lambda; \mathbf{K}}$ once

$$K_C = \begin{cases} h + \beta \# \{j \notin \Lambda : j \sim i\} & \text{if } C = \{i\} \subset \Lambda \\ \beta, & \text{if } C = \{i, j\} \subset \Lambda, i \sim j. \\ 0, & \text{otherwise} \end{cases}$$

EXERCISE 2.8. Check that $\mu_{\Lambda; \beta, h}^\emptyset$ and $\mu_{\Lambda; \beta, h}^{\text{per}}$ can also be written in this form for a suitable choice of the coefficients $\mathbf{K}$, and that these coefficients can all be taken non-negative if $h \geq 0$.

We can now state the following generalisation of [Theorem 2.16](#).

THEOREM 2.17.

Let $\mathbf{K} = (K_C)_{C \subset \Lambda}$ be such that $K_C \geq 0$ for all $C \subset \Lambda$. Then, for any $A, B \subset \Lambda$,

$$\langle \sigma_A \rangle_{\Lambda; \mathbf{K}} \geq 0, \tag{2.9}$$

$$\langle \sigma_A \sigma_B \rangle_{\Lambda; \mathbf{K}} \geq \langle \sigma_A \rangle_{\Lambda; \mathbf{K}} \langle \sigma_B \rangle_{\Lambda; \mathbf{K}}. \tag{2.10}$$

Proof.

□

The FKG Inequality

The FKG Inequality (named after Fortuin, Kasteleyn and Ginibre) states that increasing events are positively correlated.

The total order on the set $\{-1, 1\}$ induces a *partial order* on Ω : $\omega \leq \omega'$ if and only if $\omega_i \leq \omega'_i$ for all $i \in \mathbb{Z}^d$. An event $E \subset \Omega$ is *increasing* if $\omega \in E$ and $\omega \leq \omega'$ implies $\omega' \in E$. If E and F are both increasing events depending on the spins inside Λ , then again, due to the ferromagnetic nature of the model, one can expect that the occurrence of an increasing event enhances the probability of another increasing event. That is, assuming that F has positive probability:

$$\mu_{\Lambda; \beta, h}^+(E | F) \geq \mu_{\Lambda; \beta, h}^+(E).$$

Multiplying by the probability of F , this inequality can be written as:

$$\mu_{\Lambda; \beta, h}^+(E \cap F) \geq \mu_{\Lambda; \beta, h}^+(E) \mu_{\Lambda; \beta, h}^+(F). \tag{2.11}$$

A function $f : \Omega \rightarrow \mathbb{R}$ is *non-decreasing* if and only if $f(\omega) \leq f(\omega')$ for all $\omega \leq \omega'$.

A particularly useful feature of the FKG inequality is its applicability for all possible boundary conditions and arbitrary (that is, not necessarily non-negative) values of the magnetic field. They are also valid in the general setting presented in the last section, in which β and h are replaced by $\mathbf{J}$ and $\mathbf{h}$.

THEOREM 2.18 (FKG Inequality).

Let $\mathbf{J} = (J_{ij})_{i,j \in \mathbb{Z}^d}$ be a collection of non-negative real numbers and let $\mathbf{h} = (h_i)_{i \in \mathbb{Z}^d}$ be a collection of arbitrary real numbers. Let $\Lambda \Subset \mathbb{Z}^d$ and $\#$ be some arbitrary boundary condition. Then, for any pair of non-decreasing functions f and g ,

$$\langle fg \rangle_{\Lambda, \mathbf{J}, \#} \geq \langle f \rangle_{\Lambda, \mathbf{J}, \#} \langle g \rangle_{\Lambda, \mathbf{J}, \#}. \quad (2.12)$$

Remark. Actually, (2.12) can be seen as a natural extension of the following elementary result: if f and g are two non-decreasing functions from $\mathbb{R}$ to $\mathbb{R}$ and μ is a probability measure on $\mathbb{R}$, then $\langle fg \rangle_\mu \geq \langle f \rangle_\mu \langle g \rangle_\mu$. Namely, it suffices to write

$$\langle fg \rangle_\mu - \langle f \rangle_\mu \langle g \rangle_\mu = \frac{1}{2} \int (f(x) - f(y))(g(x) - g(y))\mu(dx)\mu(dy),$$

and to observe that $f(x) - f(y)$ and $g(x) - g(y)$ have the same sign, since f and g are both non-decreasing.

Inequality (2.11) follows by taking $J_{ij} = \beta$ and $h_i = h$, and $f = \mathbb{1}_E$, $g = \mathbb{1}_F$. Note also that $\langle fg \rangle_{\Lambda, \mathbf{J}, \#}^\eta \leq \langle f \rangle_{\Lambda, \mathbf{J}, \#}^\eta \langle g \rangle_{\Lambda, \mathbf{J}, \#}^\eta$ whenever f is non-decreasing and g is non-increasing (simply apply (2.12) to f and $-g$).

Consequences

LEMMA 2.19.

Let f be a non-decreasing function and $\Lambda_1 \subset \Lambda_2 \Subset \mathbb{Z}^d$. Then, for any $\beta \geq 0$ and $h \in \mathbb{R}$,

$$\langle f \rangle_{\Lambda_1; \beta, h}^+ \geq \langle f \rangle_{\Lambda_2; \beta, h}^+.$$

The same statement holds for the $-$ boundary condition and a non-increasing function f .

Proof. □

Actually, some form of monotonicity with respect to the volume can also be established for the Gibbs distributions with free boundary condition.

EXERCISE 2.9. Using the GKS inequalities, prove that, for all $\beta, h \geq 0$,

$$\langle \sigma_A \rangle_{\Lambda_1; \beta, h}^+ \geq \langle \sigma_A \rangle_{\Lambda_2; \beta, h}^+, \quad \langle \sigma_A \sigma_B \rangle_{\Lambda_1; \beta, h}^\emptyset \geq \langle \sigma_A \sigma_B \rangle_{\Lambda_2; \beta, h}^\emptyset,$$

for all $A \subset \Lambda_1 \subset \Lambda_2 \subset \mathbb{Z}^d$.

SOLUTION. Let $D \subset \mathcal{E}_{\Lambda_2}$ be the set of edges $\{i, j\}$ with $i \in \Lambda_2 \setminus \Lambda_1$, $j \in \Lambda_1$. Consider, for $s \in [0, 1]$, the Hamiltonian

$$\mathcal{H}_{\Lambda_2; \beta, h}^s(\omega) := -\beta \sum_{\{i, j\} \in \mathcal{E}_{\Lambda_2} \setminus D} \sigma_i \sigma_j - s\beta \sum_{\{i, j\} \in D} \sigma_i \sigma_j - h \sum_{i \in \Lambda_2} \sigma_i.$$

Let $\langle \cdot \rangle_{\Lambda_2; \beta, h}^s$ denote the corresponding Gibbs distribution. Observe that, when $A \subset \Lambda_1$, $\langle \sigma_A \rangle_{\Lambda_2; \beta, h}^\emptyset = \langle \sigma_A \rangle_{\Lambda_2; \beta, h}^{s=1}$ and $\langle \sigma_A \rangle_{\Lambda_1; \beta, h}^\emptyset = \langle \sigma_A \rangle_{\Lambda_2; \beta, h}^{s=0}$. The conclusion follows since $\langle \sigma_A \rangle_{\Lambda_2; \beta, h}^{s=0} \leq \langle \sigma_A \rangle_{\Lambda_2; \beta, h}^{s=1}$.

For the other claim, add a magnetic field h' acting on the spins in $\Lambda_2 \setminus \Lambda_1$ and let $h' \rightarrow \infty$. □

The next lemma shows that the Gibbs distributions with $+$ and $-$ boundary condition play an extremal role, in the sense that they maximally favor $+$, respectively $-$, spins.

LEMMA 2.20.

Let f be an arbitrary non-decreasing function. Then, for any $\beta \geq 0$ and $h \in \mathbb{R}$,

$$\langle f \rangle_{\Lambda; \beta, h}^- \leq \langle f \rangle_{\Lambda; \beta, h}^\eta \leq \langle f \rangle_{\Lambda; \beta, h}^+,$$

for any boundary condition $\eta \in \Omega$ and any $\Lambda \Subset \mathbb{Z}^d$. Similarly, if f is a local function with $\text{supp}(f) \subset \Lambda$ or $\text{supp}(f) \subset V_n$, then

$$\langle f \rangle_{\Lambda; \beta, h}^- \leq \langle f \rangle_{\Lambda; \beta, h}^\emptyset \leq \langle f \rangle_{\Lambda; \beta, h}^+, \quad \langle f \rangle_{V_{N-1}; \beta, h}^- \leq \langle f \rangle_{V_N; \beta, h}^{\text{per}} \leq \langle f \rangle_{V_{N-1}; \beta, h}^+.$$

Proof. Let

$$I(\omega) = \exp \left(\beta \sum_{\substack{i \in \Lambda, j \notin \Lambda \\ i \sim j}} \omega_i (1 - \eta_j) \right).$$

First, observe that

$$\sum_{\omega \in \Omega_\Lambda^+} e^{-\mathcal{H}_{\Lambda; \beta, h}(\omega)} = \sum_{\omega \in \Omega_\Lambda^\eta} e^{-\mathcal{H}_{\Lambda; \beta, h}(\omega)} I(\omega),$$

and, for any non-decreasing f ,

$$\sum_{\omega \in \Omega_\Lambda^+} e^{-\mathcal{H}_{\Lambda; \beta, h}(\omega)} f(\omega) \geq \sum_{\omega \in \Omega_\Lambda^\eta} e^{-\mathcal{H}_{\Lambda; \beta, h}(\omega)} I(\omega) f(\omega).$$

(The inequality is a consequence of our not assuming that $\text{supp}(f) \subset \Lambda$.) This implies that

$$\langle f \rangle_{\Lambda; \beta, h}^+ = \frac{\sum_{\omega \in \Omega_\Lambda^+} e^{-\mathcal{H}_\Lambda(\omega)} f(\omega)}{\sum_{\omega \in \Omega_\Lambda^+} e^{-\mathcal{H}_\Lambda(\omega)}} \geq \frac{\sum_{\omega \in \Omega_\Lambda^\eta} e^{-\mathcal{H}_\Lambda(\omega)} I(\omega) f(\omega)}{\sum_{\omega \in \Omega_\Lambda^\eta} e^{-\mathcal{H}_\Lambda(\omega)} I(\omega)} = \frac{\langle If \rangle_{\Lambda; \beta, h}^\eta}{\langle I \rangle_{\Lambda; \beta, h}^\eta} \geq \langle f \rangle_{\Lambda; \beta, h}^\eta,$$

where we applied the *FKG inequality* in the last step, making use of the fact that the function I is non-decreasing.

The proof for the free boundary condition is identical, using the non-decreasing function $I(\omega) = \exp \left(\beta \sum_{\substack{i \in \Lambda, j \notin \Lambda \\ i \sim j}} \omega_i \right)$.

Let us finally consider the Gibbs distribution with periodic boundary condition. In that case, we can argue as in the proof of [Lemma 2.19](#), since, for any $\omega \in \Omega_{V_{N-1}}^+$ (considering $V_N = \{0, \dots, N\}^d$ as a subset of $\mathbb{Z}^d$),

$$\mu_{V_N; \beta, h}^{\text{per}}(\omega|_{V_N} \mid \sigma_i = 1, \forall i \in \Sigma_N) = \mu_{V_{N-1}; \beta, h}^+(\omega),$$

where

$$\Sigma_N := \{i = (i_1, \dots, i_d) \in V_N : \exists 1 \leq k \leq d \text{ such that } i_k = 0\}$$

and the restriction of a configuration $\omega \in \Omega$ to a subset $S \subset \mathbb{Z}^d$ is defined by

$$\omega|_S := (\omega_i)_{i \in S}.$$

□

EXERCISE 2.10. Let $\eta, \omega \in \Omega$ be such that $\eta \leq \omega$. Let f be a non-decreasing function. Show that, for any $\beta \geq 0$ and $h \in \mathbb{R}$,

$$\langle f \rangle_{\Lambda; \beta, h}^\eta \leq \langle f \rangle_{\Lambda; \beta, h}^\omega,$$

for any $\Lambda \subset \mathbb{Z}^d$. *Hint.* Adapt the argument in the proof of [Lemma 2.20](#).

SECTION 2.7

| Phase Diagram

Now that we have seen that infinite-volume Gibbs states for a pair (β, h) can be constructed rigorously in various ways², the next problem is to determine whether these are the same Gibbs states, or whether there exist some values of the temperature and magnetic field for which the influence of the boundary condition survives in the thermodynamic limit, leading to multiple Gibbs states.

The answer to this question will be given in the next sections: it will depend on the dimension d and on the values of β and h . Contrarily to what often happens in mathematics, the lack of uniqueness is not a defect of this approach, but is actually one of its main features: lack of uniqueness means that providing a complete microscopic description of the system (that is, the set of configurations and the Hamiltonian) as well as fixing all the relevant thermodynamic parameters (β, h) is not sufficient to completely determine the macroscopic behavior of the system.

DEFINITION 2.21.

If at least two distinct Gibbs states can be constructed for a pair (β, h) , we say that there is a *first-order phase transition* at (β, h) .

We can now turn to the main result of this chapter, which establishes the *phase diagram* of the Ising model, that is, the determination for each pair (β, h) of whether there is a unique or multiple Gibbs states.

THEOREM 2.22. (i) In any $d \geq 1$, when $h \neq 0$, there is a unique Gibbs state for all values of $\beta \in \mathbb{R}^*$.

(ii) In $d = 1$, there is a unique Gibbs state at each $(\beta, h) \in \mathbb{R}^* \times \mathbb{R}$.

(iii) When $h = 0$ and $d \geq 2$, there exists $\beta_c = \beta_c(d) \in (0, \infty)$ such that:

- if $\beta < \beta_c$, the Gibbs state at $(\beta, 0)$ is unique;
- if $\beta > \beta_c$, a first-order phase transition occurs at $(\beta, 0)$: $\langle \cdot \rangle_{\beta,0}^+ \neq \langle \cdot \rangle_{\beta,0}^-$.

Proof. (i) In Section 3.7.4.

(ii) Already proved in the *one-dimensional Ising model*, once the results there are combined with [Theorem 2.27](#).

□

Remark. (i) It can be proved that uniqueness holds also at $(\beta_c, 0)$, when $d \geq 2$. The phase transition occurring as β crosses β_c (at $h = 0$) is thus continuous.

(ii) Although the above theorem claims the existence of at least two distinct Gibbs states when $d \geq 2$, $h = 0$, and $\beta > \beta_c$, it does not describe the structure of the set of Gibbs states associated to those values of (β, h) .

² For example, using + or – boundary conditions.

Two Criteria for (Non)-Uniqueness

In this subsection, we establish a link between uniqueness of the Gibbs state, the average magnetisation density, and differentiability of the pressure. We use these quantities to formulate several equivalent characterisations of uniqueness of the Gibbs state, which play a crucial role in our determination of the phase diagram. Moreover, the second criterion provides the rigorous link between the analytic and probabilistic definitions of first-order phase transition introduced earlier.

A First Characterisation of Uniqueness

The major role played by the states $\langle \cdot \rangle_{\beta, h}^+$ and $\langle \cdot \rangle_{\beta, h}^-$ is made clear by the following result.

THEOREM 2.23.

Let $(\beta, h) \in \mathbb{R}^* \times \mathbb{R}$. The following statements are equivalent:

- (i) there is a unique Gibbs state at (β, h) ;
- (ii) $\langle \cdot \rangle_{\beta, h}^+ = \langle \cdot \rangle_{\beta, h}^-$;
- (iii) $\langle \sigma_0 \rangle_{\beta, h}^+ = \langle \sigma_0 \rangle_{\beta, h}^-$.

Proof.

□

Some Properties of the Magnetisation Density

Remember that the average magnetisation density in $\Lambda \Subset \mathbb{Z}^d$ with an arbitrary boundary condition $\#$ was defined by $m_{\Lambda; \beta, h}^\# := \langle m_\Lambda \rangle_{\Lambda; \beta, h}^\#$. The uniqueness criterion developed in [Theorem 2.23](#) is expressed in terms of the averages $\langle \sigma_0 \rangle_{\beta, h}^+$ and $\langle \sigma_0 \rangle_{\beta, h}^-$. It is natural to wonder whether these quantities are related to $m_\Lambda^+(\beta, h)$ and $m_\Lambda^-(\beta, h)$. The following result shows that they in fact coincide in the thermodynamic limit.

PROPOSITION 2.24.

For any sequence $\Lambda \uparrow \mathbb{Z}^d$, the limits

$$m^+(\beta, h) := \lim_{\Lambda \uparrow \mathbb{Z}^d} m_\Lambda^+(\beta, h), \quad m^-(\beta, h) := \lim_{\Lambda \uparrow \mathbb{Z}^d} m_\Lambda^-(\beta, h)$$

exist and

$$m^+(\beta, h) = \langle \sigma_0 \rangle_{\beta, h}^+, \quad m^-(\beta, h) = \langle \sigma_0 \rangle_{\beta, h}^-.$$

Moreover, $h \mapsto m^+(\beta, h)$ is right-continuous, while $h \mapsto m^-(\beta, h)$ is left-continuous.

Proof.

□

Remark. By [Corollary 2.7](#), $m^+(\beta, h)$ and $m(\beta, h)$ are equal when $h \notin \mathfrak{B}_\beta$. Therefore, considering a sequence $h \downarrow 0$ in $\mathfrak{B}_\beta^c$,

$$m^*(\beta) = \lim_{h \downarrow 0} m(\beta, h) = \lim_{h \downarrow 0} m^+(\beta, h) = m^+(\beta, 0) = \langle \sigma_0 \rangle_{\beta, 0}^+.$$

Note also that,

$$\lim_{\|i\|_1 \rightarrow \infty} \langle \sigma_0 \sigma_i \rangle_{\beta, 0}^+ = \left(\langle \sigma_0 \rangle_{\beta, 0}^+ \right)^2 = m^*(\beta)^2, \quad \forall \beta \geq 0.$$

This observation provides a convenient approach for its explicit computation in $d = 2$, which avoids having to work with a non-zero magnetic field.

LEMMA 2.25. (i) For all $\beta \geq 0$, $h \mapsto \langle \sigma_0 \rangle_{\beta, h}^+$ is non-decreasing and right-continuous and $h \mapsto \langle \sigma_0 \rangle_{\beta, h}^-$ is non-decreasing and left-continuous.

(ii) For all $h \geq 0$, $\beta \mapsto \langle \sigma_0 \rangle_{\beta, h}^+$ is non-decreasing and, for all $h \leq 0$, $\beta \mapsto \langle \sigma_0 \rangle_{\beta, h}^-$ is non-increasing.

Proof. □

EXERCISE 2.11. Let $A \subseteq \mathbb{Z}^d$ and $h \geq 0$. Show that $\beta \mapsto \langle \sigma_A \rangle_{\beta, h}^\emptyset$ is left-continuous and $\beta \mapsto \langle \sigma_A \rangle_{\beta, h}^+$ is right-continuous.

Defining the Critical Inverse Temperature

Since $\langle \sigma_0 \rangle_{\beta, 0}^- = -\langle \sigma_0 \rangle_{\beta, 0}^+$ by symmetry, [Theorem 2.23](#) and [Proposition 2.24](#) and the following remark imply that, when $h = 0$, uniqueness is equivalent to $m^*(\beta) = 0$. Since [Lemma 2.25](#) implies that $m^*(\beta) = \langle \sigma_0 \rangle_{\beta, 0}^+$ is monotone in β , we are led naturally to the following definition.

DEFINITION 2.26 (Critical Inverse Temperature).

The critical inverse temperature is

$$\beta_c(d) := \inf\{\beta \geq 0 : m^*(\beta) > 0\} = \sup\{\beta \geq 0 : m^*(\beta) = 0\}.$$

That is, $\beta_c(d)$ is the unique value of β such that $m^*(\beta) = 0$ if $\beta < \beta_c$, and $m^*(\beta) > 0$ if $\beta > \beta_c$. Of course, one still has to determine whether $\beta_c(d)$ is non-trivial, that is, whether $0 < \beta_c(d) < \infty$.

Remark. By translation invariance, $\langle \sigma_i \rangle_{\beta, 0}^+ = \langle \sigma_0 \rangle_{\beta, 0}^+ = m^*(\beta)$ for all $i \in \mathbb{Z}^d$. This implies, using the FKG inequality, that

$$\langle \sigma_0 \sigma_i \rangle_{\beta, 0}^+ \geq \langle \sigma_0 \rangle_{\beta, 0}^+ \langle \sigma_i \rangle_{\beta, 0}^+ = m^*(\beta)^2.$$

In particular,

$$\inf_{i \in \mathbb{Z}^d} \langle \sigma_0 \sigma_i \rangle_{\beta, 0}^+ > 0, \quad \forall \beta > \beta_c.$$

Such a behaviour is referred to as long-range order. The presence of long-range order does not, however, imply that the random variables σ_i display strong correlations at large distances. Indeed, for any β ,

$$\lim_{\|i\|_1 \rightarrow \infty} \langle \sigma_0 \sigma_i \rangle_{\beta, 0}^+ - \langle \sigma_0 \rangle_{\beta, 0}^+ \langle \sigma_i \rangle_{\beta, 0}^+ = 0,$$

so that σ_0 and σ_i are always asymptotically (as $\|i\|_1 \rightarrow \infty$) uncorrelated.

A Second Characterisation of Uniqueness

The following theorem provides the promised link between the two notions of first-order phase transition introduced in [Definition 2.21](#): non-uniqueness occurs at (β, h) if and only if the pressure fails to be differentiable in h at this point. The theorem also provides the extension of the relation (2.4) to values of h at which the pressure is not differentiable. In that case, we can rely on the convexity of the pressure, which we proved in [Theorem 2.6](#), to conclude that its right- and left-derivatives with respect to h are always well defined.

THEOREM 2.27.

The following identities hold for all values of $\beta \geq 0$ and $h \in \mathbb{R}$:

$$\frac{\partial \psi}{\partial h^+}(\beta, h) = m^+(\beta, h), \quad \frac{\partial \psi}{\partial h^-}(\beta, h) = m^-(\beta, h).$$

In particular, $h \mapsto \psi(\beta, h)$ is differentiable at h if and only if there is a unique Gibbs state at (β, h) .

Proof. □

Remark. The theorem shows that the pressure is differentiable with respect to h precisely for those values of β and h at which there is a unique infinite-volume Gibbs state. The uniqueness of the infinite-volume Gibbs state also implies differentiability with respect to β . (Actually, the pressure of the Ising model on $\mathbb{Z}^d$ is always differentiable with respect to β .)

Uniqueness in Nonzero Magnetic Field

We now prove that, when $h \neq 0$, the Gibbs state associated to (β, h) is always unique, regardless of the value of β . The proof will take us on a detour, using results from complex analysis, and will allow us to establish a very strong property of the pressure of the Ising model.

We will study the existence and properties of the pressure when h takes values in the complex domains

$$H^+ := \{z \in \mathbb{C} : \Re z > 0\}, \quad H^- := \{z \in \mathbb{C} : \Re z < 0\}.$$

Since the inverse temperature $\beta > 0$ will play no particular role in this section, we will omit it from the notations at some places. For example, we will write $\psi(h)$ rather than $\psi(\beta, h)$.

THEOREM 2.28.

Let $\beta > 0$. As a function of the magnetic field h , the pressure of the Ising model in the thermodynamic limit, $\psi = \psi(h)$, can be extended from $\{h \in \mathbb{R} : h > 0\}$ (resp., $\{h \in \mathbb{R} : h < 0\}$) to an analytic function on the whole domain H^+ (resp., H^-). On $H^+ \cup H^-$, ψ can be computed using the thermodynamic limit with free boundary condition.

This result of course implies that the complex derivative of ψ with respect to h exists on H^+ and H^- . Therefore, the real partial derivative $\frac{\partial \psi}{\partial h}$ exists at each real $h \neq 0$. By [Theorem 2.27](#), this implies uniqueness of the Gibbs state for all $h \neq 0$, thus completing the proof of [Theorem 2.22](#).

Remark. The GHS inequality provides an alternative proof of the differentiability of the pressure when $h \neq 0$, avoiding complex analysis. Namely, the GHS inequality can be used to show that the magnetization $h \mapsto \langle \sigma_0 \rangle_{\beta, h}^+$ is concave, and hence continuous, on $\mathbb{R}^*$. This implies that its antiderivative (which is equal to ψ up to a constant) exists and is differentiable on $(0, +\infty)$. Of course, combined with [Theorem 2.27](#), [Theorem 2.28](#) implies the much stronger statement that $h \mapsto \langle \sigma_0 \rangle_{\beta, h}^+$ is real analytic on $\{h < 0\}$ and $\{h > 0\}$.

We have seen that, for real parameters, the thermodynamic limit of the pressure can be computed using an arbitrary boundary condition. When the magnetic field is complex, the boundary condition becomes a nuisance. It turns out that the free boundary condition is

particularly convenient. We will therefore work in finite volumes $\Lambda \subset \mathbb{Z}^d$ and study

$$\psi_\Lambda^\varnothing(h) = \frac{1}{|\Lambda|} \log Z_{\Lambda;\beta,h}^\varnothing.$$

The existence and analyticity properties of the pressure are established by taking the thermodynamic limit $\Lambda \uparrow \mathbb{Z}^d$ for this choice of boundary condition. The analytic function obtained is then the analytic continuation of the pressure to complex values of the field.³

On the one hand when the magnetic field is real, $Z_{\Lambda;\beta,h}^\varnothing$, being a finite linear combination of powers of $e^{\pm h}$, is real-analytic in h . Moreover, since

$$Z_{\Lambda;\beta,h}^\varnothing > 0 \quad \text{for all } h \in \mathbb{R}, \quad (2.13)$$

the pressure $\psi_\Lambda^\varnothing(\cdot)$ is also real-analytic in h . It is not true, however, that this real analyticity always holds in the thermodynamic limit $\Lambda \uparrow \mathbb{Z}^d$. Indeed, we have seen (using Peierls' argument) that, at low temperature, the pressure is not even differentiable at $h = 0$.

On the other hand, since the Boltzmann weights are complex numbers when $h \in \mathbb{C}$, the partition function $Z_{\Lambda;\beta,h}^\varnothing$ can very well vanish, leading to a problem even for the definition of the finite-volume pressure.

Fortunately, the celebrated *Lee-Yang Circle Theorem* will show that the partition function satisfies a remarkable property, analogous to (2.13), in suitable domains of the complex plane. This will allow us to control the analyticity of the pressure in the thermodynamic limit, as explained in the following result.

THEOREM 2.29 (Lee-Yang).

Let $\beta \geq 0$. Let $D \subset \mathbb{C}$ be open, simply connected and such that $D \cap \mathbb{R}$ is an interval of $\mathbb{R}$. Assume that, for every finite volume $\Lambda \Subset \mathbb{Z}^d$,

$$Z_{\Lambda;\beta,h}^\varnothing \neq 0, \quad \forall h \in D. \quad (2.14)$$

Then, the pressure $h \mapsto \psi(h)$ admits an analytic continuation to D .

We know from the analysis done in the previous sections that the pressure is not differentiable at $h = 0$ when $\beta > \beta_c(d)$. When this happens, the previous theorem implies that there must exist a sequence $(h_k) \subset \mathbb{C}$, tending to 0 and a sequence $\Lambda_k \uparrow \mathbb{Z}^d$ such that $Z_{\Lambda_k;\beta,h_k}^\varnothing = 0$ for all k . Therefore, even though the partition functions never vanish as long as h is real, complex zeroes approach the point $h = 0$ in the thermodynamic limit. In this sense, although values of the magnetic field with a nonzero imaginary part may be experimentally meaningless, the way the partition function behaves for such complex values of the magnetic field turns out to have profound physical consequences.

Proof. Let $\Lambda_n \uparrow \mathbb{Z}^d$. Using (2.14), [Theorem A.5](#) guarantees that one can find a function $h \mapsto \log Z_{\Lambda_n;\beta,h}^\varnothing$ analytic on D and coinciding with the quantity studied in the rest of this chapter when $h \in D \cap \mathbb{R}$ (see [Appendix A.3](#) for the existence of a branch of the logarithm with this property). One can then define

$$g_n(h) := \exp\left(|\Lambda_n|^{-1} \log Z_{\Lambda_n;\beta,h}^\varnothing\right),$$

³ If two functions analytic on a domain D coincide on a set $A \subset D$ which has an accumulation point in D , then these two functions are equal on D . Therefore, if it were possible to obtain another pressure $\tilde{\psi}$ using a different boundary condition, analytic on H^+ and H^- , then, since this pressure coincides with the one obtained with free boundary conditions on the real axis, it must coincide with it on H^+ and H^- .

which is also analytic on D . Now, when $h \in D \cap \mathbb{R}$, $g_n(h)$ coincides with $e^{\psi_{\Lambda_n}^\circ(h)}$, and [Theorem 2.6](#) thus guarantees that, for such values of h , $g_n(h) \rightarrow g(h) := e^{\psi(h)}$ as $n \rightarrow \infty$, where ψ is the pressure of the Ising model in infinite volume.

The next observation is that the sequence (g_n) is locally uniformly bounded on D , since

$$\begin{aligned} |Z_{\Lambda_n; \beta, h}^\circ| &\leq \sum_{\omega \in \Omega_{\Lambda_n}} \left| \exp\left(-\mathcal{H}_{\Lambda_n; \beta, h}^\circ(\omega)\right) \right| \\ &= \sum_{\omega \in \Omega_{\Lambda_n}} \exp\left(-\mathcal{H}_{\Lambda_n; \beta, \Re h}^\circ(\omega)\right) \leq \exp((2d\beta + |\Re h| + \log 2)|\Lambda_n|), \end{aligned}$$

and thus

$$|g_n(h)| = \exp\left(|\Lambda_n|^{-1} \log |Z_{\Lambda_n; \beta, h}^\circ|\right) \leq \exp(2d\beta + |\Re h| + \log 2), \quad \forall h \in D.$$

We are now in a position to apply *Vitali's convergence theorem* in order to conclude that $(g_n)_{n \geq 1}$ converges locally uniformly on D to an analytic function g .

Moreover, since $g_n(h) \neq 0$ for all $h \in D$ and all $n \geq 1$, *Hurwitz' theorem* implies that g has no zeroes on D . Indeed, the other possibility (that is, $g \equiv 0$ on D) is incompatible with the fact that $g = e^\psi > 0$ on $D \cap \mathbb{R}$.

Since g does not vanish on D , it follows from [Theorem A.5](#) that the latter admits an analytic logarithm in D . However, choosing again the branch that is real on $D \cap \mathbb{R}$, the function $\log g$ coincides with the pressure of the Ising model on the real axis, which proves the theorem. $\square$

THEOREM 2.30 (Lee-Yang Circle Theorem).

Condition (2.14) is satisfied when $D = H^+$ and when $D = H^-$.

The proof will involve working with the variable $z := e^{-2h}$ rather than h . But $h \in H^+$ if and only if $z \in \mathcal{U}$, where $\mathcal{U}$ is the open unit disc

$$\mathcal{U} := \{z \in \mathbb{C} : |z| < 1\}.$$

Therefore, [Theorem 2.30](#) implies that all zeroes of $Z_{\Lambda_n; \beta, h}^\circ$ (seen as a function of z) lie on the unit circle. This explains the origin of the name associated to the above result.

The Curie-Weiss Model

In statistical mechanics, a *mean-field approximation* is often used to approximate a model by a simpler one, whose global behaviour can be studied with the help of *explicit* computations. The information thus extracted can then be used as an indication of the kind of properties that can be expected from the original model. In addition, this approximation turns out to provide quantitatively correct results in sufficiently high dimensions.

The Ising model is a classical example of a model with a rich behaviour but with no explicit solution in general (the exceptions being the one-dimensional model and the two-dimensional model when $h = 0$). In this chapter, we consider its mean-field approximation, in the form of the *Curie-Weiss model*. Although it is an over-simplification of the Ising model, the Curie-Weiss model still displays a phase transition, with distinct behaviours at high and low temperature.

SECTION 3.1

| The Mean-Field Approximation

Consider a system of Ising spins living on $\mathbb{Z}^d$, described by the Ising Hamiltonian defined by

$$\mathcal{H}_\Lambda(\omega) := - \sum_{\substack{i,j \in \Lambda \\ i \sim j}} \omega_i \omega_j - h \sum_{i \in \Lambda} \omega_i, \quad \omega \in \Omega_\Lambda.$$

In that model, the spin ω_i located at vertex i interacts with the rest of the system via its neighbours. The contribution to the total energy coming from the interaction of ω_i with its $2d$ neighbours can be written as

$$-\beta \sum_{j:j \sim i} \omega_i \omega_j = -2d\beta \omega_i \cdot \frac{1}{2d} \sum_{j:j \sim i} \omega_j.$$

Written this way, one can interpret the contribution of ω_i to the total energy as an interaction of ω_i with a local magnetisation density, produced by the average of its $2d$ nearest neighbours $\frac{1}{2d} \sum_{j:j \sim i} \omega_j$. Of course, this magnetisation density is *local* and varies from one point to the other. The *mean-field approximation* consists in assuming that each local magnetisation density can be approximated by the *global* magnetisation density,

$$\frac{1}{N} \sum_{j=1}^N \omega_j,$$

where N is the number of spins in the system. The mean-field approximation of the Ising model thus amounts to do the following transformation on the Hamiltonian (up to a multiplicative constant that will be absorbed in β):

$$\text{Replace } -\beta \sum_{i \sim j} \omega_i \omega_j \text{ by } -\frac{d\beta}{N} \sum_{i,j} \omega_i \omega_j.$$

The term involving the magnetic field, on the other hand, remains unchanged.

DEFINITION 3.1 (Curie–Weiss Hamiltonian).

The *Curie–Weiss Hamiltonian* for a collection of spins $\omega = (\omega_1, \dots, \omega_N)$ at inverse temperature β and with an external magnetic field h is defined by

$$\mathcal{H}_{N;\beta,h}^{\text{CW}}(\omega) := -\frac{d\beta}{N} \sum_{i,j=1}^N \omega_i \omega_j - h \sum_{i=1}^N \omega_i.$$

In contrast to those of the Ising model, the interactions of the Curie–Weiss model are global: each spin interacts with all other spins in the same way, and the relative positions of the spins can therefore be ignored. Actually, due to this lack of geometry, one may think of this model as defined on the *complete graph* with N vertices, which has an edge between any pair of distinct vertices.

We denote by $\Omega_N := \{\pm 1\}^N$ the set of all possible configurations of the Curie–Weiss model. The Gibbs distribution on Ω_N is written

$$\mu_{N;\beta,h}^{\text{CW}}(\omega) := \frac{e^{-\mathcal{H}_{N;\beta,h}^{\text{CW}}(\omega)}}{Z_{N;\beta,h}^{\text{CW}}},$$

where

$$Z_{N;\beta,h}^{\text{CW}} := \sum_{\omega \in \Omega_N} e^{-\mathcal{H}_{N;\beta,h}^{\text{CW}}(\omega)}.$$

SECTION 3.2

| The Behaviour for Large N when $h = 0$

When $h = 0$, the Hamiltonian is invariant under the global spin flip $\omega \mapsto -\omega$, which implies that the magnetisation density

$$m_N := \frac{M_N}{N},$$

where

$$M_N := \sum_{i=1}^N \omega_i$$

has a symmetric distribution:

$$\mu_{N;\beta,0}^{\text{CW}}(m_N = -m) = \mu_{N;\beta,0}^{\text{CW}}(m_N = +m).$$

In particular, $\langle m_N \rangle_{N;\beta,0}^{\text{CW}} = 0$. We expect that the spins should be essentially independent when β is small, but that, when β is large, the most probable configurations should have most spins equal and thus be close to one of the two ground states, in which all spins are equal.

THEOREM 3.2 ($h = 0$).

Let $\beta_c = \beta_c(d) := \frac{1}{2d}$. Then, the following holds.

- (i) When $\beta \leq \beta_c$, the magnetisation concentrates at zero: for all $\epsilon > 0$, there exists $c = c(\beta, \epsilon) > 0$ such that, for large enough N ,

$$\mu_{N;\beta,0}^{\text{CW}}(m_N \in (-\epsilon, \epsilon)) \geq 1 - 2e^{-cN}.$$

- (ii) When $\beta > \beta_c$, the magnetisation is bounded away from zero. More precisely, there exists $m^{*,\text{CW}}(\beta) > 0$, called the spontaneous magnetisation, such that, for all small enough $\epsilon > 0$, there exists $b = b(\beta, \epsilon) > 0$ such that if

$$J_*(\epsilon) := (-m^{*,\text{CW}}(\beta) - \epsilon, -m^{*,\text{CW}}(\beta) + \epsilon) \cup (m^{*,\text{CW}}(\beta) - \epsilon, m^{*,\text{CW}}(\beta) + \epsilon),$$

then, for large enough N ,

$$\mu_{N;\beta,0}^{\text{CW}}(m_N \in J_*(\epsilon)) \geq 1 - 2e^{-bN}.$$

β_c is called the inverse critical temperature or inverse Curie temperature.

The Curie–Weiss model possesses a remarkable feature, which makes its analysis much easier than that of the Ising model on $\mathbb{Z}^d$: since

$$\sum_{i,j=1}^N \omega_i \omega_j = \left(\sum_{i=1}^N \omega_i \right)^2 \equiv M_N^2,$$

the Hamiltonian $\mathcal{H}_{N;\beta,0}^{\text{CW}}$ is entirely determined by the magnetization density:

$$\mathcal{H}_{N;\beta,0}^{\text{CW}} = -d\beta m_N^2 N. \quad (3.1)$$

This property will make it possible to compute explicitly the thermodynamic potentials (and other quantities) associated to the Curie–Weiss model.

DEFINITION 3.3.

Let $e(m) := -dm^2$ and

$$s(m) := -\frac{1-m}{2} \log \frac{1-m}{2} - \frac{1+m}{2} \log \frac{1+m}{2}.$$

Then

$$f_\beta^{\text{CW}}(m) := \beta e(m) - s(m)$$

is called the *free energy* of the Curie–Weiss model.

PROPOSITION 3.4.

For any β ,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \log Z_{N;\beta,0}^{\text{CW}} = - \min_{m \in [-1,1]} f_\beta^{\text{CW}}(m). \quad (3.2)$$

Moreover, for any interval $J \subset [-1, 1]$,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \log \mu_{N;\beta,0}^{\text{CW}}(m_N \in J) = - \inf_{m \in J} I_\beta^{\text{CW}}(m), \quad (3.3)$$

where

$$I_\beta^{\text{CW}}(m) := f_\beta^{\text{CW}}(m) - \min_{\tilde{m} \in [-1,1]} f_\beta^{\text{CW}}(\tilde{m}).$$

Proof. Observe that for a fixed N , m_N is a random variable taking values in the set

$$\mathcal{A}_N := \left\{ -1 + \frac{2k}{N} : k = 0, \dots, N \right\} \subset [-1, 1].$$

Let $J \subset [-1, 1]$ be an interval. Then,

$$\mu_{N;\beta,0}^{\text{CW}}(m_N \in J) = \sum_{m \in J \cap \mathcal{A}_N} \mu_{N;\beta,0}^{\text{CW}}(m_N = m).$$

Since there are exactly $\binom{N}{\frac{1+m}{2}N}$ configurations $\omega \in \Omega_N$ which have $m_N(\omega) = m$, one can express explicitly the distribution of m_N using (3.1):

$$\mu_{N;\beta,0}^{\text{CW}}(m_N = m) = \sum_{\omega \in \Omega_N : m_N(\omega) = m} \frac{e^{-\mathcal{H}_{N;\beta,0}^{\text{CW}}(\omega)}}{Z_{N;\beta,0}^{\text{CW}}} = \frac{1}{Z_{N;\beta,0}^{\text{CW}}} \binom{N}{\frac{1+m}{2}N} e^{d\beta m^2 N}.$$

In the same way,

$$Z_{N;\beta,0}^{\text{CW}} = \sum_{m \in \mathcal{A}_N} \binom{N}{\frac{1+m}{2}N} e^{d\beta m^2 N}.$$

Since we are interested in its behaviour on the exponential scale and since it is a sum of only $|\mathcal{A}_N| = N + 1$ positive terms, $Z_{N;\beta,0}^{\text{CW}}$ can be estimated by keeping only its dominant term:

$$\max_{m \in \mathcal{A}_N} \binom{N}{\frac{1+m}{2}N} e^{d\beta m^2 N} \leq Z_{N;\beta,0}^{\text{CW}} \leq (N + 1) \max_{m \in \mathcal{A}_N} \binom{N}{\frac{1+m}{2}N} e^{d\beta m^2 N}.$$

To study the large N behaviour of the binomial factors, we use *Stirling's formula*. The latter implies the existence of two constants $c_-, c_+ > 0$ such that, for all $m \in \mathcal{A}_N$,

$$c_- N^{-\frac{1}{2}} e^{Ns(m)} \leq \binom{N}{\frac{1+m}{2}N} \leq c_+ e^{Ns(m)}. \quad (3.4)$$

We can thus compute an upper bound as follows:

$$Z_{N;\beta,0}^{\text{CW}} \leq c_+(N + 1) \exp \left(N \max_{m \in \mathcal{A}_N} \{d\beta m^2 + s(m)\} \right) \leq c_+(N + 1) \exp \left(-N \min_{m \in [-1,1]} f_\beta^{\text{CW}}(m) \right).$$

which yields

$$\limsup_{N \rightarrow \infty} \frac{1}{N} \log Z_{N;\beta,0}^{\text{CW}} \leq - \min_{m \in [-1,1]} f_{\beta}^{\text{CW}}(m).$$

For the lower bound, we first use the continuity of $m \mapsto \{d\beta m^2 + s(m)\}$ on $[-1, 1]$ and consider some $m' \in [-1, 1]$ for which $f_{\beta}^{\text{CW}}(m') = \min_m f_{\beta}^{\text{CW}}(m)$. Fix $\epsilon > 0$, and choose some $m \in \mathcal{A}_N$ such that

$$\left| f_{\beta}^{\text{CW}}(m) - f_{\beta}^{\text{CW}}(m') \right| \leq \epsilon,$$

which is always possible once N is large enough. We then have

$$Z_{N;\beta,0}^{\text{CW}} \geq c_- N^{-\frac{1}{2}} \exp\left(-N\left(f_{\beta}^{\text{CW}}(m') + \epsilon\right)\right).$$

This yields

$$\liminf_{N \rightarrow \infty} \frac{1}{N} \log Z_{N;\beta,0}^{\text{CW}} \geq - \min_{m \in [-1,1]} f_{\beta}^{\text{CW}}(m) - \epsilon.$$

Since ϵ was arbitrary, (3.2) follows. A similar computation can be done for the sum over $m \in J \cap \mathcal{A}_N$,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \log \sum_{m \in J \cap \mathcal{A}_N} \binom{N}{\frac{1+m}{2}N} e^{d\beta m^2 N} = - \inf_{m \in J} f_{\beta}^{\text{CW}}(m),$$

and we get (3.3). $\square$

Verification of (3.4). $\square$

Proof of Theorem 3.2. As discussed after the statement of [Proposition 3.4](#), we must locate the zeros of I_{β}^{CW} . Since the latter is smooth and $I_{\beta}^{\text{CW}} \geq 0$, the zeros correspond to the solutions of $\frac{\partial I_{\beta}^{\text{CW}}}{\partial m} = 0$. After a straightforward computation, we easily see that this condition is equivalent to the *mean-field equation*:

$$\tanh(2d\beta m) = m. \quad (3.5)$$

Since $\lim_{m \rightarrow \pm\infty} \tanh(\beta m) = \pm 1$, there always exists at least one solution and, as an analysis of the graph of $m \mapsto \tanh(\beta m)$ shows (see below), the number of solutions of (3.5) depends on whether $2d\beta$ is larger or smaller than 1, that is, whether β is larger or smaller than β_c .

On the one hand, when $\beta \leq \beta_c$, (3.5) has a unique solution, given by $m = 0$. On the other hand, when $\beta > \beta_c$, there are two additional non-trivial solutions, $+m^{*,\text{CW}}(\beta)$ and $-m^{*,\text{CW}}(\beta)$ (which depend on β).

The trivial solution $m = 0$ is a local maximum of I_{β}^{CW} , whereas $+m^{*,\text{CW}}(\beta)$ and $-m^{*,\text{CW}}(\beta)$ are global minima. Combined with (3.3), this analysis proves the theorem. $\square$

EXERCISE 3.1. Let $\zeta : \mathbb{N} \rightarrow \mathbb{R}^+$. Consider the following modification of the Curie-Weiss Hamiltonian (with $h = 0$):

$$\tilde{\mathcal{H}}_{N;\beta,0}(\omega) := -\frac{\beta}{\zeta(N)} \sum_{i,j=1}^N \omega_i \omega_j.$$

Denote by $\tilde{\mu}_{N;\beta}$ the corresponding Gibbs distribution. Show that the following holds.

- (i) If $\lim_{N \rightarrow \infty} \frac{\zeta(N)}{N} = \infty$, then m_N tends to 0 in probability for all $\beta \geq 0$.
- (ii) If $\lim_{N \rightarrow \infty} \frac{\zeta(N)}{N} = 0$, then $|m_N|$ tends to 1 in probability for all $\beta > 0$.

This shows that the only scaling leading to a non-trivial dependence in β is when $\zeta(N)$ is of the order of N .

SOLUTION. If one writes $\tilde{\mathcal{H}}_{N;\beta,0} = \mathcal{H}_{N;\beta(N),0}^{\text{CW}}$ where $\beta(N) := \frac{N\beta}{\zeta(N)}$, then either $\beta(N) \uparrow +\infty$ or $\beta(N) \downarrow 0$. The conclusion now follows from our previous analysis. $\square$

SECTION 3.3

| The Behaviour for Large N when $h \neq 0$

THEOREM 3.5.

The pressure

$$\psi_{\beta}^{\text{CW}}(h) := \lim_{N \rightarrow \infty} \frac{1}{N} \log Z_{N;\beta,h}^{\text{CW}}$$

exists and is convex in h . Moreover, it equals the Legendre transform of the free energy:

$$\psi_{\beta}^{\text{CW}}(h) = \max_{m \in [-1,1]} \left\{ hm - f_{\beta}^{\text{CW}}(m) \right\}.$$

Proof.

□

SECTION 3.4

| The Hubbard–Stratonovich Transformation

Observe first that, for any $\alpha > 0$, a simple integration yields

$$\exp(\alpha x^2) = \frac{1}{\sqrt{\pi\alpha}} \int_{-\infty}^{\infty} \exp\left(-\frac{y^2}{\alpha} + 2yx\right) dy.$$

This can be used to express the interactions among spins in the Boltzmann weight as

$$\exp\left(\frac{d\beta}{N} \left(\sum_{i=1}^N \omega_i\right)^2\right) = \sqrt{\frac{N}{\pi d\beta}} \int_{-\infty}^{\infty} \exp\left(-\frac{N}{d\beta} y^2 + 2y \sum_{i=1}^N \omega_i\right) dy.$$

The advantage of this reformulation is that the quadratic term in the spin variables has been replaced by a linear one. As a consequence, the sum over configurations, in the partition function, can now be performed as in the previous subsection:

$$\begin{aligned} Z_{N;\beta,h}^{\text{CW}} &= \sum_{\omega \in \Omega_N} e^{-\mathcal{H}_{N;\beta,h}^{\text{CW}}(\omega)} \\ &= \sqrt{\frac{N}{\pi d\beta}} \int_{-\infty}^{\infty} e^{-\frac{Ny^2}{d\beta}} \prod_{i=1}^N \sum_{\omega_i=\pm 1} \exp((2y+h)\omega_i) dy = \sqrt{\frac{N}{\pi d\beta}} \int_{-\infty}^{\infty} e^{-N\varphi_{\beta,h}(y)} dy, \end{aligned}$$

where

$$\varphi_{\beta,h}(y) := \frac{y^2}{d\beta} - \log(2 \cosh(2y+h)).$$

EXERCISE 3.2. Show that

$$\lim_{N \rightarrow \infty} \sqrt{N} \int_{-\infty}^{\infty} e^{-N(\varphi_{\beta,h}(y) - \min_y \varphi_{\beta,h}(y))} dy > 0.$$

Hint. Use second-order Taylor expansions of $\varphi_{\beta,h}$ around its minima.

SOLUTION. Let us write $\varphi(y) \equiv \varphi_{\beta,h}(y)$. Notice that, since $\beta > 0$, $\varphi(y) \uparrow +\infty$ as $y \rightarrow \pm\infty$, sufficiently fast to ensure that

$$\int_{-\infty}^{+\infty} e^{-c\varphi(y)} dy < \infty$$

for all $c > 0$. Depending on β , φ has either one or two global minima. For simplicity, consider the case in which there is a unique global minimum y_* . Let $\tilde{\varphi}(y) := \varphi(y) - \varphi(y_*)$, $B_\epsilon(y_*) := [y_* - \epsilon, y_* + \epsilon]$ and write

$$\int_{-\infty}^{\infty} e^{-N(\varphi_{\beta,h}(y) - \min_y \varphi_{\beta,h}(y))} dy \geq \int_{B_\epsilon(y_*)} e^{-N\tilde{\varphi}(y)} dy.$$

Let $c > 0$ be such that $\tilde{\varphi}(y) \leq c(y - y_*)^2$ for all $y \in B_\epsilon(y_*)$. Then

$$\sqrt{N} \int_{B_\epsilon(y_*)} e^{-N\tilde{\varphi}(y)} dy \geq \sqrt{N} \int_{B_\epsilon(y_*)} e^{-cN(y-y_*)^2} dy = \frac{1}{\sqrt{2c}} \int_{-\epsilon\sqrt{2cN}}^{+\epsilon\sqrt{2cN}} e^{-\frac{x^2}{2}} dx,$$

and this last expression converges to $\sqrt{\frac{\pi}{c}}$ when $N \rightarrow \infty$. $\square$

Infinite-Volume Gibbs Measures

A model is defined by first considering the set Ω_0 , called the *single-spin space*, which describes all the possible states of one spin. The *spin configurations* on a (possible infinite) subset $S \subseteq \mathbb{Z}^d$ are defined as:

$$\Omega_S := \Omega_0^S = \{(\omega_i)_{i \in S} : \omega_i \in \Omega_0, \forall i \in S\}.$$

When $S = \mathbb{Z}^d$, we simply write $\Omega \equiv \Omega_{\mathbb{Z}^d}$. Then, for each finite subset $\Lambda \subseteq \mathbb{Z}^d$, the energy of a configuration in Λ is determined by a *Hamiltonian* $\mathcal{H}_\Lambda : \Omega \rightarrow \mathbb{R}$.

For the *Ising model*, $\Omega_0 = \{+1, -1\}$. The nearest-neighbour version corresponds to

$$\mathcal{H}_\Lambda(\omega) = -\beta \sum_{\{i,j\} \in \mathcal{E}_\Lambda^b} \omega_i \omega_j - h \sum_{i \in \Lambda} \omega_i,$$

where $\mathcal{E}_\Lambda^b$ is the set of nearest-neighbour edges of $\mathbb{Z}^d$ with at least one endpoint in Λ . We will also consider a long-range version of this model:

$$\mathcal{H}_\Lambda(\omega) = - \sum_{\{i,j\} \cap \Lambda \neq \emptyset} J_{ij} \omega_i \omega_j - h \sum_{i \in \Lambda} \omega_i,$$

where $J_{ij} \rightarrow 0$ (sufficiently fast) when $\|j - i\|_1 \rightarrow \infty$.

SECTION 4.1

| The Problem with Infinite Systems

By considering for example the + boundary condition, we started in a finite volume $\Lambda \subseteq \mathbb{Z}^d$, and defined the Gibbs distribution unambiguously by

$$\mu_{\Lambda; \beta, h}^+(\omega) = \frac{e^{-\mathcal{H}_{\Lambda; \beta, h}(\omega)}}{\mathbf{Z}_{\Lambda; \beta, h}^+}, \quad \omega \in \Omega_\Lambda^+.$$

Then, to describe the Ising model on the infinite lattice, we introduced the thermodynamic limit. We considered a sequence of subsets $\Lambda_n \uparrow \mathbb{Z}^d$ and showed, for each local function f , existence of the limit

$$\langle f \rangle_{\beta, h}^+ = \lim_{n \rightarrow \infty} \langle f \rangle_{\Lambda_n; \beta, h}^+.$$

This defined a linear functional $\langle \cdot \rangle_{\beta, h}^+$ on local functions, which was called an *infinite-volume Gibbs state*.

SECTION 4.2

| Events and Probability Measures on Ω

THEOREM 4.1 (Kolmogorov's Extension Theorem).

Let $\{\mu_\Lambda\}_{\Lambda \in \mathbb{Z}^d}$, $\mu_\Lambda \in \mathcal{M}_1(\Omega_\Lambda)$, be consistent in the sense that

$$\forall \Lambda \in \mathbb{Z}^d : \mu_\Lambda = \mu_\Lambda \circ (\Pi_\Lambda^\Lambda)^{-1}, \quad \forall \Delta \subset \Lambda.$$

Then there exists a unique $\mu \in \mathcal{M}_1(\Omega)$ such that $\mu|_\Lambda = \mu_\Lambda$ for all $\Lambda \in \mathbb{Z}^d$.

SECTION 4.3

Specifications and Measures

DEFINITION 4.2.

Let $\Lambda \subseteq \mathbb{Z}^d$. A *probability kernel* from $\mathcal{F}_{\Lambda^c}$ to $\mathcal{F}$ is a map $\pi_\Lambda : \mathcal{F} \times \Omega \rightarrow [0, 1]$ with the following properties:

- (i) for each $\omega \in \Omega$, $\pi_\Lambda(\cdot \mid \omega)$ is a probability measure on $(\Omega, \mathcal{F})$;
- (ii) for each $A \in \mathcal{F}$, $\pi_\Lambda(A \mid \cdot)$ is $\mathcal{F}_{\Lambda^c}$ -measurable.

If, moreover,

$$\pi_\Lambda(B \mid \omega) = \mathbb{1}_B(\omega), \quad \forall B \in \mathcal{F}_{\Lambda^c},$$

for all $\omega \in \Omega$, π_Λ is said to be *proper*.

EXERCISE 4.1. Let $\Delta \subset \Lambda \in \mathbb{Z}^d$. Show that $\pi_\Lambda \pi_\Delta$ is a proper probability kernel from $\mathcal{F}_{\Lambda^c}$ to $\mathcal{F}$.

SOLUTION. By definition,

$$\pi_\Lambda \pi_\Delta(A \mid \eta) = \sum_{\omega_\Delta \in \Omega_\Delta} \pi_\Lambda(\omega_\Delta \mid \eta) \pi_\Delta(A \mid \omega_\Delta \eta_{\Lambda^c}),$$

which only depends on η_{Λ^c} . This also immediately implies that $\pi_\Lambda \pi_\Delta$ is proper. $\square$

DEFINITION 4.3.

A *specification* is a family $\pi = \{\pi_\Lambda\}_{\Lambda \in \mathbb{Z}^d}$ of proper probability kernels that is *consistent*, in the sense that

$$\pi_\Lambda \pi_\Delta = \pi_\Lambda, \quad \forall \Delta \subset \Lambda \in \mathbb{Z}^d.$$

EXERCISE 4.2. Show that, for every bounded measurable function f , every measure $\mu \in \mathcal{M}_1(\Omega)$ and every kernel π_Λ , $\mu \pi_\Lambda(f) = \mu(\pi_\Lambda f)$.

SOLUTION. If $f = \mathbb{1}_A$,

$$\mu \pi_\Lambda(\mathbb{1}_A) = \int \pi_\Lambda(A \mid \omega) \mu(d\omega) = \int \pi_\Lambda \mathbb{1}_A(\omega) \mu(d\omega) = \mu(\pi_\Lambda \mathbb{1}_A).$$

For the general case, just approximate f by a sequence of simple functions of the form $\sum_i a_i \mathbb{1}_{A_i}$. $\square$

EXERCISE 4.3. For each $i \in \mathbb{Z}^d$, let ρ_i be a probability distribution on $\{\pm 1\}$. For each $\Lambda \in \mathbb{Z}^d$, define the product distribution ρ^Λ on Ω_Λ by

$$\rho^\Lambda(\omega_\Lambda) := \prod_{i \in \Lambda} \rho_i(\omega_i).$$

For $\tau_\Lambda \in \Omega_\Lambda$ and $\eta \in \Omega$, let

$$\pi_\Lambda(\tau_\Lambda \mid \eta) := \rho^\Lambda(\tau_\Lambda).$$

- (i) Show that $\pi = \{\pi_\Lambda\}_{\Lambda \in \mathbb{Z}^d}$ is a specification.
- (ii) Show that the product measure $\rho^{\mathbb{Z}^d}$ is the unique probability measure specified by π : $\mathcal{G}(\pi) = \{\rho^{\mathbb{Z}^d}\}$.

SOLUTION. (ii) Observe that, for any $A \in \mathcal{F}$,

$$\rho\pi_\Lambda(A) = \int \pi_\Lambda(A \mid \omega)\rho(d\omega) = \int \mathbf{1}_A(\tau_\Lambda\omega_{\Lambda^c})\rho^\Lambda(d\tau_\Lambda)\rho(d\omega_{\Lambda^c}) = \rho(A),$$

so that $\rho \in \mathcal{G}(\pi)$. To prove uniqueness, let $\mu \in \mathcal{G}(\pi)$ and consider an arbitrary cylinder $C = \Pi_\Lambda^{-1}(E)$ with base Λ . Then, one must have

$$\mu(C) = \mu\pi_\Lambda(C) = \int \pi_\Lambda(\Pi_\Lambda^{-1}(E) \mid \omega)\mu(d\omega) = \int \rho^\Lambda(E)\mu(d\omega) = \rho^\Lambda(E),$$

and therefore μ must coincide with ρ on all cylinders, which implies that $\mu = \rho$. $\square$

DEFINITION 4.4.

If, for each finite $B \Subset \mathbb{Z}^d$, $\Phi_B : \Omega \rightarrow \mathbb{R}$ is $\mathcal{F}_B$ -measurable, then the collection $\Phi = \{\Phi_B\}_{B \Subset \mathbb{Z}^d}$ is called a *potential*. The *Hamiltonian* in the box $\Lambda \Subset \mathbb{Z}^d$ associated to the potential Φ is defined by

$$\mathcal{H}_{\Lambda;\Phi}(\omega) := \sum_{B \Subset \mathbb{Z}^d : B \cap \Lambda \neq \emptyset} \Phi_B(\omega), \quad \forall \omega \in \Omega.$$

We now proceed to define a specification $\pi^\Phi = \{\pi_\Lambda^\Phi\}_{\Lambda \Subset \mathbb{Z}^d}$ such that $\pi_\Lambda^\Phi(\cdot \mid \omega)$ gives to each configuration $\tau_\Lambda\omega_{\Lambda^c}$ a probability proportional to the Boltzmann weight prescribed by equilibrium statistical mechanics:

$$\pi_\Lambda^\Phi(\tau_\Lambda \mid \omega) := \frac{1}{Z_{\Lambda;\Phi}^\omega} e^{-\mathcal{H}_{\Lambda;\Phi}(\tau_\Lambda\omega_{\Lambda^c})},$$

where we have written explicitly the dependence on ω_{Λ^c} , and where the partition function $Z_{\Lambda;\Phi}^\omega$ is given by

$$Z_{\Lambda;\Phi}^\omega := \sum_{\tau_\Lambda \in \Omega_\Lambda} \exp(-\mathcal{H}_{\Lambda;\Phi}(\tau_\Lambda\omega_{\Lambda^c})).$$

LEMMA 4.5.

$\pi^\Phi = \{\pi_\Lambda^\Phi\}_{\Lambda \Subset \mathbb{Z}^d}$ is a specification.

DEFINITION 4.6.

The specification π^Φ associated to a potential Φ is said to be *Gibbsian*. A probability measure μ compatible with the Gibbsian specification π^Φ is said to be an *infinite-volume Gibbs measure* (or simply a *Gibbs measure*) associated to the potential Φ .

DEFINITION 4.7.

If $\mathcal{G}(\Phi)$ contains at least two distinct Gibbs measures, $|\mathcal{G}(\Phi)| > 1$, we say that there is a *first-order phase transition* for the potential Φ .

SECTION 4.4

| Existence

EXERCISE 4.4. Let η^- denote the configuration in which all spins are -1 , and let $\eta^{-,i}$ denote the configuration in which all spins are -1 , except at the vertex i , at which it is $+1$. For $\Lambda \Subset \mathbb{Z}^d$, let

$$\pi_\Lambda(A \mid \omega) := \begin{cases} \frac{1}{|\Lambda|} \sum_{i \in \Lambda} \mathbb{1}_A(\eta^{-,i}) & \text{if } \omega_{\Lambda^c} = \eta_{\Lambda^c}^- \\ \mathbb{1}_A(\eta_\Lambda^- \omega_{\Lambda^c}) & \text{otherwise} \end{cases}.$$

Show that $\pi = \{\pi_\Lambda\}_{\Lambda \Subset \mathbb{Z}^d}$ is a specification and explain why it describes a system consisting of a single $+$ spin, located anywhere on $\mathbb{Z}^d$, in a sea of $-$ spins. Show that π is not quasilocal and that $\mathcal{G}(\pi) = \emptyset$.

SOLUTION. Assume there exists $\mu \in \mathcal{G}(\pi)$. Notice that

$$\mu(N^+ = 0) = \mu(\{\eta^-\}) = \mu\pi_\Lambda(\{\eta^-\}) = \int \pi_\Lambda(\{\eta^-\} \mid \omega) \mu(d\omega) = 0.$$

Then, $\mu(N^+ = 1) = \sum_{i \in \mathbb{Z}^d} \mu(\{\eta^{-,i}\})$. However, for all $\Lambda \subseteq \mathbb{Z}^d$ containing i ,

$$\mu(\{\eta^{-,i}\}) = \mu\pi_\Lambda(\{\eta^{-,i}\}) \leq \frac{1}{|\Lambda|},$$

so that $\mu(\{\eta^{-,i}\}) = 0$. We conclude that $\mu(N^+ = 1) = 0$. Finally,

$$\mu(N^+ \geq 2) \leq \sum_{i \neq j} \mu(\{\omega_i = \omega_j = +1\}) = 0,$$

since

$$\mu(\{\omega_i = \omega_j = +1\}) = \mu\pi_{\{i,j\}}(\{\omega_i = \omega_j = +1\}) = 0$$

for all $i \neq j$. All this implies that $\mu(\Omega) = 0$, which contradicts the assumption that μ is a probability measure. $\square$

SECTION 4.5

| Uniqueness

Dobrushin's Uniqueness Theorem

Our first uniqueness criterion will be formulated in terms of the one-vertex kernels $\pi_{\{i\}}(\cdot \mid \omega)$, which for simplicity will be denoted by π_i . Each $\pi_i(\cdot \mid \omega)$ should be considered as a distribution for the spin at vertex i , with boundary condition ω . We will measure the dependence of $\pi_i(\cdot \mid \omega)$ on the value of the boundary condition ω at other vertices. We will measure the proximity between two such distributions using the *total variation distance*:

$$\|\pi_i(\cdot \mid \omega) - \pi_i(\cdot \mid \omega')\|_{\text{TV}} := \frac{1}{2} \sum_{\eta_i = \pm 1} |\pi_i(\eta_i \mid \omega) - \pi_i(\eta_i \mid \omega')|.$$

We can then introduce

$$c_{ij}(\pi) := \sup_{\omega, \omega' \in \Omega: \omega_k = \omega'_k \forall k \neq j} \|\pi_i(\cdot \mid \omega) - \pi_i(\cdot \mid \omega')\|_{\text{TV}},$$

and

$$c(\pi) := \sup_{i \in \mathbb{Z}^d} \sum_{j \in \mathbb{Z}^d} c_{ij}(\pi).$$

THEOREM 4.8.

Let π be a quasilocal specification satisfying Dobrushin's Condition of Weak Dependence: $c(\pi) < 1$. Then the probability measure specified by π is unique: $|\mathcal{G}(\pi)| = 1$.

Application to Gibbsian Specifications

EXERCISE 4.5. The (nearest-neighbor) Potts antiferromagnet on $\mathbb{Z}^d$ at inverse temperature $\beta \geq 0$ has single spin space $\Omega_0 := \{0, \dots, q-1\}$ and is associated to the potential

$$\Phi_B(\omega) = \begin{cases} +\beta \delta_{\omega_i, \omega_j} & \text{if } B = \{i, j\}, i \sim j \\ 0 & \text{otherwise} \end{cases}.$$

Show that this model has a unique Gibbs measure for all $\beta \in \mathbb{R}^*$, provided that $q > 6d$.

SOLUTION. Clearly, $c_{ij}(\pi) = 0$ whenever $j \not\sim i$. Let $j \sim i$ and consider two configurations ω, η such that $\omega_k = \eta_k$ for all $k \neq j$. When $s \in \{0, \dots, q-1\} \setminus \{\omega_j, \eta_j\}$, $\pi_i(\sigma_i = s \mid \eta) = \pi_i(\sigma_i = s \mid \omega)$. Let us therefore assume that $\omega_j = s \neq \eta_j$. In this case,

$$\pi_i(\sigma_i = s \mid \eta) - \pi_i(\sigma_i = s \mid \omega) = \frac{e^{-\beta \#\{k \sim i: \eta_k = s\}}}{Z_{\{i\}}^\eta} \left\{ 1 - \frac{Z_{\{i\}}^\eta}{Z_{\{i\}}^\omega} e^{-\beta} \right\}.$$

Now, observe that

$$\frac{Z_{\{i\}}^\eta}{Z_{\{i\}}^\omega} e^{-\beta} = \langle e^{-\beta(\delta_{\sigma_i, \omega_j} - \delta_{\sigma_i, \eta_j} + 1)} \rangle_{\{i\}}^\omega \in [e^{-2\beta}, 1].$$

Therefore,

$$|\pi_i(\sigma_i = s \mid \eta) - \pi_i(\sigma_i = s \mid \omega)| \leq \frac{1}{Z_{\{i\}}^\eta} \leq \frac{1}{q - 2d}.$$

This yields

$$c_{ij}(\pi) \leq \frac{2}{q-2d} \implies c(\pi) \leq \frac{4d}{q-2d},$$

which is indeed smaller than 1 as soon as $q > 6d$. □

COROLLARY 4.9.

Let $\Phi = \{\Phi_B\}_{B \in \mathbb{Z}^d}$ be an absolutely summable potential satisfying

$$b := \sup_{i \in \mathbb{Z}^d} \sum_{B \ni i} (|B| - 1) \|\Phi_B\|_\infty < \infty,$$

and let $\beta_0 := \frac{1}{2b}$. Then, for all $\beta < \beta_0$, there is a unique measure compatible with $\pi^{\beta\Phi}$.

SECTION 4.6

| Translation Invariant Gibbs Measures

DEFINITION 4.10.

$\mu \in \mathcal{M}_1(\Omega)$ is *translation invariant* if $\theta_j \mu = \mu$ for all $j \in \mathbb{Z}^d$.

SECTION 4.7

| Convexity and Extremal Gibbs Measures

THEOREM 4.11.

 $\mathcal{G}(\pi)$ is convex.*Proof.* Let $\mu = \lambda v_1 + (1 - \lambda)v_2$, with $v_1, v_2 \in \mathcal{G}(\pi)$. For all $\Lambda \in \mathbb{Z}^d$,

$$\mu\pi_\Lambda = \lambda v_1\pi_\Lambda + (1 - \lambda)v_2\pi_\Lambda = \lambda v_1 + (1 - \lambda)v_2 = \mu,$$

and so $\mu \in \mathcal{G}(\pi)$. □

Properties of Extremal Gibbs Measures

THEOREM 4.12.

Let π be a specification. Let $\mu \in \mathcal{G}(\pi)$. The following conditions are equivalent characterisations of extremality:

- (i) μ is extremal;
- (ii) μ is trivial on $\mathcal{T}_\infty$: if $A \in \mathcal{T}_\infty$, then $\mu(A)$ is either 1 or 0;
- (iii) all $\mathcal{T}_\infty$ -measurable functions are μ -a.s. constant;
- (iv) μ has short-range correlations: for all $A \in \mathcal{F}$ (or, equivalently, for all $A \in \mathcal{C}$),

$$\lim_{\Lambda \uparrow \mathbb{Z}^d} \sup_{B \in \mathcal{F}_{\Lambda^c}} |\mu(A \cap B) - \mu(A)\mu(B)| = 0. \quad (4.1)$$

Let $\Lambda \subseteq \mathbb{Z}^d$. We define the restriction $r_\Lambda : \mathcal{M}_1(\Omega, \mathcal{F}) \rightarrow \mathcal{M}_1(\Omega, \mathcal{F}_{\Lambda^c})$ by

$$r_\Lambda \mu(B) := \mu(B), \quad \forall B \in \mathcal{F}_{\Lambda^c}.$$

Observe that if $g : \Omega \rightarrow \mathbb{R}$ is $\mathcal{F}_{\Lambda^c}$ -measurable, then $r_\Lambda \mu(g) = \mu(g)$. Using a specification π , one can define for each $\Lambda \subseteq \mathbb{Z}^d$ the extension $t_\Lambda^\pi : \mathcal{M}_1(\Omega, \mathcal{F}_{\Lambda^c}) \rightarrow \mathcal{M}_1(\Omega, \mathcal{F})$ by

$$t_\Lambda^\pi \nu(A) := \nu\pi_\Lambda(A), \quad \forall A \in \mathcal{F}.$$

Note that the composition of t_Λ^π with r_Λ is such that $t_\Lambda^\pi r_\Lambda : \mathcal{M}_1(\Omega, \mathcal{F}) \rightarrow \mathcal{M}_1(\Omega, \mathcal{F})$.

PROPOSITION 4.13.

 $\mu \in \mathcal{G}(\pi)$ if and only if $\mu = t_\Lambda^\pi r_\Lambda \mu$ for all $\Lambda \subseteq \mathbb{Z}^d$.*Proof.* Composing t_Λ^π with r_Λ gives, for all $A \in \mathcal{F}$,

$$t_\Lambda^\pi r_\Lambda \mu(A) = (r_\Lambda \mu)\pi_\Lambda(A) = \int \pi_\Lambda(A \mid \omega) r_\Lambda \mu(d\omega) = \int \pi_\Lambda(A \mid \omega) \mu(d\omega) = \mu\pi_\Lambda(A).$$

In the third identity, we used the $\mathcal{F}_{\Lambda^c}$ -measurability of $\pi_\Lambda(A \mid \cdot)$. □

LEMMA 4.14.

Let $\Lambda \subseteq \mathbb{Z}^d$.

- (i) Let $\mu \in \mathcal{M}_1(\Omega, \mathcal{F})$ and let $f : \Omega \rightarrow \mathbb{R}^*$ be an $\mathcal{F}$ -measurable function such that $\mu(f) = 1$. Then

$$r_\Lambda(f\mu) = \mu(f \mid \mathcal{F}_{\Lambda^c})r_\Lambda \mu.$$

- (ii) Let $\nu \in \mathcal{M}_1(\Omega, \mathcal{F}_{\Lambda^c})$ and let $g : \Omega \rightarrow \mathbb{R}^*$ be an $\mathcal{F}_{\Lambda^c}$ -measurable function such that $\nu(g) = 1$. Then

$$t_{\Lambda}^{\pi}(g\nu) = g \cdot t_{\Lambda}^{\pi}\nu.$$

Proof. (i) Take $B \in \mathcal{F}_{\Lambda^c}$ and use the definition of conditional expectation:

$$r_{\Lambda}(f\mu)(B) = \int_B f(\omega)\mu(d\omega) = \int_B \mu(f \mid \mathcal{F}_{\Lambda^c})(\omega)\mu(d\omega) = \int_B \mu(f \mid \mathcal{F}_{\Lambda^c})(\omega)r_{\Lambda}\mu(d\omega).$$

- (ii) Take $A \in \mathcal{F}$ and compute:

$$t_{\Lambda}^{\pi}(g\nu)(A) = (g\nu)\pi_{\Lambda}(A) = g\nu\pi_{\Lambda}(A) = g \cdot t_{\Lambda}^{\pi}\nu(A).$$

□

PROPOSITION 4.15.

Let π be a specification.

- (i) Let $\mu \in \mathcal{G}(\pi)$. Let $f : \Omega \rightarrow \mathbb{R}_{\geq 0}$, $\mathcal{F}$ -measurable, such that $\mu(f) = 1$. Then $f\mu \in \mathcal{G}(\pi)$ if and only if f is equal μ -almost everywhere to a $\mathcal{T}_{\infty}$ -measurable function.
- (ii) Let $\mu, \nu \in \mathcal{G}(\pi)$ be two probability measures that coincide on $\mathcal{T}_{\infty}$: $\mu(A) = \nu(A)$ for all $A \in \mathcal{T}_{\infty}$. Then $\mu = \nu$.

EXERCISE 4.6. Consider the two-dimensional Ising model with $h = 0$ and $\beta > \beta_c(2)$. Take any $\lambda \in (0, 1)$ and consider the (non-extremal) Gibbs measure

$$\mu = \lambda\mu_{\beta,0}^+ + (1 - \lambda)\mu_{\beta,0}^-.$$

Show that μ does not satisfy (4.1), by taking $A = \{\sigma_0 = 1\}$, $B_i = \{\sigma_i = 1\}$ and verifying that

$$\liminf_{\|i\|_1 \rightarrow \infty} |\mu(A \cap B_i) - \mu(A)\mu(B_i)| > 0.$$

SOLUTION. Since

$$\mathbf{1}_A = \frac{1 + \sigma_0}{2}, \quad \mathbf{1}_{B_i} = \frac{1 + \sigma_i}{2},$$

we have

$$\mu(A \cap B_i) - \mu(A)\mu(B_i) = \frac{1}{4}(\mu(\sigma_0\sigma_i) - \mu(\sigma_0)\mu(\sigma_i)).$$

By symmetry,

$$\mu_{\beta,0}^+(\sigma_0\sigma_i) = \mu_{\beta,0}^-(\sigma_0\sigma_i), \quad \mu(\sigma_0) = (2\lambda - 1)\mu_{\beta,0}^+(\sigma_0), \quad \mu(\sigma_i) = (2\lambda - 1)\mu_{\beta,0}^+(\sigma_i).$$

By the FKG inequality,

$$\mu_{\beta,0}^+(\sigma_0\sigma_i) \geq \mu_{\beta,0}^+(\sigma_0)\mu_{\beta,0}^+(\sigma_i).$$

We therefore conclude that

$$\mu(\sigma_0\sigma_i) - \mu(\sigma_0)\mu(\sigma_i) \geq (1 - (2\lambda - 1)^2) \left(\mu_{\beta,0}^+(\sigma_0) \right)^2,$$

which is positive for all $\beta > \beta_c(2)$ and all $\lambda \in (0, 1)$. □

LEMMA 4.16.

Distinct extremal measures $\mu, \nu \in \text{ex}\mathcal{G}(\pi)$ are singular: there exists a tail event $A \in \mathcal{T}_{\infty}$ such that $\mu(A) = 0$ and $\nu(A) = 1$.

Proof. If $\mu, \nu \in \mathcal{G}(\pi)$ are distinct, then [Proposition 4.15](#) shows that there must exist $A \in \mathcal{T}_\infty$ such that $\mu(A) \neq \nu(A)$. But if μ and ν are extremal, they are trivial on $\mathcal{T}_\infty$ ([Theorem 4.12](#)), so either $\mu(A) = 0$ and $\nu(A) = 1$, or $\mu(A^c) = 0$ and $\nu(A^c) = 1$. $\square$

Extremal Gibbs Measures and the Thermodynamic Limit

THEOREM 4.17.

Let $\mu \in \text{ex}\mathcal{G}(\pi)$. Then, for μ -almost all ω ,

$$\pi_{B(n)}(\cdot \mid \omega) \Rightarrow \mu.$$

Proof. We need to prove that, for μ -almost all ω ,

$$\pi_{B(n)}(C \mid \omega) \xrightarrow{n \rightarrow \infty} \mu(C), \quad \forall C \in \mathcal{C}. \quad (4.2)$$

Let $C \in \mathcal{C}$. On the one hand (see (6.23)), there exists $\Omega_{n,C}$, $\mu(\Omega_{n,C}) = 1$, such that

$$\pi_{B(n)}(C \mid \omega) = \mu(C \mid \mathcal{F}_{B(n)^c})(\omega) \quad \text{for all } \omega \in \Omega_{n,C}.$$

On the other hand, the extremality of μ ([Theorem 4.12](#)) guarantees that there exists Ω_C , $\mu(\Omega_C) = 1$, such that

$$\mu(C) = \mu(C \mid \mathcal{T}_\infty)(\omega) \quad \text{for all } \omega \in \Omega_C.$$

Using (6.64) with $f = \mathbb{1}_C$, there also exists $\tilde{\Omega}_C$, $\mu(\tilde{\Omega}_C) = 1$, such that

$$\mu(C \mid \mathcal{F}_{B(n)^c})(\omega) \rightarrow \mu(C \mid \mathcal{T}_\infty)(\omega) \quad \forall \omega \in \tilde{\Omega}_C.$$

Therefore, for all ω that belong to the countable intersection of all the sets Ω_C , $\tilde{\Omega}_C$, and $\Omega_{n,C}$, which has μ -measure 1, (4.2) holds. $\square$

EXAMPLE 4.18. Let us consider the 3-dimensional Ising model, with $\beta > \beta_c(3)$ and $h = 0$, in the box $\mathbb{B}(n)$. The sequence of finite-volume Gibbs distributions with Dobrushin boundary condition admits a converging subsequence, defining a Gibbs measure $\mu_{\beta,0}^{\text{Dob}}$ satisfying, for any $\epsilon > 0$,

$$\langle \sigma_{(0,0,0)} \sigma_{(0,0,-1)} \rangle_{\beta,0}^{\text{Dob}} \leq -1 + \epsilon, \quad (4.3)$$

once β is large enough (see [Theorem 3.60](#)). Let us denote by $\mu_{\beta,0}^{\text{Dob}}$ the corresponding Gibbs measure. Applying a global spin flip, we obtain another Gibbs measure,

$$\mu_{\beta,0}^{-\text{Dob}} := \tau_f \mu_{\beta,0}^{\text{Dob}},$$

also satisfying (4.3). Since $\mathcal{G}(\beta, 0)$ is convex,

$$\mu := \frac{1}{2} \mu_{\beta,0}^{\text{Dob}} + \frac{1}{2} \mu_{\beta,0}^{-\text{Dob}} \in \mathcal{G}(\beta, 0).$$

We show that it cannot be obtained as a thermodynamic limit. Notice that $\langle \sigma_i \rangle_\mu = 0$ for all $i \in \mathbb{Z}^d$ and that one has, for any $\epsilon > 0$,

$$\langle \sigma_{(0,0,0)} \sigma_{(0,0,-1)} \rangle_\mu \leq -1 + \epsilon, \quad (4.4)$$

once β is large enough. Suppose there exists a sequence $(\mu_{\mathbb{B}(n_k); \beta, 0}^{\eta_k})_{k \geq 1}$ converging to μ . By the *FKG inequality*, for all $k \geq 1$,

$$\langle \sigma_{(0,0,0)} \sigma_{(0,0,-1)} \rangle_{\mathbb{B}(n_k); \beta, 0}^{\eta_k} \geq \langle \sigma_{(0,0,0)} \rangle_{\mathbb{B}(n_k); \beta, 0}^{\eta_k} \langle \sigma_{(0,0,-1)} \rangle_{\mathbb{B}(n_k); \beta, 0}^{\eta_k}.$$

and thus

$$\langle \sigma_{(0,0,0)} \sigma_{(0,0,-1)} \rangle_\mu \geq \langle \sigma_{(0,0,0)} \rangle_\mu \langle \sigma_{(0,0,-1)} \rangle_\mu = 0.$$

This contradicts (4.4).

LEMMA 4.19.

$\mu_{\beta,h}^+$ and $\mu_{\beta,h}^-$ are extremal.

SECTION 4.8

| The Variational Principle

Formulation in the Finite Case

LEMMA 4.20 (Variational Principle Finite Version).

For all $\mu_\Lambda \in \mathcal{M}_1(\Omega_\Lambda)$,

$$\mathcal{W}_\Lambda(\mu_\Lambda) \leq \mathcal{W}_\Lambda(\mu_\Lambda^{\text{Gibbs}}).$$

Moreover, $\mu_\Lambda^{\text{Gibbs}}$ is the unique maximiser of $\mathcal{W}_\Lambda(\cdot)$.

Proof. Since $\log(\cdot)$ is concave, Jensen's inequality gives, for all $\mu_\Lambda \in \mathcal{M}_1(\Omega_\Lambda)$,

$$\mathcal{W}_\Lambda(\mu_\Lambda) = \sum_{\omega_\Lambda \in \Omega_\Lambda} \mu_\Lambda(\omega_\Lambda) \log \frac{e^{-\beta \mathcal{H}_\Lambda(\omega_\Lambda)}}{\mu_\Lambda(\omega_\Lambda)} \leq \log \sum_{\omega_\Lambda \in \Omega_\Lambda} e^{-\beta \mathcal{H}_\Lambda(\omega_\Lambda)} = \log Z_\Lambda,$$

and equality holds if and only if $\frac{e^{-\beta \mathcal{H}_\Lambda(\omega_\Lambda)}}{\mu_\Lambda(\omega_\Lambda)}$ is constant, that is, if $\mu_\Lambda = \mu_\Lambda^{\text{Gibbs}}$. $\square$

EXERCISE 4.7. Let $\mathcal{H}_{V_n; \beta, h}^{\text{per}}$ be the Hamiltonian of the Ising model in $V_n := \{0, \dots, n-1\}^d$ with periodic boundary condition. Show that, among all product probability measures $\mu_{V_n} = \bigotimes_{i \in V_n} \rho_i$ on $\{\pm 1\}^{V_n}$ (where all ρ_i are equal), the unique measure maximising

$$\mathcal{W}_{V_n}(\mu_{V_n}) := S_{V_n}(\mu_{V_n}) - \beta \langle \mathcal{H}_{V_n; \beta, h}^{\text{per}} \rangle_{\mu_{V_n}}$$

is the measure such that $\rho_i = \nu$ for all $i \in V_n$, where ν is the probability measure on $\{\pm 1\}$ with mean m satisfying $m = \tanh(2d\beta m + h)$.

SOLUTION. A simple computation yields

$$\mathcal{W}_{V_n}(\mu_{V_n}) = |V_n| \left\{ d\beta m^2 + hm - \frac{1+m}{2} \log \frac{1+m}{2} - \frac{1-m}{2} \log \frac{1-m}{2} \right\},$$

where we have introduced $m := \langle \sigma_i \rangle_{\rho_i}$. It is now a matter of straightforward calculus to show that the unique maximum is attained when m satisfies $m = \tanh(2d\beta m + h)$. $\square$

Specific Entropy and Energy Density

Variational Principle for Gibbs Measures

PROPOSITION 4.21.

Let Φ be absolutely convergent and translation invariant. Let $\nu \in \mathcal{G}_\theta(\beta\Phi)$. Then, for all $\mu \in \mathcal{M}_{1,\theta}(\Omega)$, the Gibbs free energy

$$h(\mu \mid \Phi) := \lim_{n \rightarrow \infty} \frac{1}{|\mathbb{B}(n)|} \mathcal{H}_{\mathbb{B}(n)}(\mu \mid \nu) \quad (4.5)$$

exists and does not depend on ν (only on Φ). Moreover, $h(\mu \mid \Phi)$ is non-negative and satisfies

$$h(\mu \mid \Phi) = \beta \psi(\Phi) - \{s(\mu) - \beta \langle u_\Phi \rangle_\mu\}.$$

Proof. If $\nu \in \mathcal{G}_\theta(\beta\Phi)$, then $\mathcal{H}_{\mathbb{B}(n)}(\mu \mid \nu) = \mathcal{H}_{\mathbb{B}(n)}(\mu \mid \pi_{\mathbb{B}(n)}^{\beta\Phi})$. Therefore, to show the existence of the limit (4.5), we can use Lemma 6.80 with $\nu_n = \nu$. As earlier, we choose $\tilde{\nu}_n = \delta_\omega$, for which we know that the limit of the second sequence in (6.99) exists and equals $\beta \psi(\Phi) - \{s(\mu) - \beta \langle u_\Phi \rangle_\mu\}$, as seen after (6.101). $\square$

THEOREM 4.22 (Variational Principle).

Let $\mu \in \mathcal{M}_{1,\theta}(\Omega)$. Then

$$\mu \in \mathcal{G}_\theta(\beta\Phi) \iff h(\mu \mid \Phi) = 0.$$

THEOREM 4.23.

Let Φ be an absolutely summable, translation-invariant potential. Define the affine functional $\mathcal{W} : \mathcal{M}_{1,\theta}(\Omega) \rightarrow \mathbb{R}$ by

$$\mu \mapsto \mathcal{W}(\mu) := s(\mu) - \beta \langle u_\Phi \rangle_\mu.$$

Then the maximisers of $\mathcal{W}(\cdot)$ are the translation-invariant Gibbs measures compatible with $\pi^{\beta\Phi}$.

The Ideal Gas

SECTION 5.1

| Uniform Distribution on Energy Shells and Boltzmann Entropy

SECTION 5.2

| Statistical Ensembles and Thermodynamic Potentials

Point Processes

| Configuration Space

SECTION 6.2

| Probability Measures

EXAMPLE 6.1 (Binomial Point Process). Let ν be a probability measure on $\mathcal{X}$, $m \in \mathbb{N}$, and $X_1, \dots, X_m$ i.i.d. $\mathcal{X}$ -valued random variables with distribution ν . Then $Z = \sum_{j=1}^m \delta_{X_j}$ is a point process (the measurability of $Z : \Omega \rightarrow \mathcal{N}$ follows from *Corollary 2.6*). Its distribution satisfies, for every $\Delta \in \mathcal{X}$ and $k \in \mathbb{N}_0$:

$$\begin{aligned} \mathbb{P}(N_\Delta = k) &= \mathbb{P}(\{\omega \in \Omega : Z(\omega, \Delta) = k\}) \\ &= \mathbb{P}\left(\left\{\mathbf{x} \in \mathcal{X}^m : \sum_{j=1}^m \delta_{x_j}(\Delta) = k\right\}\right) \\ &= \mathbb{P}\left(\sum_{j=1}^m \mathbb{1}_{\{X_j \in \Delta\}} = k\right) = \binom{m}{k} \nu(\Delta)^k (1 - \nu(\Delta))^{m-k}. \end{aligned}$$

Z is the *binomial point process* with sample size m , sampling distribution ν , and law $\mathbb{P}$.

SECTION 6.3

| Observables

SECTION 6.4

| Poisson Point Process

Gibbs Measures in Finite Volume

SECTION 7.1

| Energy Functions and Interaction Potentials

SECTION 7.2

| Boundary Conditions

SECTION 7.3

| Grand-Canonical Gibbs Measure

SECTION 7.4

| The Pressure and its Derivatives

Phase Transition for the Widom-Rowlinson Model

Mathematical Backgrounds

SECTION A.1

| Convex Functions

DEFINITION A.1 (Convexity).

SECTION A.2

| Probability

Random Variables and Vectors

THEOREM A.2 (Jensen Inequality).

Proof.

□

SECTION A.3

| Complex Analysis

Let $D \subset \mathbb{C}$ be a domain (that is, open and connected). Remember that a function $f : D \rightarrow \mathbb{C}$ is *holomorphic* if

$$f'(z) := \lim_{w \rightarrow z} \frac{f(w) - f(z)}{w - z}$$

exists and is finite at each $z \in D$. It is well known that holomorphic functions have derivatives of all orders, and that f is holomorphic if and only if it is *analytic*, that is, if and only if it can be represented at each point $z_0 \in D$ by a convergent Taylor series:

$$f(z) = \sum_{n \geq 0} a_n (z - z_0)^n,$$

where $a_n = \frac{1}{n!} f^{(n)}(z_0)$ and z belongs to a small disc around z_0 . Therefore, holomorphic and analytic should be considered as synonyms in this section. We start with the following fundamental result of complex analysis.

THEOREM A.3 (Cauchy's Integral Theorem).

Let $D \subset \mathbb{C}$ be open and simply connected, and let f be holomorphic on D . Then

$$\oint_{\gamma} f(\xi) d\xi = 0,$$

for all closed paths $\gamma \subset D$.

Proof.

□

COROLLARY A.4.

Let $D \subset \mathbb{C}$ be open and simply connected, and let f be holomorphic on D . Then, there exists a function F , holomorphic on D , such that $F' = f$.

Proof. We fix some point $z_0 \in D$. Since D is open and connected, any other $z \in D$ can be joined from z_0 by a continuous path $\gamma_z \subset D$. Let

$$F(z) := \int_{\gamma_z} f(\xi) d\xi.$$

By [Theorem A.3](#), this definition does not depend on the choice of the path γ_z . Choose $r > 0$ small enough to ensure that the disc $B(z, r) := \{w \in \mathbb{C} : |w - z| \leq r\} \subset D$. Then,

$$F(w) = F(z) + \int_{[z,w]} f(\xi) d\xi, \quad \forall w \in B(z, r),$$

where $[z, w]$ is the straight line segment connecting z to w . But f being holomorphic implies in particular that $f(\xi) = f(z) + O(|\xi - z|)$ for all $\xi \in B(z, r)$, and so

$$\lim_{w \rightarrow z} \frac{F(w) - F(z)}{w - z} = \lim_{w \rightarrow z} \frac{1}{w - z} \int_{[z,w]} f(\xi) d\xi = f(z).$$

This implies that F is holomorphic and that $F' = f$.

□

THEOREM A.5.

Let f be a holomorphic function on a simply connected open set $D \subset \mathbb{C}$, which has no zeroes on D .

Then, there exists a function g analytic on D , called a branch of the logarithm of f on D , such that $f = e^g$.

Proof. Our assumptions imply that f'/f is holomorphic on D . [Corollary A.4](#) thus implies the existence of a function F , holomorphic on D , such that $F' = f'/f$. In particular,

$$(fe^{-F})' = f'e^{-F} - fF'e^{-F} \equiv 0.$$

Therefore, there exists $c \in \mathbb{C}$ such that $fe^{-F} = e^c$, or equivalently $f = e^{F+c}$. $\square$

Remark. (i) Let g be a branch of the logarithm of f on D . Then $\Re g = \log |f|$. Indeed,

$$|f| = |e^g| = e^{\Re g} \cdot |e^{i\Im g}| = e^{\Re g}.$$

(ii) Let g_1 and g_2 be two branches of the logarithm of f on D . Since, for each $z \in D$,

$$e^{g_2(z)-g_1(z)} = \frac{e^{g_2(z)}}{e^{g_1(z)}} = \frac{f(z)}{f(z)} = 1,$$

we conclude that $g_2(z) = g_1(z) + 2ik(z)\pi$ for some $k(z) \in \mathbb{Z}$. However, $z \mapsto k(z) = (g_2(z) - g_1(z))/2i\pi$ is continuous and integer-valued; it is therefore constant on D . This implies that $g_2 = g_1 + 2ik\pi$ for some $k \in \mathbb{Z}$.

(iii) Assume that the domain D in [Theorem A.5](#) is such that $D \cap \mathbb{R}$ is connected. Suppose also that $f(z) \in \mathbb{R}_{>0}$ for $z \in D \cap \mathbb{R}$. Then there is a branch g of the logarithm of f on D such that $g(z) \in \mathbb{R}$ for all $z \in D \cap \mathbb{R}$; in particular, g coincides with the usual logarithm of f (seen as a real function) on $D \cap \mathbb{R}$. Indeed, it suffices to observe that the function F in the proof can be constructed by starting from a point $z_0 \in D \cap \mathbb{R}$ at which one can fix $g(z_0) = \log |f(z_0)|$ and use the fact that $F(z) = \int_{z_0}^z f'(x)/f(x) dx$ for $z \in D \cap \mathbb{R}$.

It is well known that the limit of a sequence of analytic functions need not be analytic. Let us see how additional conditions can be imposed to guarantee that the limiting function is also analytic.

Remember that a family $\mathcal{A}$ of functions on $\mathbb{C}$ is *locally uniformly bounded* on a set $D \subset \mathbb{C}$ if, for each $z \in D$, there exists a real number M and a neighborhood $\mathcal{U}$ of z such that $|f(w)| \leq M$ for all $w \in \mathcal{U}$ and all $f \in \mathcal{A}$.

THEOREM A.6 (Vitali Convergence Theorem).

Let D be an open, connected subset of $\mathbb{C}$ and $(f_n)_{n \geq 1}$ be a sequence of analytic functions on D , which are locally uniformly bounded and converge on a set having a cluster point in D . Then the sequence $(f_n)_{n \geq 1}$ converges locally uniformly on D to an analytic function.

Proof. $\square$

THEOREM A.7 (Hurwitz Theorem).

Let D be an open subset of $\mathbb{C}$ and $(f_n)_{n \geq 1}$ be a sequence of analytic functions, which converge, locally uniformly, on D to an analytic function f . If $f_n(z) \neq 0$, for all $z \in D$ and for all n , then either f vanishes identically, or f is never zero on D .

Proof. $\square$

The following theorem is the complex counterpart to [Theorem B.7](#). (Notice that, in the complex case, no control is needed on the series of the derivatives.)

THEOREM A.8 (Weierstrass' Theorem on Uniformly Convergent Series of Analytic Functions).
 Let $D \subset \mathbb{C}$ be a domain. For each k , let $f_k : D \rightarrow \mathbb{C}$ be an analytic function. If the series

$$f(z) := \sum_k f_k(z)$$

is uniformly convergent on every compact subset $K \subset D$, then it defines an analytic function on D . Moreover, for each $n \in \mathbb{N}$, $\sum_k f_k^{(n)}$ converges uniformly on every compact $K \subset D$ and

$$f^{(n)}(z) = \sum_k f_k^{(n)}(z), \quad \forall z \in D.$$

Let $U, V \subset \mathbb{C}$. A continuous function $F : U \times V \rightarrow \mathbb{C}$ is said to be *analytic* on $U \times V$ if $F(\cdot, z)$ is analytic on U for any fixed $z \in V$ and $F(z, \cdot)$ is analytic on V for any fixed $z \in U$.

THEOREM A.9 (Implicit Function Theorem).

Let $(\omega, z) \mapsto F(\omega, z)$ be an analytic function on an open domain $U \times V \subset \mathbb{C}^2$. Let $(\omega_0, z_0) \in U \times V$ be such that $F(\omega_0, z_0) = 0$ and $\frac{\partial F}{\partial z}(\omega_0, z_0) \neq 0$. Then there exists an open subset $U_0 \subset U$ containing ω_0 and an analytic map $\varphi : U_0 \rightarrow V$ such that $\varphi(\omega_0) = z_0$ and

$$F(\omega, \varphi(\omega)) = 0 \quad \text{for all } \omega \in U_0.$$

Exercises

EXERCISE B.1. Consider the Ising model in $\mathbb{Z}^2$ on the graph $(V_N, \mathcal{E}_N^{\text{per}})$ where $V_N = \{1, \dots, N\}^2$ with periodic boundary conditions.

- (i) Compute explicitly $\lim_{N \rightarrow \infty} \langle \frac{1}{N^2} \sum_{i \in V_N} \sigma_i \rangle_{N,0,h}^{\text{per}}$ and $\lim_{N \rightarrow \infty} \langle \frac{1}{N^2} \sum_{i \in V_N} \sigma_i \rangle_{N,\infty,h}^{\text{per}}$.
- (ii) Let $N = 2$. Compute explicitly $\langle \frac{1}{4} |\sum_{i \in V_2} \sigma_i| \rangle_{2,\beta,h}^{\text{per}}$.
- (iii) Draw the following functions: $\beta \mapsto \langle \frac{1}{4} |\sum_{i \in V_2} \sigma_i| \rangle_{2,\beta,0}^{\text{per}}$ and $h \mapsto \langle \frac{1}{4} |\sum_{i \in V_2} \sigma_i| \rangle_{2,1,h}^{\text{per}}$.

SOLUTION. The Hamiltonian for the Ising model with periodic boundary conditions is given by

$$\mathcal{H}_{V_n;\beta,h}^{\text{per}} = -\beta \sum_{\{i,j\} \in \mathcal{E}_{V_n}^{\text{per}}} \sigma_i \sigma_j - h \sum_{i \in V_n} \sigma_i.$$

And the Gibbs distribution is

$$\mu_{V_n;\beta,h}^{\text{per}} = \frac{1}{Z_{V_n;\beta,h}^{\text{per}}} \exp(-\mathcal{H}_{V_n;\beta,h}^{\text{per}}) = \frac{1}{Z_{V_n;\beta,h}^{\text{per}}} \exp\left(\beta \sum_{\{i,j\} \in \mathcal{E}_{V_n}^{\text{per}}} \sigma_i \sigma_j + h \sum_{i \in V_n} \sigma_i\right),$$

where

$$Z_{V_n;\beta,h}^{\text{per}} = \sum_{\omega \in \Omega_{V_n}} \exp(-\mathcal{H}_{V_n;\beta,h}^{\text{per}}(\omega)), \quad \omega \in \{-1, +1\}^{V_n}.$$

- (i) When $\beta = 0$, the Hamiltonian simplifies to

$$\mathcal{H}_{V_n;0,h}^{\text{per}} = -h \sum_{i \in V_n} \sigma_i.$$

Then the Gibbs distribution becomes

$$\mu_{V_n;0,h}^{\text{per}} = \frac{1}{Z_{V_n;0,h}^{\text{per}}} \exp\left(h \sum_{i \in V_n} \sigma_i\right),$$

where

$$Z_{V_n;0,h}^{\text{per}} = \sum_{\omega \in \Omega_{V_n}} \exp\left(h \sum_{i \in V_n} \sigma_i(\omega)\right), \quad \omega \in \{-1, +1\}^{V_n}.$$

Since the spins are no longer interacting, for each σ_i , we calculate the normalisation constant as

$$Z_i^{\text{per}} = e^{h \cdot (+1)} + e^{h \cdot (-1)} = e^h + e^{-h}.$$

Then the expected value is calculated as

$$\langle \sigma_i \rangle = (+1) \cdot \mathbb{P}(\sigma_i = +1) + (-1) \cdot \mathbb{P}(\sigma_i = -1) = \frac{e^h}{e^h + e^{-h}} - \frac{e^{-h}}{e^h + e^{-h}} = \tanh h.$$

Since all spins are *i.i.d.* in this regime,

$$\lim_{N \rightarrow \infty} \left\langle \frac{1}{N^2} \sum_{i \in V_N} \sigma_i \right\rangle_{N,0,h}^{\text{per}} = \langle \sigma_i \rangle = \tanh h.$$

When $\beta = \infty$, we have to consider the ground states that minimise the Hamiltonian. The term $-\beta \sum_{\{i,j\} \in \mathcal{E}_{V_n}^{\text{per}}} \sigma_i \sigma_j$ is minimised when $\sigma_i \sigma_j = \pm 1$, and the term $-h \sum_{i \in V_n} \sigma_i$ depends on the value of h : if $h > 0$, it favours $\sigma_i = +1$; if $h < 0$, it favours $\sigma_i = -1$. Therefore, we consider different cases based on the value of h .

- $h = 0$. The term $-h \sum_{i \in V_n} \sigma_i$ vanishes, and there are two ground states: $\sigma_i = +1, \forall i$ and $\sigma_i = -1, \forall i$. And we have

$$\lim_{\beta \rightarrow \infty} \mu_{\Lambda; \beta}(\omega) = \begin{cases} \frac{1}{2}, & \omega \in \{-1, +1\} \\ 0, & \text{otherwise} \end{cases}.$$

- $h > 0$. The ground state is $\sigma_i = +1, \forall i$.
- $h < 0$. The ground state is $\sigma_i = -1, \forall i$.

In conclusion, at zero temperature we have

$$\lim_{N \rightarrow \infty} \left\langle \frac{1}{N^2} \sum_{i \in V_N} \sigma_i \right\rangle_{N,\infty,h}^{\text{per}} = \begin{cases} +1, & h > 0 \\ 0, & h = 0 \\ -1, & h < 0 \end{cases}.$$

(ii) $N = 2$, then there are $2^4 = 16$ possible spin configurations.

- $M = +4$: all spins are +1, 1 configuration.

$$\mathcal{H} = -\beta \cdot 8 \cdot (+1) - h \cdot (+4) = -8\beta - 4h.$$

- $M = +2$: 3 spins are +1, 1 spin is -1, 4 configurations.

$$\mathcal{H} = 0 - h \cdot (+2) = -2h.$$

- $M = 0$: 2 spins are +1, 2 spins are -1, 6 configurations.

$$\mathcal{H} = \frac{-\beta \cdot (-4) + 0}{2} = 2\beta.$$

- $M = -2$: 1 spin is +1, 3 spins are -1, 4 configurations.

$$\mathcal{H} = 0 - h \cdot (-2) = 2.$$

- $M = -4$: all spins are -1, 1 configuration.

$$\mathcal{H} = -\beta \cdot 8 \cdot (-1) - h \cdot (-4) = -8\beta + 4h.$$

The partition function is calculated by

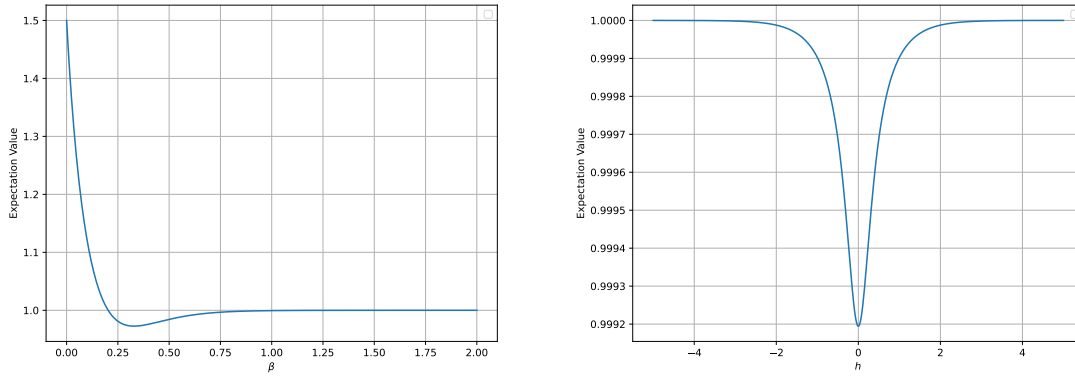
$$e^{8\beta+4h} + 4e^{2h} + 6e^{-2\beta} + 4e^{-2h} + 8e^{8\beta-4h}.$$

Therefore

$$\begin{aligned} \left\langle \frac{1}{4} \left| \sum_{i \in V_2} \sigma_i \right| \right\rangle_{2,\beta,h}^{\text{per}} &= \frac{1}{4} \langle |M| \rangle \\ &= \frac{1}{4Z} \sum |M| \cdot \#M \cdot e^{-\mathcal{H}(M)} = \frac{e^{8\beta+4h} + 2e^{2h} + 2e^{-2h} + e^{8\beta-4h}}{e^{8\beta+4h} + 4e^{2h} + 6e^{-2\beta} + 4e^{-2h} + 8e^{8\beta-4h}}. \end{aligned}$$

(iii) When $h = 0$,

$$\left\langle \frac{1}{4} \left| \sum_{i \in V_2} \sigma_i \right| \right\rangle_{2,\beta,0}^{\text{per}} = \frac{2e^{8\beta} + 4}{2e^{8\beta} + 8 - 6e^{-2\beta}}.$$



When $\beta = 1$,

$$\left\langle \frac{1}{4} \left| \sum_{i \in V_2} \sigma_i \right| \right\rangle_{2,1,h}^{\text{per}} = \frac{e^{8+4h} + 2e^{2h} + 2e^{-2h} + e^{8-4h}}{e^{8+4h} + 4e^{2h} + 6e^{-2} + 4e^{-2h} + e^{8-4h}}.$$

So we should draw the functions

$$\beta \mapsto \frac{2e^{8\beta} + 4}{2e^{8\beta} + 8 - 6e^{-2\beta}},$$

and

$$h \mapsto \frac{e^{8+4h} + 2e^{2h} + 2e^{-2h} + e^{8-4h}}{e^{8+4h} + 4e^{2h} + 6e^{-2} + 4e^{-2h} + e^{8-4h}},$$

where the images are shown above.

□

EXERCISE B.2. Imagine that N students are sitting at a circular table in an oriental restaurant. Each student can either eat (state $+1$) or wait (state -1). In order to eat one needs two chopsticks: chopsticks are shared between the two neighbouring students, one with the left and one with the right one. Thus, if a student is eating, his left and right neighbours cannot be eating at the same time.

- (i) Evaluate explicitly the number of allowed configurations $\mathcal{N}_N$ of the N students at the table, for $N \in \{2, 3, 4, 5\}$. Verify in this set that $\mathcal{N}_N = \mathcal{N}_{N-1} + \mathcal{N}_{N-2}$.
- (ii) We model the problem with a $1-d$ spin chain with N sites, carrying a spin variable $\sigma_i \in \{-1, +1\}$ for $i \in \{1, \dots, N\}$, and with periodic boundary conditions. We denote with $\mathcal{N}_N(s)$ the number of acceptable configurations such that $\sigma_N = s$, with $s \in \{-1, +1\}$. Show that:

$$\mathcal{N}_N = \mathcal{N}_N(+1) + \mathcal{N}_N(-1),$$

and that:

$$\mathcal{N}_N(-1) = \mathcal{N}_{N-1} + \mathcal{N}_{N-2}(+1).$$

Hence, prove the recursive relation:

$$\mathcal{N}_N = \mathcal{N}_{N-1} + \mathcal{N}_{N-2}.$$

Are the $\mathcal{N}_i$ the Fibonacci numbers?

- (iii) Show that $\mathcal{N}_N$ equals the cardinality of the minimal-energy states (i.e., *ground states*) of a one-dimensional Ising model defined by the Hamiltonian:

$$H(\sigma) = - \sum_{i=1}^N [J\sigma_i\sigma_{i+1} + h\sigma_i]$$

with $J = -1$; $h = -2$ and periodic boundary conditions (i.e., $\sigma_{N+1} \equiv \sigma_1$). Evaluate the ground state energy $E_0(N) := \min_{\sigma} H(\sigma)$.

- (iv) Show that:

$$\min_{\beta \rightarrow \infty} e^{\beta E_0(N)} Z_{\beta;N} = \mathcal{N}_N,$$

where $Z_{\beta;N} := \sum_{\sigma} e^{-\beta H(\sigma)}$.

- (v) Find the expression of the infinite volume pressure of the system.

SOLUTION. (i) We consider each case listed below for $N \in \{2, 3, 4, 5\}$.

- $N = 2$: $2^2 = 4$ configurations with 3 valid $((+1, -1), (-1, +1), (-1, -1))$ and 1 invalid $(+1, +1)$. Then $\mathcal{N}_2 = 3$.
- $N = 3$: $2^3 = 8$ configurations with 4 valid $((+1, -1, -1), (-1, +1, -1), (-1, -1, +1), (-1, -1, -1))$ and 4 invalid $((+1, +1, +1), (+1, +1, -1), (+1, -1, +1), (-1, +1, +1))$. Then $\mathcal{N}_3 = 4$.
- $N = 4$: $2^4 = 16$ configurations with 7 valid $((-1, -1, -1, -1), (+1, -1, -1, -1), (-1, +1, -1, -1), (-1, -1, +1, -1), (-1, -1, -1, +1), (+1, -1, +1, -1), (-1, +1, -1, +1))$ and all others are invalid. Then $\mathcal{N}_4 = 7$.
- $N = 5$: $2^5 = 32$ configurations with 11 valid $((-1, -1, -1, -1, -1), (+1, -1, -1, -1, -1), (-1, +1, -1, -1, -1), (-1, -1, +1, -1, -1), (-1, -1, -1, +1, -1), (-1, -1, -1, -1, +1), (+1, -1, +1, -1, -1), (+1, -1, -1, +1, -1), (-1, +1, -1, +1, -1), (-1, +1, -1, -1, +1), (-1, -1, +1, -1, +1))$ and all others are invalid. Then $\mathcal{N}_5 = 11$.

Since $\mathcal{N}_4 = \mathcal{N}_2 + \mathcal{N}_3$ and $\mathcal{N}_5 = \mathcal{N}_3 + \mathcal{N}_4$, it is straightforward to see that the recurrence relation $\mathcal{N}_N = \mathcal{N}_{N-1} + \mathcal{N}_{N-2}$ holds for $N > 3$.

- (ii) Since we only have two kinds of configurations, the total number of acceptable configurations should be the sum of those ending with +1 and those ending with -1, i.e.,

$$\mathcal{N}_N = \mathcal{N}_N(+1) + \mathcal{N}_N(-1). \quad (\text{B.1})$$

Suppose now configurations of length N ends with $\sigma_N = -1$, then σ_{N-1} could be either -1 or +1, and the number of configurations is $\mathcal{N}_{N-1}$. Due to the circular nature, $\sigma_N = -1$ is adjacent to σ_1 . We consider the following two cases:

- $\sigma_{N-1} = -1$. Always valid and the number of configurations is $\mathcal{N}_{N-1}$, where σ_{N-2} could be either -1 or +1;
- $\sigma_{N-1} = +1$. σ_{N-2} must be -1.

Combining both cases gives

$$\mathcal{N}_N(-1) = \mathcal{N}_{N-1} + \mathcal{N}_{N-2}(+1). \quad (\text{B.2})$$

From (B.1) and (B.2), we have

$$\mathcal{N}_{N-1} + \mathcal{N}_{N-2} = \mathcal{N}_N(-1) - \mathcal{N}_{N-2}(+1) + \mathcal{N}_{N-2} = \mathcal{N}_N(-1) + \mathcal{N}_{N-2}(-1).$$

Thus we have to prove

$$\mathcal{N}_N = \mathcal{N}_N(-1) + \mathcal{N}_{N-2}(-1) \implies \mathcal{N}_N(+1) = \mathcal{N}_{N-2}(-1).$$

Since $\sigma_N = -1$, σ_{N-1} must be $+1$, and the number of configurations is $\mathcal{N}_{N-2}(-1)$, which proves the recursive relation

$$\mathcal{N}_N = \mathcal{N}_{N-1} + \mathcal{N}_{N-2}.$$

We notice that the sequence $\mathcal{N}_N$ satisfies the Fibonacci recursion but with different initial conditions. Therefore, $\mathcal{N}_N$ are not the standard Fibonacci numbers.

(iii) Substitute $J = -1$ and $h = -2$ into the Hamiltonian gives

$$\mathcal{H}(\sigma) = - \sum_{i=1}^N [-\sigma_i \sigma_{i+1} - 2\sigma_i] = \sum_{i=1}^N [\sigma_i \sigma_{i+1} + 2\sigma_i].$$

For each i , we calculate that

$$\sigma_i \sigma_{i+1} + 2\sigma_i = \begin{cases} 3, & \sigma_i = 1, \sigma_{i+1} = 1 \\ 1, & \sigma_i = 1, \sigma_{i+1} = -1 \\ -3, & \sigma_i = -1, \sigma_{i+1} = 1 \\ -1, & \sigma_i = -1, \sigma_{i+1} = -1 \end{cases}.$$

Therefore, to minimise energy, configurations must avoid adjacent $+1$ spins, but we have to check whether “some” $+1$ spins may increase or decrease the ground state energy. Suppose that $\sigma_i = -1$ for all i , $\mathcal{H}(\sigma) = \sum_{i=1}^N (-1) = -N$. Now, if we add one $+1$ spin, say $\sigma_k = +1$, and all others are -1 , we have

$$\sigma_{k-1} \sigma_k + 2\sigma_{k-1} = -3, \quad \sigma_k \sigma_{k+1} + 2\sigma_k = 1,$$

then

$$\mathcal{H}(\sigma) = -3 + 1 + (N - 2) \cdot (-1) = -N.$$

More generally, let m be the number of non-adjacent $+1$ spins, then

$$\mathcal{H}(\sigma) = m \cdot (-2) + (N - 2m) \cdot (-1) = -N.$$

Hence the ground state energy $E_0(N) := \min_{\sigma} \mathcal{H}(\sigma) = -N$.

(iv) We rewrite the expression of Hamiltonian as

$$\mathcal{H}(\sigma) = E_0(N) + \Delta E(\sigma).$$

Then the partition function is

$$\begin{aligned} Z_{\beta;N} &= \sum_{\sigma} e^{-\beta \mathcal{H}(\sigma)} \\ &= e^{-\beta E_0(N)} \sum_{\sigma} e^{-\beta \Delta E(\sigma)} \\ &= e^{-\beta E_0(N)} \left(\sum_{\sigma \in \mathcal{G}} e^{-\beta \Delta E(\sigma)} + \sum_{\sigma \notin \mathcal{G}} e^{-\beta \Delta E(\sigma)} \right) \\ &= e^{-\beta E_0(N)} \left(\mathcal{N}_N + \sum_{\sigma \notin \mathcal{G}} e^{-\beta \Delta E(\sigma)} \right), \end{aligned}$$

where $\mathcal{G}$ denotes the ground states. Then

$$\min_{\beta \rightarrow \infty} e^{\beta E_0(N)} Z_{\beta;N} = e^{\beta E_0(N)} e^{-\beta E_0(N)} \left(\mathcal{N}_N + \underbrace{\sum_{\sigma \notin \mathcal{G}} e^{-\beta \Delta E(\sigma)}}_{\rightarrow 0 \text{ as } \beta \rightarrow \infty} \right) = \mathcal{N}_N.$$

(v) Given that

$$\mathcal{H}(\sigma) = - \sum_{i=1}^N [-\sigma_i \sigma_{i+1} - 2\sigma_i] = \sum_{i=1}^N [\sigma_i \sigma_{i+1} + 2\sigma_i].$$

Note that in [Theorem 2.8](#):

$$\psi(\beta, h) = \log \left\{ e^\beta \cosh(h) + \sqrt{e^{2\beta} \cosh^2(h) - 2 \sinh(2\beta)} \right\},$$

we have assumed that $\beta \geq 0$ and $J = 1$. Thus we define

$$\mathcal{H}'(\sigma') = - \sum_{i=1}^N [\sigma'_i \sigma'_{i+1} + 2\sigma'_i],$$

where now $J = 1$ and $h = 2$. Therefore the pressure

$$\psi(\beta, 2) = \log \left\{ e^\beta \cosh(2) + \sqrt{e^{2\beta} \cosh^2(2) - 2 \sinh(2\beta)} \right\}.$$

□

EXERCISE B.3. If we denote with $\mathcal{H}_\Lambda^{\text{Ising}}(\omega)$, the finite volume Hamiltonian of the Ising model with null magnetic field in $\Lambda \subset \mathbb{Z}^2$, let us consider a new model, whose Hamiltonian is defined by:

$$\mathcal{H}_\Lambda^{\text{anti}}(\omega) := -\mathcal{H}_\Lambda^{\text{Ising}}(\omega).$$

- (i) Show that there exists a finite $\beta_c > 0$, such that for $\beta > \beta_c$, there exist two distinct infinite volume Gibbs states. How are these states related to the Gibbs states of the Ising model?
- (ii) Describe the typical configurations under these states.

SOLUTION. (i) The Hamiltonian of the Ising model is given by

$$\mathcal{H}_\Lambda^{\text{Ising}}(\omega) = - \sum_{\{i,j\} \in \mathcal{E}_\Lambda^b} \omega_i \omega_j - h \sum_{i \in \Lambda} \omega_i, \quad \omega \in \Omega_\Lambda.$$

Since we are considering null magnetic field case ($h = 0$), we have

$$\mathcal{H}_\Lambda^{\text{anti}}(\omega) := -\mathcal{H}_\Lambda^{\text{Ising}}(\omega) = \sum_{\{i,j\} \in \mathcal{E}_\Lambda^b} \omega_i \omega_j, \quad \omega \in \Omega_\Lambda.$$

In $\mathbb{Z}^2$, the lattice can be partitioned into two disjoint sublattices:

$$A = \{i = (i_1, i_2) \in \mathbb{Z}^2 : i_1 + i_2 \text{ is even}\}, \quad B = \{i = (i_1, i_2) \in \mathbb{Z}^2 : i_1 + i_2 \text{ is odd}\}.$$

We define a spin transformation $\mathcal{T} : \Omega_\Lambda \rightarrow \Omega_\Lambda$ such that

$$\omega'_i = \begin{cases} \omega_i, & i \in A \\ -\omega_i, & i \in B \end{cases}.$$

Then, without loss of generality, we consider an edge $\{i, j\} \in \mathcal{E}_\Lambda^b$ where $i \in A$ and $j \in B$, we have

$$\mathcal{H}_\Lambda^{\text{anti}}(\omega) = \sum_{\{i,j\} \in \mathcal{E}_\Lambda^b} \omega_i \omega_j = - \sum_{\{i,j\} \in \mathcal{E}_\Lambda^b} \omega'_i \omega'_j = -\mathcal{H}_\Lambda^{\text{Ising}}(\omega').$$

We write the canonical Gibbs distributions:

$$\mu_{\Lambda, \beta}^{\text{anti}}(\omega) = \frac{1}{Z_{\Lambda, \beta}^{\text{anti}}} \exp(-\beta \mathcal{H}_\Lambda^{\text{anti}}(\omega)),$$

$$\mu_{\Lambda;\beta}^{\text{Ising}}(\omega') = \frac{1}{Z_{\Lambda;\beta}^{\text{Ising}}} \exp\left(-\beta \mathcal{H}_{\Lambda}^{\text{Ising}}(\omega')\right),$$

$$\mu_{\Lambda;-\beta}^{\text{Ising}}(\omega') = \frac{1}{Z_{\Lambda;-\beta}^{\text{Ising}}} \exp\left(\beta \mathcal{H}_{\Lambda}^{\text{Ising}}(\omega')\right).$$

Notice that the Hamiltonian $\mathcal{H}_{\Lambda}^{\text{Ising}}(\omega')$ is symmetric under global spin flip, i.e., $\mathcal{H}_{\Lambda}^{\text{Ising}}(-\omega') = \mathcal{H}_{\Lambda}^{\text{Ising}}(\omega')$, the partition function satisfies

$$Z_{\Lambda;\beta}^{\text{Ising}} = \sum_{\omega' \in \Omega} \exp\left(-\beta \mathcal{H}_{\Lambda}^{\text{Ising}}(\omega')\right) = \sum_{\omega' \in \Omega} \exp\left(\beta \mathcal{H}_{\Lambda}^{\text{Ising}}(\omega')\right) = Z_{\Lambda;-\beta}^{\text{Ising}}.$$

And by definition

$$Z_{\Lambda;\beta}^{\text{anti}} = \sum_{\omega \in \Omega} \exp\left(-\beta \mathcal{H}_{\Lambda}^{\text{anti}}(\omega)\right) = \sum_{\omega \in \Omega} \exp\left(\beta \mathcal{H}_{\Lambda}^{\text{Ising}}(\omega)\right) = Z_{\Lambda;-\beta}^{\text{Ising}} = Z_{\Lambda;\beta}^{\text{Ising}}.$$

Then the expectation is calculated as

$$\begin{aligned} \langle f(\omega) \rangle_{\mu_{\Lambda;\beta}^{\text{anti}}} &= \sum_{\omega \in \Omega} f(\omega) \mu_{\Lambda;\beta}^{\text{anti}} \\ &= \sum_{\omega \in \Omega} \frac{1}{Z_{\Lambda;\beta}^{\text{anti}}} \exp\left(\beta \mathcal{H}_{\Lambda}^{\text{Ising}}(\omega')\right) \\ &= \sum_{\omega' \in \Omega} f(\mathcal{T}^{-1}(\omega')) \frac{Z_{\Lambda;-\beta}^{\text{Ising}}}{Z_{\Lambda;\beta}^{\text{anti}}} \mu_{\Lambda;-\beta}^{\text{Ising}}(\omega') \\ &= \sum_{\omega' \in \Omega} f(\mathcal{T}^{-1}(\omega')) \mu_{\Lambda;\beta}^{\text{Ising}}(\omega') = \langle f(\mathcal{T}^{-1}(\omega')) \rangle_{\mu_{\Lambda;\beta}^{\text{Ising}}}. \end{aligned}$$

Thus under the spin transformation $\mathcal{T}$, we have the Gibbs states:

$$\langle f(\omega) \rangle_{G_1} = \langle f(\mathcal{T}^{-1}(\omega')) \rangle_{\mu_{\beta}^{+}}, \quad \langle f(\omega) \rangle_{G_2} = \langle f(\mathcal{T}^{-1}(\omega')) \rangle_{\mu_{\beta}^{-}}.$$

By [Theorem 2.22](#), when $\beta > \beta_c$, the Ising model has two distinct infinite-volume Gibbs states, i.e., $\mu_{\beta}^{+} \neq \mu_{\beta}^{-}$, it follows that $G_1 \neq G_2$.

In conclusion, the anti model has at least two distinct infinite-volume Gibbs states, and they are related to the Ising model Gibbs states via the spin transformation $\mathcal{T}$.

- (ii) For the Ising model, we have two distinct infinite-volume Gibbs states: μ_{β}^{+} and μ_{β}^{-} , and the spins exhibit long-range order. Using the spin transformation $\mathcal{T}$, we map them to the anti model:

- G_1 (corresponding to μ_{β}^{+}): spins on sublattice A remain +1, and spins on sublattice B become -1;
- G_2 (corresponding to μ_{β}^{-}): spins on sublattice A become -1, and spins on sublattice B remain +1.

Therefore, the anti system exhibits long-range antiferromagnetic order and spontaneous symmetry breaking at low temperatures. □

EXERCISE B.4. Let us consider a finite region $\Lambda \subset \mathbb{Z}^2$ and a finite volume measure with periodic boundary conditions:

$$\nu_{\beta;\Lambda}^{\text{Per}}(\sigma) := \frac{1}{Z_{\Lambda}^{\text{Per}}} e^{-\mathcal{H}_{\beta,h;\Lambda}^{\text{Per}}(\sigma)},$$

where

$$\mathcal{H}_{\beta,h;\Lambda}^{\text{Per}}(\sigma) := -h\beta \sum_{i \in \Lambda} \sigma_i - \sum_{i \in \Lambda} \ln \left[\cosh \left(\beta \sum_{j \in \Lambda: d(i,j)=1} \sigma_j + \beta h \right) \right],$$

with $\sigma \in \{-1, +1\}^\Lambda$, $h \in \mathbb{R}$, $\beta \geq 0$ and Z_Λ^{Per} a normalization.

(i) Calculate the limit:

$$H_{\Lambda;h}^{\text{Per}}(\sigma) := \lim_{\beta \rightarrow \infty} \frac{1}{\beta} \mathcal{H}_{\beta,h;\Lambda}^{\text{Per}}(\sigma).$$

(ii) Find the configurations that minimize $H_{\Lambda;0}^{\text{Per}}(\sigma)$.

(iii) Find the configurations that minimize $H_{\Lambda;h}^{\text{Per}}(\sigma)$.

(iv) For any $h \geq 0$, calculate the limiting measure $\nu_{\infty;\Lambda}^{\text{Per}}(\sigma)$, defined as:

$$\nu_{\infty;\Lambda}^{\text{Per}}(\sigma) = \lim_{\beta \rightarrow \infty} \nu_{\beta;\Lambda}^{\text{Per}}(\sigma).$$

SOLUTION. (i) Given that the Hamiltonian

$$\mathcal{H}_{\beta,h;\Lambda}^{\text{Per}}(\sigma) := -h\beta \sum_{i \in \Lambda} \sigma_i - \sum_{i \in \Lambda} \ln \left[\cosh \left(\beta \sum_{j \in \Lambda: d(i,j)=1} \sigma_j + \beta h \right) \right],$$

we have

$$\frac{1}{\beta} \mathcal{H}_{\beta,h;\Lambda}^{\text{Per}}(\sigma) = -h \sum_{i \in \Lambda} \sigma_i - \frac{1}{\beta} \sum_{i \in \Lambda} \ln \left[\cosh \left(\beta \sum_{j \in \Lambda: d(i,j)=1} \sigma_j + \beta h \right) \right].$$

Since $-h \sum_{i \in \Lambda} \sigma_i$ is independent of β , to calculate $H_{\Lambda;h}^{\text{Per}}(\sigma)$, we focus on the $\frac{1}{\beta} \sum_{i \in \Lambda} \ln \cosh \left(\beta \sum_{j \in \Lambda: d(i,j)=1} \sigma_j + \beta h \right)$ term. Define

$$S_i := \sum_{j \in \Lambda: d(i,j)=1} \sigma_j, \quad X_i := S_i + h,$$

the term simplifies to

$$\frac{1}{\beta} \sum_{i \in \Lambda} \ln \cosh(\beta X_i) = \sum_{i \in \Lambda} \frac{1}{\beta} \ln \cosh(\beta X_i).$$

And the behaviour of $\ln \cosh(\beta X_i)$ is as follows:

- $X_i = 0$: $\ln \cosh(\beta X_i) = 0$;
- $X_i > 0$:

$$\cosh(\beta X_i) = \frac{e^{\beta X_i} + e^{-\beta X_i}}{2} \stackrel{\beta \rightarrow \infty}{\approx} \frac{e^{\beta X_i}}{2} \implies \ln \cosh(\beta X_i) \approx \beta X_i - \ln 2;$$

- $X_i < 0$:

$$\cosh(\beta X_i) \stackrel{\beta \rightarrow \infty}{\approx} \frac{e^{-\beta |X_i|}}{2} \implies \ln \cosh(\beta X_i) \approx \beta |X_i| - \ln 2.$$

Then the limit is

$$\frac{1}{\beta} \ln \cosh(\beta X_i) = \begin{cases} 0, & X_i = 0 \\ |X_i|, & X_i \neq 0 \end{cases} = |X_i|.$$

Therefore, plug everything back, we have

$$\begin{aligned} H_{\Lambda;h}^{\text{Per}}(\sigma) &= -h \sum_{i \in \Lambda} \sigma_i - \frac{1}{\beta} \sum_{i \in \Lambda} \ln \cosh(\beta X_i) \\ &= -h \sum_{i \in \Lambda} \sigma_i - \sum_{i \in \Lambda} |X_i| \end{aligned}$$

$$= -h \sum_{i \in \Lambda} \sigma_i - \sum_{i \in \Lambda} |S_i + h| = -h \sum_{i \in \Lambda} \sigma_i - \left| \sum_{j \in \Lambda: d(i,j)=1} \sigma_j + h \right|.$$

(ii) When $h = 0$, from (i), the Hamiltonian simplifies to

$$H_{\Lambda,0}^{\text{Per}}(\sigma) = - \left| \sum_{j \in \Lambda: d(i,j)=1} \sigma_j \right| = -|S_i|.$$

Since $\lambda \subset \mathbb{Z}^2$, $S_i \in \{4, 2, 0, -2, -4\}$. To maximise $|S_i|$, we need $S_i = \pm 4$ and hence the configurations that minimise $H_{\Lambda,0}^{\text{Per}}(\sigma)$ are those where all spins are +1 or all spins are -1.

(iii) From (i), the Hamiltonian is

$$H_{\Lambda,h}^{\text{Per}}(\sigma) = -h \sum_{i \in \Lambda} \sigma_i - \sum_{i \in \Lambda} |S_i + h|.$$

Then we consider the following cases:

- $h > 0$: the first term is minimised when $\sigma_i = +1$, and the second term is minimised when $S_i = 4$, i.e., $\sigma_i = +1$. Then the configuration where all spins are +1 minimises $H_{\Lambda,h}^{\text{Per}}(\sigma)$;
- $h < 0$: the first term is minimised when $\sigma_i = -1$, and the second term is minimised when $S_i = -4$, i.e., $\sigma_i = -1$. Then the configuration where all spins are -1 minimises $H_{\Lambda,h}^{\text{Per}}(\sigma)$;
- $h = 0$: same as (ii), the configurations that minimise $H_{\Lambda,0}^{\text{Per}}(\sigma)$ are those where all spins are +1 or all spins are -1.

(iv) From (i), we have $\mathcal{H}_{\beta,h;\Lambda}^{\text{Per}}(\sigma) \approx \beta H_{\Lambda,h}^{\text{Per}}(\sigma)$ as $\beta \rightarrow \infty$. The partition function is

$$Z_{\Lambda}^{\text{Per}} = \sum_{\sigma \in \{-1,+1\}^{\Lambda}} e^{-\mathcal{H}_{\beta,h;\Lambda}^{\text{Per}}(\sigma)} \stackrel{\beta \rightarrow \infty}{\approx} \sum_{\sigma \in \{-1,+1\}^{\Lambda}} e^{-\beta H_{\Lambda,h}^{\text{Per}}(\sigma)}.$$

As $\beta \rightarrow \infty$, the exponential becomes extremely small unless $H_{\Lambda,h}^{\text{Per}}(\sigma)$ attains its minimal value, and we denote such a configuration as σ^* . Then the measure becomes

$$\nu_{\beta;\Lambda}^{\text{Per}}(\sigma) := \frac{1}{Z_{\Lambda}^{\text{Per}}} e^{-\mathcal{H}_{\beta,h;\Lambda}^{\text{Per}}(\sigma)} \approx \frac{e^{-\beta H_{\Lambda,h}^{\text{Per}}(\sigma)}}{e^{-\beta H_{\Lambda,h}^{\text{Per}}(\sigma^*)}} = \exp\left(-\beta \left(H_{\Lambda,h}^{\text{Per}}(\sigma) - H_{\Lambda,h}^{\text{Per}}(\sigma^*)\right)\right).$$

- When $\sigma = \sigma^*$: $\nu_{\beta;\Lambda}^{\text{Per}}(\sigma^*) = e^{-\beta(0)} = 1$.
- When $\sigma \neq \sigma^*$: $\nu_{\beta;\Lambda}^{\text{Per}}(\sigma^*) = e^{-\beta \Delta H} \rightarrow 0$ as $\beta \rightarrow \infty$.

Recall that from (ii), when $h = 0$, the configurations that minimise $H_{\Lambda,0}^{\text{Per}}(\sigma)$ are those where all spins are +1 or all spins are -1; from (iii), when $h > 0$, the configuration where all spins are +1 minimises $H_{\Lambda,h}^{\text{Per}}(\sigma)$. Therefore

$$\nu_{\infty;\Lambda}^{\text{Per}}(\sigma) = \lim_{\beta \rightarrow \infty} \nu_{\beta;\Lambda}^{\text{Per}}(\sigma) = \begin{cases} \delta_{\sigma=+1}, & h > 0 \\ \frac{1}{2}(\delta_{\sigma=+1} + \delta_{\sigma=-1}), & h = 0 \end{cases},$$

where δ denotes the Dirac delta measure. □

EXERCISE B.5. Let $\eta, \omega \in \Omega$ be such that $\eta \leq \omega$. Let f be a nondecreasing function. Show that for any $\beta > 0, h \in \mathbb{R}$,

$$\langle f \rangle_{\Lambda, \beta, h}^\eta \leq \langle f \rangle_{\Lambda, \beta, h}^\omega,$$

where $\Lambda \subset \mathbb{Z}^d$.

SOLUTION. The Gibbs distribution of the Ising mode with boundary condition η is given by

$$\mu_{\Lambda, \beta, h}^\eta(\sigma) = \frac{1}{\mathbf{Z}_{\Lambda, \beta, h}^\eta} \exp\left(-\beta \mathcal{H}_{\Lambda, h}^\eta(\sigma)\right), \quad \sigma \in \Omega_\Lambda^\eta,$$

and the Hamiltonian is

$$\mathcal{H}_{\Lambda, h}^\eta(\sigma) = - \sum_{\substack{i, j \in \Lambda \\ i \sim j}} \sigma_i \sigma_j - \sum_{\substack{i \in \Lambda, j \notin \Lambda \\ i \sim j}} \sigma_i \eta_j - h \sum_{i \in \Lambda} \sigma_i.$$

Similarly, the Gibbs distribution with boundary condition ω is

$$\mu_{\Lambda, \beta, h}^\omega(\sigma) = \frac{1}{\mathbf{Z}_{\Lambda, \beta, h}^\omega} \exp\left(-\beta \mathcal{H}_{\Lambda, h}^\omega(\sigma)\right),$$

with Hamiltonian

$$\mathcal{H}_{\Lambda, h}^\omega(\sigma) = - \sum_{\substack{i, j \in \Lambda \\ i \sim j}} \sigma_i \sigma_j - \sum_{\substack{i \in \Lambda, j \notin \Lambda \\ i \sim j}} \sigma_i \omega_j - h \sum_{i \in \Lambda} \sigma_i.$$

Then the difference between the Hamiltonians is

$$\Delta \mathcal{H}(\sigma) := \mathcal{H}_{\Lambda, h}^\eta(\sigma) - \mathcal{H}_{\Lambda, h}^\omega(\sigma) = - \sum_{\substack{i \in \Lambda, j \notin \Lambda \\ i \sim j}} \sigma_i (\eta_j - \omega_j).$$

Since $\eta \leq \omega$ and $\eta, \omega \in \Omega$, we have $\eta_j - \omega_j \leq 0$ for all $j \notin \Lambda$. Define the non-negative function $I(\sigma)$ such that

$$\begin{aligned} I(\sigma) &= \exp(\Delta \mathcal{H}(\sigma)) \\ &= \exp\left(- \sum_{\substack{i \in \Lambda, j \notin \Lambda \\ i \sim j}} \sigma_i (\eta_j - \omega_j)\right) \\ &= \exp\left(\sum_{i \in \Lambda} \sigma_i \varphi_i\right) \quad (\varphi_i := - \sum_{\substack{j \notin \Lambda \\ i \sim j}} (\eta_j - \omega_j)) \\ &= \prod_{i \in \Lambda} \exp(\sigma_i \varphi_i). \end{aligned}$$

Observe that

$$\mathbf{Z}_{\Lambda, \beta, h}^\omega = \sum_{\sigma \in \Omega_\Lambda^\omega} e^{-\beta \mathcal{H}_{\Lambda, h}^\omega(\sigma)} = \sum_{\sigma \in \Omega_\Lambda^\eta} e^{-\beta \mathcal{H}_{\Lambda, h}^\eta(\sigma)} I(\sigma) = \mathbf{Z}_{\Lambda, \beta, h}^\eta \langle I \rangle_{\Lambda, \beta, h}^\eta,$$

and for any non-decreasing $f(\sigma)$, the expectation is

$$\begin{aligned} \langle f \rangle_{\Lambda, \beta, h}^\omega &= \frac{1}{\mathbf{Z}_{\Lambda, \beta, h}^\omega} \sum_{\sigma \in \Omega_\Lambda^\omega} e^{-\beta \mathcal{H}_{\Lambda, h}^\omega(\sigma)} f(\sigma) \\ &= \frac{1}{\mathbf{Z}_{\Lambda, \beta, h}^\omega} \sum_{\sigma \in \Omega_\Lambda^\eta} e^{-\beta \mathcal{H}_{\Lambda, h}^\eta(\sigma)} I(\sigma) f(\sigma) \\ &= \frac{\mathbf{Z}_{\Lambda, \beta, h}^\eta}{\mathbf{Z}_{\Lambda, \beta, h}^\omega} \langle I f \rangle_{\Lambda, \beta, h}^\eta \end{aligned}$$

$$\begin{aligned}
&= \frac{\langle If \rangle_{\Lambda; \beta, h}^\eta}{\langle I \rangle_{\Lambda; \beta, h}^\eta} \\
&\geq \frac{\langle I \rangle_{\Lambda; \beta, h}^\eta \langle f \rangle_{\Lambda; \beta, h}^\eta}{\langle I \rangle_{\Lambda; \beta, h}^\eta} \quad (\text{FKG inequality}) \\
&= \langle f \rangle_{\Lambda; \beta, h}^\eta.
\end{aligned}$$

Therefore, we have shown that

$$\langle f \rangle_{\Lambda; \beta, h}^\eta \leq \langle f \rangle_{\Lambda; \beta, h}^\omega, \quad \Lambda \subset \mathbb{Z}^d.$$

□

EXERCISE B.6. Let $\zeta : \mathbb{N} \rightarrow (0, \infty)$. Consider the following modification of the Curie-Weiss Hamiltonian with $h = 0$:

$$H_N(\omega) = -\frac{1}{\zeta(N)} \sum_{i,j=1}^N \omega_i \omega_j.$$

Denote by m_N the average magnetization. Show the following statements:

- (i) If $\lim_{N \rightarrow \infty} \frac{\zeta(N)}{N} = \infty$, then m_N tends to 0 in probability for all $\beta > 0$.
- (ii) If $\lim_{N \rightarrow \infty} \frac{\zeta(N)}{N} = 0$, then $|m_N|$ tends to 1 in probability for all $\beta > 0$.

SOLUTION. Since

$$\sum_{i,j=1}^N \omega_i \omega_j = \left(\sum_{i=1}^N \omega_i \right)^2 = N^2 m_N^2,$$

the Hamiltonian becomes

$$H_N(\omega) = -\frac{N^2 m_N^2}{\zeta(N)}.$$

The Gibbs measure is given by

$$\mu_N(\omega) = \frac{1}{Z_N} e^{-H_N(\omega)} = \frac{1}{Z_N} \exp\left(\frac{N^2 m_N^2}{\zeta(N)}\right), \quad (\text{B.3})$$

and Z_N is the partition function such that

$$Z_N = \sum_{\omega \in \Omega_N} \exp\left(\frac{N^2 m_N^2}{\zeta(N)}\right) = \sum_{m \in J \cap \mathcal{A}_N} \binom{N}{\frac{1+m}{2}N} \exp\left(\frac{N^2 m_N^2}{\zeta(N)}\right),$$

where

$$\mathcal{A}_N := \left\{ -1 + \frac{2k}{N} : k = 0, \dots, N \right\} \subset J \subset [-1, 1].$$

By *Stirling's formula*, for large N , the binomial coefficient can be approximated as

$$\binom{N}{\frac{1+m}{2}N} \approx N s(m),$$

where

$$s(m) = -\frac{1-m}{2} \log \frac{1-m}{2} - \frac{1+m}{2} \log \frac{1+m}{2}.$$

Since

$$\frac{ds(m)}{dm} = \frac{1}{2} \log \frac{1-m}{1+m},$$

the function is maximised when $m = 0$. Combining these results with (B.3), we have

$$\mu_N(m_N = m) \approx \frac{1}{Z_N} \exp\left(Ns(m) + \frac{N^2 m_N^2}{\zeta(N)}\right).$$

Let

$$\Phi_N(m) := Ns(m) + \frac{N^2 m_N^2}{\zeta(N)},$$

we aim to analyse the behaviour of $\Phi_N(m)$ as $N \rightarrow \infty$.

- (i) If $\lim_{N \rightarrow \infty} \frac{\zeta(N)}{N} = \infty$, then $\frac{N}{\zeta(N)} \rightarrow 0$, which implies $\frac{N^2 m_N^2}{\zeta(N)} \rightarrow 0$ as $N \rightarrow \infty$. Then $\Phi_N(m) \rightarrow Ns(m)$ as $N \rightarrow \infty$. Recall that $s(m)$ is maximised when $m = 0$. For any $\epsilon > 0$,

$$\mu_N(|m_N| \geq \epsilon) \approx \exp(N(s(\epsilon) - s(0))) \rightarrow 0$$

as $N \rightarrow \infty$. Therefore, $m_N \rightarrow 0$ in probability.

- (ii) When $\lim_{N \rightarrow \infty} \frac{\zeta(N)}{N} = 0$, similarly, $\frac{N}{\zeta(N)} \rightarrow \infty$ and the term $\frac{N^2 m_N^2}{\zeta(N)}$ dominates $\Phi_N(m)$, which is maximised when $|m| = 1$. Then for any $\delta \in (0, 1)$,

$$\mu_N(|m_N| \leq 1 - \delta) \approx \exp\left(-\frac{N^2(1 - \delta)^2}{\zeta(N)}\right) \rightarrow 0$$

as $N \rightarrow \infty$. Therefore, $|m_N| \rightarrow 1$ in probability.

□

EXERCISE B.7. Let the Curie-Weiss Hamiltonian be given by

$$H_{N,0}(\omega) = -\frac{d}{N} \sum_{i,j=1}^N \omega_i \omega_j$$

for $d > 0, N \in \mathbb{N}$ and $\omega = (\omega_1, \dots, \omega_N) \in \{-1, +1\}^N$. Denote by $\mu_{N,\beta}(\omega)$ the corresponding Gibbs measure and by $\langle \cdot \rangle_{N,\beta}$ the expectation.

- (i) Show that for any $k \in \{1, \dots, N\}$ we can write

$$\langle \omega_1 \cdots \omega_k \rangle_{N,\beta} = \frac{W_N(k)}{W_N(0)}$$

where $W_N(k) = \int_{-\infty}^{\infty} e^{-N\left(\frac{C}{\beta}x^2 - \ln(\cosh(x))\right)} (\tanh(x))^k dx$, determine C .

- (ii) Show that $\langle \omega_1 \cdots \omega_k \rangle_{N,\beta} = 0$ for $k \in 2\mathbb{N} + 1$.
 (iii) Demonstrate that

- (i) For $\beta < \frac{1}{2d}$ we have,

$$\lim_{N \rightarrow \infty} N^{k/2} \langle \omega_1 \cdots \omega_k \rangle_{N,\beta} = C_1(k-1)!!(C(\beta))^{k/2},$$

for some constant $C_1 > 0$ (independent of β). Determine $C(\beta)$.

- (ii) For $\beta = \frac{1}{2d}$ we have,

$$\lim_{N \rightarrow \infty} N^{k/4} \langle \omega_1 \cdots \omega_k \rangle_{N,\beta} = C_2 \frac{\int_{-\infty}^{\infty} x^k e^{-\frac{1}{12}x^4} dx}{\int_{-\infty}^{\infty} e^{-\frac{1}{12}x^4} dx},$$

for some constant $C_2 > 0$.

SOLUTION. (i) Given that

$$H_{N,0}(\omega) = -\frac{d}{N} \sum_{i,j=1}^N \omega_i \omega_j,$$

the Gibbs measure is

$$\mu_N = \frac{1}{Z_N} \exp\left(\beta \frac{d}{N} \sum_{i,j=1}^N \omega_i \omega_j\right) = \frac{1}{Z_N} \exp\left(\beta \frac{d}{N} \left(\sum_{i=1}^N \omega_i\right)^2\right).$$

By the *Hubbard–Stratonovich transformation*, the Boltzmann weight is

$$\exp\left(\beta \frac{d}{N} \left(\sum_{i=1}^N \omega_i\right)^2\right) = \sqrt{\frac{N}{\pi\beta d}} \int_{-\infty}^{\infty} \exp\left(-\frac{Ny^2}{\beta d} + 2y \sum_{i=1}^N \omega_i\right) dy,$$

then

$$\begin{aligned} Z_N &= \sum_{\omega \in \{-1,+1\}^N} \sqrt{\frac{N}{\pi\beta d}} \int_{-\infty}^{\infty} \exp\left(-\frac{Ny^2}{\beta d} + 2y \sum_{i=1}^N \omega_i\right) dy \\ &= \sqrt{\frac{N}{\pi\beta d}} \int_{-\infty}^{\infty} \exp\left(-\frac{Ny^2}{\beta d}\right) \left(\sum_{\omega_i=\pm 1} e^{2y\omega_i}\right)^N dy \\ &= \sqrt{\frac{N}{\pi\beta d}} \int_{-\infty}^{\infty} \exp\left(-\frac{Ny^2}{\beta d}\right) (2 \cosh(2y))^N dy. \end{aligned}$$

We can write

$$\begin{aligned} \langle \omega_1 \cdots \omega_k \rangle_{N,\beta} &= \frac{1}{Z_N} \sum_{\omega \in \{-1,+1\}^N} \omega_1 \cdots \omega_k \exp\left(\beta \frac{d}{N} \left(\sum_{i=1}^N \omega_i\right)^2\right) \\ &= \frac{1}{Z_N} \sqrt{\frac{N}{\pi\beta d}} \int_{-\infty}^{\infty} \sum_{\omega \in \{-1,+1\}^N} \omega_1 \cdots \omega_k \exp\left(-\frac{Ny^2}{\beta d} + 2y \sum_{i=1}^N \omega_i\right) dy \\ &= \frac{1}{Z_N} \sqrt{\frac{N}{\pi\beta d}} \int_{-\infty}^{\infty} e^{-\frac{Ny^2}{\beta d}} \sum_{\omega \in \{-1,+1\}^N} \omega_1 \cdots \omega_k e^{2y \sum_{i=1}^N \omega_i} dy \\ &= \frac{1}{Z_N} \sqrt{\frac{N}{\pi\beta d}} \int_{-\infty}^{\infty} e^{-\frac{Ny^2}{\beta d}} \left(\prod_{i=1}^k \sum_{\omega_i=\pm 1} \omega_i e^{2y\omega_i}\right) \left(\prod_{i=k+1}^N \sum_{\omega_i=\pm 1} \omega_i e^{2y\omega_i}\right) dy \\ &= \frac{1}{Z_N} \sqrt{\frac{N}{\pi\beta d}} \int_{-\infty}^{\infty} e^{-\frac{Ny^2}{\beta d}} (2 \sinh(2y))^k (2 \cosh(2y))^{N-k} dy \\ &= \frac{1}{Z_N} \sqrt{\frac{N}{\pi\beta d}} \int_{-\infty}^{\infty} e^{-\frac{Ny^2}{\beta d}} (2 \cosh(2y))^N (\tanh(2y))^k dy \\ &= \frac{\sqrt{\frac{N}{\pi\beta d}} \int_{-\infty}^{\infty} e^{-\frac{Ny^2}{\beta d}} (2 \cosh(2y))^N (\tanh(2y))^k dy}{\sqrt{\frac{N}{\pi\beta d}} \int_{-\infty}^{\infty} \exp\left(-\frac{Ny^2}{\beta d}\right) (2 \cosh(2y))^N dy} \\ &= \frac{\int_{-\infty}^{\infty} e^{-\frac{Ny^2}{\beta d}} (\tanh(2y))^k dy}{\int_{-\infty}^{\infty} e^{-\frac{Ny^2}{\beta d}} dy} \\ &= \frac{\int_{-\infty}^{\infty} e^{-\frac{Nx^2}{4\beta d}} (\tanh x)^k dx}{\int_{-\infty}^{\infty} e^{-\frac{Nx^2}{4\beta d}} dx} \quad (x := 2y) \end{aligned}$$

$$\begin{aligned}
&= \frac{\int_{-\infty}^{\infty} e^{-\frac{Nx^2}{4\beta d}} (\cosh x)^N (\tanh x)^k dx}{\int_{-\infty}^{\infty} e^{-\frac{Nx^2}{4\beta d}} (\cosh x)^N dx} \\
&= \frac{\int_{-\infty}^{\infty} e^{-\frac{Nx^2}{4\beta d}} e^{N \ln \cosh x} (\tanh x)^k dx}{\int_{-\infty}^{\infty} e^{-\frac{Nx^2}{4\beta d}} e^{N \ln \cosh x} dx} \\
&= \frac{\int_{-\infty}^{\infty} e^{-N\left(\frac{x^2}{4\beta d} - \ln \cosh x\right)} (\tanh x)^k dx}{\int_{-\infty}^{\infty} e^{-N\left(\frac{x^2}{4\beta d} - \ln \cosh x\right)} dx} = \frac{W_N(k)}{W_N(0)},
\end{aligned}$$

where

$$W_N(k) = \int_{-\infty}^{\infty} e^{-N\left(\frac{C}{\beta}x^2 - \ln \cosh x\right)} (\tanh x)^k dx,$$

and $C = \frac{1}{4d}$.

(ii) By (i),

$$\langle \omega_1 \cdots \omega_k \rangle_{N,\beta} = \frac{\int_{-\infty}^{\infty} e^{-N\left(\frac{x^2}{4\beta d} - \ln \cosh x\right)} (\tanh x)^k dx}{\int_{-\infty}^{\infty} e^{-N\left(\frac{x^2}{4\beta d} - \ln \cosh x\right)} dx},$$

observe that the integrand in the numerator is an odd function: x^2 and $\cosh x$ are even functions, then the exponential part is an even function, and $(\tanh x)^k$ is an odd function when k is odd. Therefore, the numerator integral is zero, and hence $\langle \omega_1 \cdots \omega_k \rangle_{N,\beta} = 0$ for $k \in 2\mathbb{N} + 1$.

(iii) Given that

$$\langle \omega_1 \cdots \omega_k \rangle_{N,\beta} = \frac{W_N(k)}{W_N(0)}$$

where $W_N(k) = \int_{-\infty}^{\infty} e^{-N\left(\frac{C}{\beta}x^2 - \ln \cosh x\right)} (\tanh x)^k dx$. Let

$$\Phi_\beta(x) := \frac{C}{\beta}x^2 - \ln \cosh x,$$

where $C = \frac{1}{4d}$, then $W_N(k) = \int_{-\infty}^{\infty} e^{-N\Phi_\beta(x)} (\tanh x)^k dx$. We compute

$$\begin{aligned}
\Phi'_\beta(x) &= \frac{2C}{\beta}x - \tanh x, & \Phi''_\beta(x) &= \frac{2C}{\beta} - (\operatorname{sech} x)^2, \\
\Phi^{(3)}_\beta(x) &= 2 \tanh x (\operatorname{sech} x)^2, & \Phi^{(4)}_\beta(x) &= 2(\operatorname{sech} x)^4 - 4(\tanh x)^2 (\operatorname{sech} x)^2.
\end{aligned}$$

(i) When $\beta < \frac{1}{2d}$, we have $\Phi'_\beta(0) = 0$ and $\Phi''_\beta(0) = \frac{1}{2d\beta} - 1 > 0$, then $x_* = 0$ is a local minimum. Since $\Phi_\beta(x)$ is convex near $x_* = 0$ and symmetric, $x_* = 0$ is the unique global minimum. By the Taylor expansion,

$$\Phi_\beta(x) \approx \Phi_\beta(0) + \frac{1}{2}\Phi''_\beta(0)x^2 + O(x^2), \quad (\tanh x)^2 \approx x^2 + O(x^4).$$

Then we approximate

$$W_N(k) \approx \int_{-\infty}^{\infty} e^{-N\left(\Phi_\beta(0) + \frac{1}{2}\Phi''_\beta(0)x^2\right)} x^k dx, \quad W_N(0) \approx \int_{-\infty}^{\infty} e^{-N\left(\Phi_\beta(0) + \frac{1}{2}\Phi''_\beta(0)x^2\right)} dx.$$

where the common factor $e^{-N\Phi_\beta(0)}$ cancels out when taking the ratio, so we focus on $\int_{-\infty}^{\infty} e^{-N\left(\frac{1}{2}\Phi''_\beta(0)x^2\right)} x^k dx$ term. And we have

$$\int_{-\infty}^{\infty} e^{-ax^2} x^k dx = \frac{(k-1)!!}{(2a)^{\frac{k}{2}}} \sqrt{\frac{\pi}{a}}$$

for even k . Then

$$W_N(k) \approx e^{-N\Phi_\beta(0)} \frac{(k-1)!!}{(N\Phi_\beta''(0))^{\frac{k}{2}}} \sqrt{\frac{2\pi}{N\Phi_\beta''(0)}}, \quad W_N(0) \approx e^{-N\Phi_\beta(0)} \sqrt{\frac{2\pi}{N\Phi_\beta''(0)}}.$$

The expectation value is

$$\langle \omega_1 \cdots \omega_k \rangle_{N,\beta} = \frac{W_N(k)}{W_N(0)} = \frac{(k-1)!!}{(N\Phi_\beta''(0))^{\frac{k}{2}}} = \frac{(k-1)!!}{(N(\frac{1}{2d\beta} - 1))^{\frac{k}{2}}} = \frac{(k-1)!!}{N^{\frac{k}{2}}} (C(\beta))^{\frac{k}{2}}.$$

where we define

$$C(\beta) := \frac{1}{\Phi_\beta''(0)} = \frac{2d\beta}{1 - 2d\beta}.$$

Multiplying both sides by $N^{\frac{k}{2}}$, we have shown that

$$\lim_{N \rightarrow \infty} N^{\frac{k}{2}} \langle \omega_1 \cdots \omega_k \rangle_{N,\beta} = C_1(k-1)!!(C(\beta))^{\frac{k}{2}},$$

for $\beta < \frac{1}{2d}$ and some constant $C_1 > 0$.

(ii) Similarly, we compute

$$\Phi_\beta^{(3)}(0) = 0, \quad \Phi_\beta^{(4)}(0) = 2.$$

By the *Taylor expansion*,

$$\Phi_\beta(x) \approx \Phi_\beta(0) + \frac{1}{4!} \Phi_\beta^{(4)}(x) x^4 + O(x^4).$$

By *Theorem 1*, the numerator integral

$$\int_{-\infty}^{\infty} e^{-Nf(x)} x^k dx \approx e^{-N\Phi_\beta(0)} N^{-\frac{k+1}{4}} \int_{-\infty}^{\infty} e^{-\frac{1}{12}x^4} x^k dx.$$

And the denominator becomes

$$e^{-N\Phi_\beta(0)} N^{-\frac{1}{4}} \int_{-\infty}^{\infty} e^{-\frac{1}{12}x^4} dx.$$

Then the ratio is

$$\langle \omega_1 \cdots \omega_k \rangle_{N,\beta} \approx N^{-\frac{k}{4}} \frac{\int_{-\infty}^{\infty} e^{-\frac{1}{12}x^4} x^k dx}{\int_{-\infty}^{\infty} e^{-\frac{1}{12}x^4} dx}.$$

In conclusion, we have proved

$$\lim_{N \rightarrow \infty} N^{\frac{k}{4}} \langle \omega_1 \cdots \omega_k \rangle_{N,\beta} = C_2 \frac{\int_{-\infty}^{\infty} x^k e^{-\frac{1}{12}x^4} dx}{\int_{-\infty}^{\infty} e^{-\frac{1}{12}x^4} dx},$$

for some constant $C_2 > 0$.

□

EXERCISE B.8. (i) Show that for $\beta < \frac{1}{2d}$ we have that

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N \omega_i \xrightarrow{d} Z$$

in distribution as $N \rightarrow \infty$ where $Z \sim N(0, \sigma^2(\beta))$. Determine $\sigma^2(\beta)$.

(ii) Show that for $\beta = \frac{1}{2d}$ we have that

$$\frac{1}{N^{3/4}} \sum_{i=1}^N \omega_i \xrightarrow{d} \tilde{Z}$$

in distribution as $N \rightarrow \infty$ where $\tilde{Z}$ has density $f(x) = Ce^{-\frac{1}{12}x^4}$ and $C > 0$ is some constant.

SOLUTION. The free energy is

$$f_{\beta}^{\text{CW}}(m) = \beta e(m) - s(m),$$

where

$$e(m) = -dm^2, \quad s(m) = -\frac{1-m}{2} \log \frac{1-m}{2} - \frac{1+m}{2} \log \frac{1+m}{2}.$$

(i) For $\beta < \frac{1}{2d}$, the function $f_{\beta}^{\text{CW}}(m)$ attains minimum at $m = 0$. We compute

$$\frac{d^2 s(m)}{dm^2} = -\frac{1}{1-m^2}, \quad \frac{d^2 e(m)}{dm^2} = -2d,$$

then at $m = 0$,

$$f_{\beta}^{\text{CW}}(m) = f_{\beta}^{\text{CW}}(0) + \frac{1}{2!} f_{\beta}^{\text{CW}''}(0) m^2 + O(m^2),$$

and

$$\mu_N^{\text{CW}}(m_N = m) \approx \exp\left(-N\left(f_{\beta}^{\text{CW}}(m) - f_{\beta}^{\text{CW}}(0)\right)\right) \approx \exp\left(-\frac{1}{2} N f_{\beta}^{\text{CW}''}(0) m^2\right).$$

Then

$$\text{Var}(m_N) = \frac{1}{N f_{\beta}^{\text{CW}''}(0)} = \frac{1}{N(1-2d\beta)},$$

and

$$\text{Var}(M_N) = N^2 \text{Var}(m_N) = \frac{N}{1-2d\beta}.$$

Therefore,

$$\sigma^2(\beta) = \text{Var}\left(\frac{M_N}{\sqrt{N}}\right) = \frac{1}{1-2d\beta}.$$

The k -th moment of $\frac{M_N}{\sqrt{N}}$ is

$$\mathbb{E}\left[\left(\frac{M_N}{\sqrt{N}}\right)^k\right] = \frac{1}{\sqrt{N}} \mathbb{E}[M_N^k],$$

which are finite for all t . The cumulant generating function of M_N is

$$f_N(t) = \log \mathbb{E}[e^{tM_N}] = N(\Psi_N(t) - \Psi_N(0)),$$

where

$$\Psi_N(t) = \frac{1}{N} \log Z_N^{\text{CW}}(t).$$

For $\beta < \frac{1}{2d}$, the higher-order cumulants vanish as $N \rightarrow \infty$. Since the moments converge, *Theorem 2* applies, and we conclude that

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N \omega_i \xrightarrow{d} Z$$

in distribution as $N \rightarrow \infty$.

(ii) If $\beta = \frac{1}{2d}$, $f_{\beta}^{\text{CW}''}(0) = 0$. Compute the fourth derivative

$$f^{(4)}(0) = \beta e^{(4)}(0) - s^{(4)}(0) = -s^{(4)}(0) = 2.$$

Then the *Taylor expansion*

$$f_\beta^{\text{CW}}(m) = f_\beta(0) + \frac{1}{4!} f_\beta^{(4)}(0) m^4 + O(m^4) = -\log 2 + \frac{1}{12} m^4 + O(m^4),$$

and

$$\mu_N^{\text{CW}}(m_N \approx m) \approx \exp\left(-N\left(f_\beta^{\text{CW}}(m) - f_\beta^{\text{CW}}(0)\right)\right) \approx \exp\left(-\frac{1}{12} N m^4\right).$$

Let $N m^4 := x^4$, and $X_N := N^{\frac{3}{4}} m_N$, then

$$\mu_N^{\text{CW}}(X_N \approx x) \approx \exp\left(-\frac{x^4}{12}\right) \frac{dx}{N^{\frac{1}{4}}},$$

and the limiting density of X_N is

$$f(x) = C e^{-\frac{x^4}{12}},$$

where $C > 0$ is a normalisation constant. Since

$$\mathbb{E}[X_N^k] = N^{\frac{3k}{4}} \mathbb{E}[m_N^k] = N^{\frac{3k}{4}} \int_{-\infty}^{\infty} x^k \frac{1}{N^{\frac{1}{4}}} e^{-\frac{x^4}{12}} dx,$$

the moments are finite and satisfy:

$$\mathbb{E}[|X|^k] = \int_{-\infty}^{\infty} |x|^k f(x) dx = \int_{-\infty}^{\infty} |x|^k C e^{-\frac{x^4}{12}} dx < \infty,$$

where $e^{-\frac{x^4}{12}}$ decays faster than any exponential of x^2 , by *Theorem 2*, we verify that

$$\frac{1}{N^{\frac{3}{4}}} \sum_{i=1}^N \omega_i \xrightarrow{d} \tilde{Z}$$

in distribution as $N \rightarrow \infty$.

□

EXERCISE B.9. Let us consider the spin space $\Omega_0 = \{0, \dots, q-1\}$, $q \in \mathbb{N}$, and associated to the potential:

$$\Phi_B(\omega) = \begin{cases} \beta \delta_{\omega_i, \omega_j} & \text{if } B = \{i, j\}, i \sim j \\ 0 & \text{otherwise.} \end{cases}$$

The spin configuration space is $\Omega = \{0, \dots, q-1\}^{\mathbb{Z}^d}$. Prove explicitly that the model has a unique Gibbs measure for all $\beta > 0$ provided that $q > 6d$.

PROOF. For any $i, j \in \mathbb{Z}^d$, we have

$$c_{ij}(\pi) = \sup_{\omega, \omega' \in \Omega: \omega_k = \omega'_k, \forall k \neq j} \|\pi_i(\cdot | \omega) - \pi_i(\cdot | \omega')\|_{\text{TV}},$$

where

$$\|\pi_i(\cdot | \omega) - \pi_i(\cdot | \omega')\|_{\text{TV}} = \frac{1}{2} \sum_{\eta_i = \pm 1} |\pi_i(\eta_i | \omega) - \pi_i(\eta_i | \omega')|$$

is the total variation distance. When $j \sim i$, let ω, ω' be two configurations such that $\omega_k = \omega'_k$ for all $k \neq j$. Given that

$$\pi_\Lambda(\tau_\Lambda | \omega) = \frac{1}{Z_\Lambda^\omega} \exp(-\mathcal{H}_\Lambda(\tau_\Lambda \omega_\Lambda^c)) = \frac{1}{Z_\Lambda^\omega} \exp\left(-\sum_{B \in \mathbb{Z}^d: B \cap \Lambda \neq \emptyset} \Phi_B(\tau_\Lambda \omega_\Lambda^c)\right),$$

then for each $i \in \mathbb{Z}^d$, the conditional probability is

$$\pi_i(\sigma_i = s \mid \omega) = \frac{1}{Z_i^\omega} \exp\left(-\beta \sum_{j \sim i} \delta_{s, \omega_j}\right),$$

where $\omega'_j \neq s = \omega_j \in \{0, \dots, q-1\} \setminus \{\omega_j, \omega'_j\}$, and Z_i^ω is the local partition function:

$$Z_i^\omega = \sum_{r=0}^{q-1} \exp\left(-\beta \sum_{j \sim i} \delta_{r, \omega_j}\right).$$

And we aim to calculate

$$\Delta = \frac{1}{2} \sum_{s=0}^{q-1} |\pi_i(s \mid \omega) - \pi_i(s \mid \omega')|,$$

where

$$\pi_i(s \mid \omega) = \frac{1}{Z_i^\omega} \exp\left(-\beta \sum_{k \sim i} \delta_{s, \omega_k}\right), \quad \pi_i(s \mid \omega') = \frac{1}{Z_i^{\omega'}} \exp\left(-\beta \sum_{k \sim i} \delta_{s, \omega'_k}\right).$$

Since ω and ω' differ only at j , the sums differ only in the term involving ω_j , then

$$\sum_{k \sim i} \delta_{s, \omega_k} = n_s^\omega + \delta_{s, \omega_j}, \quad \sum_{k \sim i} \delta_{s, \omega'_k} = n_s^\omega + \delta_{s, \omega'_j},$$

where n_s^ω is the number of neighbours of i (excluding j) that are in state s . Then the difference in the numerators is

$$\begin{aligned} |A_s^\omega - A_s^{\omega'}| &= \left| \exp\left(-\beta(n_s^\omega + \delta_{s, \omega_j})\right) - \exp\left(-\beta(n_s^\omega + \delta_{s, \omega'_j})\right) \right| \\ &= \exp(-\beta n_s^\omega) \left| \exp(-\beta \delta_{s, \omega_j}) - \exp(-\beta \delta_{s, \omega'_j}) \right| = 0 \end{aligned}$$

unless $s = \omega_j$ or $s = \omega'_j$. The local partition functions are

$$Z_i^\omega = \sum_{r=0}^{q-1} A_r^\omega, \quad Z_i^{\omega'} = \sum_{r=0}^{q-1} A_r^{\omega'}.$$

Since $\omega_k = \omega'_k$ for all $k \neq j$, the terms A_r^ω and $A_r^{\omega'}$ are equal for $r \neq \omega_j = \omega'_j$. Then the total variation distance is

$$\begin{aligned} \Delta &= \frac{1}{2} \sum_{s=0}^{q-1} \left| \frac{A_s^\omega}{Z_i^\omega} - \frac{A_s^{\omega'}}{Z_i^{\omega'}} \right| \\ &\leq \frac{1}{2} \sum_{s=0}^{q-1} \left| \frac{A_s^\omega}{Z_i^\omega} - \frac{A_s^{\omega'}}{Z_i^\omega} \right| + \frac{1}{2} \sum_{s=0}^{q-1} \left| \frac{A_s^{\omega'}}{Z_i^\omega} - \frac{A_s^{\omega'}}{Z_i^{\omega'}} \right| \\ &= \frac{1}{2Z_i^\omega} \sum_{s=0}^{q-1} |A_s^\omega - A_s^{\omega'}| + \frac{1}{2} \left| \frac{1}{Z_i^\omega} - \frac{1}{Z_i^{\omega'}} \right| \sum_{s=0}^{q-1} A_s^{\omega'}. \end{aligned}$$

For $s \notin \{\omega_j, \omega'_j\}$, we have $A_s^\omega = A_s^{\omega'}$. Therefore,

$$Z_i^\omega - Z_i^{\omega'} = (A_{\omega_j}^\omega - A_{\omega_j}^{\omega'}) + (A_{\omega'_j}^\omega - A_{\omega'_j}^{\omega'}) = 0.$$

Since $\sum_s A_s^{\omega'} = Z_i^{\omega'}$ and $Z_i^\omega = Z_i^{\omega'}$, the second term simplifies to zero, and the bound simplifies to

$$\Delta \leq \frac{1}{Z_i^\omega} \sum_{s=0}^{q-1} |A_s^\omega - A_s^{\omega'}| = \frac{1}{Z_i^\omega} \left(|A_{\omega_j}^\omega - A_{\omega_j}^{\omega'}| + |A_{\omega'_j}^\omega - A_{\omega'_j}^{\omega'}| \right).$$

For the numerator

$$\begin{aligned} \left| A_{\omega_j}^\omega - A_{\omega_j'}^\omega \right| &= \exp\left(-\beta n_{\omega_j}^\omega\right) \left| \exp\left(-\beta \delta_{\omega_j, \omega_j'}\right) - \exp\left(-\beta \delta_{\omega_j, \omega_j'}\right) \right| \\ &= \exp\left(-\beta n_{\omega_j}^\omega\right) |e^{-\beta} - 1| = \exp\left(-\beta n_{\omega_j}^\omega\right) (1 - e^{-\beta}) \leq 1 - e^{-\beta}. \end{aligned}$$

And similarly

$$\left| A_{\omega_j'}^\omega - A_{\omega_j''}^\omega \right| = \exp\left(-\beta n_{\omega_j'}^\omega\right) (1 - e^{-\beta}) \leq 1 - e^{-\beta}.$$

For the denominator

$$\begin{aligned} Z_i^\omega &= \sum_{r=0}^{q-1} \exp\left(-\beta \left(n_r^\omega + \delta_{r, \omega_j}\right)\right) \\ &\geq \sum_{r \notin \{\omega_j, \omega_j'\}} \exp\left(-\beta n_r^\omega\right) + \exp\left(-\beta (n_r^\omega + 1)\right) \\ &\geq (q-2) \exp(-\beta d) + 2 \exp(-\beta (d+1)) \\ &\geq (q-2) \exp(-\beta d) + 2 \exp(-\beta d) \\ &= q \exp(-\beta d) \geq (q-2d) \exp(-\beta d), \end{aligned}$$

where d is the degree of each site and $n_r^\omega \leq d$. Then, since $1 - e^{-\beta} \leq 1$ and $e^{\beta d} \geq 1$, we have

$$\Delta \leq \frac{2(1 - e^{-\beta})}{(q-2d) \exp(-\beta d)} \leq \frac{2e^{\beta d}}{q-2d} \leq \frac{2}{q-2d} \implies c(\pi) \leq \frac{4d}{q-2d},$$

which is indeed smaller than 1 as soon as $q > 6d$, and by *Dobrushin's Uniqueness Theorem*, the Gibbs measure is unique.

If $j \neq i$, changing the spin at site j does not affect the conditional probability at site i , because the interaction is only between nearest-neighbours. Therefore, $\pi_i(s \mid \omega) = \pi_i(s \mid \omega')$ and hence $c_{ij}(\pi) = 0 < 1$. $\square$

EXERCISE B.10. Consider the two-dimensional Ising model with $h = 0$ and $\beta > \beta_c(2)$. Take any $\lambda \in (0, 1)$ and consider the (non-extremal) Gibbs measure

$$\mu = \lambda \mu_{\beta,0}^+ + (1 - \lambda) \mu_{\beta,0}^-.$$

Show that μ does not satisfy

$$\lim_{\Lambda \rightarrow \mathbb{Z}^d} \sup_{B \in \mathcal{F}_{\Lambda^c}} |\mu(A \cap B) - \mu(A)\mu(B)| = 0$$

for some suitable A, B . Interpret the result.

SOLUTION. We choose $A = \{\sigma_0 = 1\}$, the event that the spin at the origin is +1, and $B_i = \{\sigma_i = 1\}$, the event that the spin at site i is +1, where $i \in \mathbb{Z}^2$. Then

$$\begin{aligned} \mu(A) &= \mu(\sigma_0 = 1) = \lambda \mu_{\beta,0}^+(\sigma_0 = 1) + (1 - \lambda) \mu_{\beta,0}^-(\sigma_0 = 1), \\ \mu(B_i) &= \mu(\sigma_i = 1) = \lambda \mu_{\beta,0}^+(\sigma_i = 1) + (1 - \lambda) \mu_{\beta,0}^-(\sigma_i = 1). \end{aligned}$$

Since the Ising model with $h = 0$ and $\beta > \beta_c(2)$ exhibits spontaneous magnetisation $m^*(\beta) > 0$, we have

$$\mu_{\beta,0}^+(\sigma_i) = m^*(\beta), \quad \mu_{\beta,0}^-(\sigma_i) = -m^*(\beta), \quad \forall i \in \mathbb{Z}^2.$$

Then

$$\mu_{\beta,0}^+(\sigma_i = 1) = \frac{1 + m^*(\beta)}{2}, \quad \mu_{\beta,0}^-(\sigma_i = 1) = \frac{1 - m^*(\beta)}{2}.$$

Then we can compute

$$D = \lambda \frac{1 + m^*(\beta)}{2} + (1 - \lambda) \frac{1 - m^*(\beta)}{2} = \frac{1}{2} + \left(\lambda - \frac{1}{2} \right) m^*(\beta),$$

similarly, the model is translation invariant,

$$\mu(B_i) = \frac{1}{2} + \left(\lambda - \frac{1}{2} \right) m^*(\beta) = \mu(A).$$

We have

$$\mu_{\beta,0}^+(\sigma_0 = 1, \sigma_i = 1) = \frac{1 + \mu_{\beta,0}^+(\sigma_0 \sigma_i)}{2}, \quad \mu_{\beta,0}^-(\sigma_0 = 1, \sigma_i = 1) = \frac{1 - \mu_{\beta,0}^-(\sigma_0 \sigma_i)}{2}.$$

The model is symmetric under spin flip: $\mu_{\beta,0}^+(\sigma_0 \sigma_i) = \mu_{\beta,0}^-(\sigma_0 \sigma_i)$. Then

$$\mu_{\beta,0}^+(\sigma_0 \sigma_i) = \mu_{\beta,0}^-(\sigma_0 \sigma_i) = \left(\mu_{\beta,0}^+(\sigma_0) \right)^2 + \text{cov}_{\mu_{\beta,0}^+}(\sigma_0, \sigma_i) = (m^*(\beta))^2 + C(i),$$

where $C(i)$ is the covariance between σ_0 and σ_i , and

$$\lim_{\|i\| \rightarrow \infty} C(i) = \text{Var}_{\mu_{\beta,0}^+}(\sigma_0) = \mu_{\beta,0}^+(\sigma_0^2) - \left(\mu_{\beta,0}^+(\sigma_0) \right)^2 = 1 - (m^*(\beta))^2.$$

Then

$$\mu_{\beta,0}^+(\sigma_0 = 1, \sigma_i = 1) = \mu_{\beta,0}^-(\sigma_0 = 1, \sigma_i = 1) = \frac{1 + (m^*(\beta))^2 + C(i)}{2}.$$

Therefore, $\mu(A \cap B_i)$ becomes

$$\lambda \frac{1 + (m^*(\beta))^2 + C(i)}{2} + (1 - \lambda) \frac{1 + (m^*(\beta))^2 + C(i)}{2} = \frac{1 + (m^*(\beta))^2 + C(i)}{2}.$$

Then let $s = \lambda - \frac{1}{2}$, we have

$$\begin{aligned} \mu(A \cap B_i) - \mu(A)\mu(B_i) &= \frac{1 + (m^*(\beta))^2 + C(i)}{2} - \left(\frac{1}{2} + s m^*(\beta) \right)^2 \\ &= \frac{1 + (m^*(\beta))^2}{2} - \left(\frac{1}{4} + s m^*(\beta) + s^2 (m^*(\beta))^2 \right) \\ &= (m^*(\beta))^2 \left(\frac{1}{2} - s^2 \right) - s m^*(\beta) + \frac{1}{4} = \frac{1}{4} \left(\mu_{\beta,0}^+(\sigma_0 \sigma_i) - \mu_{\beta,0}^+(\sigma_0) \mu_{\beta,0}^+(\sigma_i) \right). \end{aligned}$$

By FKG inequality, $\mu_{\beta,0}^+(\sigma_0 \sigma_i) \geq \mu_{\beta,0}^+(\sigma_0) \mu_{\beta,0}^+(\sigma_i)$. Then

$$\mu_{\beta,0}^+(\sigma_0 \sigma_i) - \mu_{\beta,0}^+(\sigma_0) \mu_{\beta,0}^+(\sigma_i) \geq (1 - (2\lambda - 1)^2) \left(\mu_{\beta,0}^+(\sigma_0) \right)^2,$$

which is positive for all $\beta > \beta_c(2)$ and all $\lambda \in (0, 1)$. The result shows that at low temperatures ($\beta > \beta_c(2)$), the system exhibits spontaneous magnetisation and long-range order. Under the measure μ , the spins at distant sites remain correlated, even as the distance between them becomes infinite. $\square$

EXERCISE B.11. Let $H_{V_n, h}^{\text{per}}$ be the Hamiltonian of the Ising model in $V_n = \{0, \dots, n-1\}^d$ with periodic boundary condition. Show that among all product probability measures $\mu_{V_n} = \otimes_{i \in V_n} \rho_i$ on $\{\pm 1\}^{V_n}$ (where all ρ_i are equal), the unique measure maximizing

$$W_{V_n}(\mu_{V_n}) = S_{V_n}(\mu_{V_n}) - \beta \langle H_{V_n, h}^{\text{per}} \rangle_{\mu_{V_n}}$$

is the measure such that $\rho_i = \nu$ for all $i \in V_n$, where ν is the probability measure on $\{\pm 1\}$ with mean m satisfying $m = \tanh(2d\beta m + \beta h)$. Interpret the result.

SOLUTION. We can write

$$\begin{aligned} S_{V_n}(\mu_{V_n}) &= - \sum_{\omega \in \{\pm 1\}^{V_n}} \mu_{V_n}(\omega) \log \mu_{V_n}(\omega) \\ &= - \sum_{i \in V_n} \sum_{\omega_i = \pm 1} \rho(\omega_i) \log \rho(\omega_i) \\ &= |V_n| \left(- \sum_{s = \pm 1} \rho(s) \log \rho(s) \right) = |V_n| S(\rho), \end{aligned}$$

where $S(\rho) = - \sum_{s = \pm 1} \rho(s) \log \rho(s)$. Let m be the mean of ρ , then

$$m = \langle \omega_i \rangle_\rho = \rho(+1) - \rho(-1) \implies \langle \omega_i \omega_j \rangle_{\mu_{V_n}} = \langle \omega_i \rangle_{\mu_{V_n}} \langle \omega_j \rangle_{\mu_{V_n}} = m^2.$$

Then the expected Hamiltonian is

$$\langle H_{V_n, h}^{\text{per}} \rangle_{\mu_{V_n}} = - \sum_{\langle i, j \rangle} m^2 - h \sum_{i \in V_n} m = -|V_n| dm^2 - h|V_n|m,$$

where the total number of unique nearest-neighbour pairs is $|V_n|d$. Then

$$W_{V_n}(\mu_{V_n}) = |V_n| (S(\rho) - \beta(dm^2 + hm)).$$

Given that

$$\rho(+1) = \frac{1+m}{2}, \quad \rho(-1) = \frac{1-m}{2},$$

we have

$$S(\rho) = - \left(\frac{1+m}{2} \log \frac{1+m}{2} + \frac{1-m}{2} \log \frac{1-m}{2} \right),$$

and

$$W_{V_n}(\mu_{V_n}) = |V_n| \left(- \left(\frac{1+m}{2} \log \frac{1+m}{2} + \frac{1-m}{2} \log \frac{1-m}{2} \right) - \beta(dm^2 + hm) \right).$$

We define

$$W(m) := - \left(\frac{1+m}{2} \log \frac{1+m}{2} + \frac{1-m}{2} \log \frac{1-m}{2} \right) - \beta(dm^2 + hm),$$

and calculate

$$W'(m) = s'(m) - \beta(2dm + h),$$

where

$$s(m) = - \left(\frac{1+m}{2} \log \frac{1+m}{2} + \frac{1-m}{2} \log \frac{1-m}{2} \right),$$

and

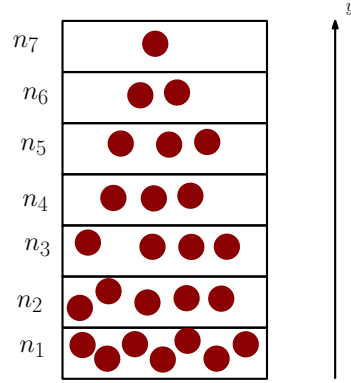
$$s'(m) = - \frac{1}{2} \left(\log \frac{1+m}{1-m} \right).$$

So

$$W'(m) = - \frac{1}{2} \left(\log \frac{1+m}{1-m} \right) - \beta(2dm + h) = 0,$$

which leads to $m = \tanh(2d\beta m + \beta h)$. This result indicates that the maximising product measure corresponds to the mean-field approximation of the Ising model. When we approximate the interacting spin system by independent spins in a mean field, the optimal choice of the mean field is precisely the one that satisfies the mean-field equation. $\square$

EXERCISE B.12. We consider a large number of particles N that can be in different *states* $i = 1, \dots, k$ such that a particle in state i has energy $E_i > 0$. This is a very simplified model for N air molecules in our atmosphere, where the k boxes (i.e., states) are meant to model different heights above the ground, as shown in the picture. The number of particles in box i is denoted with n_i .



- (i) Derive the distribution of the particles for which the number of possible arrangements is maximal under the constraints:

$$\sum_{j=1}^k E_j n_j = E, \quad \sum_{j=1}^k n_j = N.$$

Hint. Since N and the n_j are very large, you can use *Stirling's formula*: $\ln N! \sim N \ln N$.

- (ii) What does the result of point (i) imply for the concrete situation of the atmosphere, where $E_j \propto j$?

SOLUTION. (i) Suppose we choose n_1 particles for state 1, n_2 particles for state 2, and so forth, the number of arrangements is

$$\begin{aligned} & \binom{N}{n_1} \binom{N-n_1}{n_2} \dots \binom{N-\sum_{i=1}^{k-1} n_i}{n_k} \\ &= \frac{N!}{n_1!(N-n_1)!} \cdot \frac{(N-n_1)!}{n_2!(N-n_1-n_2)!} \dots \frac{(N-\sum_{i=1}^{k-1} n_i)!}{n_k!(N-\sum_{i=1}^k n_i)!} = \frac{N!}{n_1!n_2! \dots n_k!}, \end{aligned} \quad (\text{B.4})$$

where $\sum_{i=1}^k n_i = N$. By *Stirling's formula*,

$$\ln N! \sim N \ln N, \quad \ln n_i! \sim n_i \ln n_i.$$

Taking the logarithm of (B.4):

$$\ln(N!) - \sum_{i=1}^k \ln(n_i!) \sim N \ln N - \sum_{i=1}^k n_i \ln n_i.$$

We set up a Lagrangian

$$\mathcal{L}(n_1, \dots, n_k, \alpha, \beta) = N \ln N - \sum_{i=1}^k n_i \ln n_i + \alpha \left(\sum_{i=1}^k n_i - N \right) + \beta \left(\sum_{j=1}^k E_j n_j - E \right). \quad (\text{B.5})$$

Differentiate (B.5) with respect to each n_i :

$$\frac{\partial \mathcal{L}}{\partial n_i} = -\ln n_i - 1 + \alpha + \beta E_i. \quad (\text{B.6})$$

Set the derivative in (B.6) to zero, then $n_i = \exp(\alpha - 1 + \beta E_i)$. Apply the constraints, we get the distribution of the particles:

$$\sum_{i=1}^k n_i = \exp(\alpha - 1) \sum_{i=1}^k \exp(\beta E_i) = N \implies n_i = \frac{N \exp(\beta E_i)}{\sum_{j=1}^k \exp(\beta E_j)},$$

where the multiplier β satisfies

$$\sum_{i=1}^k E_i n_i = \frac{N}{\sum_{j=1}^k \exp(\beta E_j)} \sum_{i=1}^k E_i \exp(\beta E_i) = E.$$

(ii) When $E_j \propto j$, we suppose $E_j = \gamma j$, where γ is a constant. Then

$$n_i = \frac{N \exp(\beta \gamma i)}{\sum_{j=1}^k \exp(\beta \gamma j)},$$

where the multiplier β satisfies

$$\sum_{i=1}^k E_i n_i = \frac{N}{\sum_{j=1}^k \exp(\beta \gamma j)} \sum_{i=1}^k \gamma i \exp(\beta \gamma i) = E.$$

Since the denominator of n_i is a normalisation factor, we can write $n_i \propto \exp(\beta \gamma i)$. The number of particles in state i decays exponentially with i ($\beta \gamma < 0$ according to the figure). Therefore under equilibrium conditions, the atmosphere's density profile falls off exponentially with altitude. $\square$

EXERCISE B.13. Let $\mathbf{X} = (X_1, X_2, \dots, X_n)$ be a random vector uniformly distributed on the set $\Omega_n := \{\mathbf{x} \in \mathbb{R}^n : x_i \geq 0, \sum_{i=1}^n x_i = n\mathcal{E}\}$, where $\mathcal{E} \in (0, \infty)$ is a fixed number. Find the limit distribution of x_1 as $n \rightarrow \infty$.

SOLUTION. Consider the normalised vector $U_i := \frac{X_i}{n\mathcal{E}}$, then $\mathbf{U} = (U_1, U_2, \dots, U_n)$ is uniformly distributed on the $(n-1)$ -dimensional simplex:

$$\left\{ \mathbf{u} = (u_1, u_2, \dots, u_n) \mid u_i \geq 0, \sum_{i=1}^n u_i = 1 \right\}.$$

The distribution of $\mathbf{U}$ is a Dirichlet distribution with parameters all equal to 1, i.e., $\mathbf{U} \sim \text{Dirichlet}(1, 1, \dots, 1)$. The marginal distribution is $U_1 \sim \text{Beta}(1, n-1)$. By the definition of the Beta function,

$$B(1, n-1) = \frac{\Gamma(1)\Gamma(n-1)}{\Gamma(n)} = \frac{(n-2)!}{(n-1)!} = \frac{1}{n-1}.$$

Then the probability density function is

$$f_{U_1}(u) = \frac{1}{B(1, n-1)} (1-u)^{n-2} = (n-1)(1-u)^{n-2}, \quad 0 \leq u \leq 1,$$

and the cumulative distribution function is

$$\begin{aligned} F_{U_1}(u) &= \mathbb{P}(U_1 < u) \\ &= (n-1) \int_0^u (1-t)^{n-2} dt \end{aligned}$$

$$\begin{aligned}
&= (n-1) \left[\frac{-(1-t)^{n-1}}{n-1} \right]_0^u \\
&= [-(1-t)^{n-1}]_0^u = 1 - (1-u)^{n-1}, \quad 0 \leq u \leq 1.
\end{aligned}$$

Let $V := nU_1$, then

$$F_{V_1}(v) = \mathbb{P}(V_1 \leq v) = \mathbb{P}\left(U_1 \leq \frac{v}{n}\right) = 1 - \left(1 - \frac{v}{n}\right)^{n-1}.$$

Therefore

$$\lim_{n \rightarrow \infty} F_{V_1}(v) = 1 - e^{-v}, \quad v \geq 0,$$

which is an exponential distribution with parameter 1. Since $X_1 = n\mathcal{E}U_1 = \mathcal{E}V_1$, the limit distribution of X_1 as $n \rightarrow \infty$ is

$$X_1 \xrightarrow{d} \exp\left(\frac{1}{\mathcal{E}}\right).$$

□

EXERCISE B.14. Given $L > 0$, $W := [-\frac{L}{2}, \frac{L}{2}]^d \subset \mathbb{R}^d$, $m \in \mathbb{N}$, let Q be the uniform distribution on W , and $X_1, \dots, X_m$ i.i.d. random variables with law Q . Moreover, let $\eta := \delta_{X_1} + \dots + \delta_{X_m}$ be the corresponding binomial point process. Consider $A, B \subset W$ two disjoint Borel sets.

- (i) Determine the joint distribution of $\eta(A)$ and $\eta(B)$ and $\mathbb{E}[\eta(A)\eta(B)]$.
- (ii) Are $\eta(A)$ and $\eta(B)$ independent?
- (iii) Let $(L_m)_{m \in \mathbb{N}}$ be a sequence in $(0, \infty)$, $W_m := [-\frac{L_m}{2}, \frac{L_m}{2}]^d$ and η_m the corresponding binomial point process. We consider sequences such that: $\lim_{m \rightarrow \infty} \frac{m}{L_m^d} = \lambda > 0$. For any bounded Borel sets $A \subset \mathbb{R}^d$ and $k \in \mathbb{N}_0$, determine the limit:

$$\lim_{m \rightarrow \infty} \mathbb{P}(\eta_m(A) = k).$$

SOLUTION. (i) Since $\eta := \delta_{X_1} + \dots + \delta_{X_m}$, we have

$$\eta(A) = \sum_{i=1}^m \mathbb{1}_{\{X_i \in A\}}, \quad \eta(B) = \sum_{i=1}^m \mathbb{1}_{\{X_i \in B\}}.$$

Given that $A, B \subset W$ are disjoint Borel sets, $(\eta(A), \eta(B))$ follows a multinomial distribution with parameter m and probabilities $(Q(A), Q(B), 1 - Q(A) - Q(B))$. Then for $k_A, k_B \in \{0, \dots, m\}$ where $k_A + k_B \leq m$, the joint distribution is

$$\mathbb{P}(\eta(A) = k_A, \eta(B) = k_B) = \frac{m!}{k_A! k_B! (m - k_A - k_B)!} (Q(A))^{k_A} (Q(B))^{k_B} (1 - Q(A) - Q(B))^{m - k_A - k_B}.$$

And

$$\begin{aligned}
\mathbb{E}[\eta(A)\eta(B)] &= \mathbb{E}\left[\sum_{i=1}^m \mathbb{1}_{\{X_i \in A\}} \sum_{i=1}^m \mathbb{1}_{\{X_i \in B\}}\right] \\
&= \mathbb{E}\left[\sum_{i=1}^m \sum_{\substack{j=1 \\ j \neq i}}^m \mathbb{1}_{\{X_i \in A\}} \mathbb{1}_{\{X_j \in B\}}\right] \\
&= \sum_{i=1}^m \sum_{\substack{j=1 \\ j \neq i}}^m \mathbb{E}[\mathbb{1}_{\{X_i \in A\}} \mathbb{1}_{\{X_j \in B\}}] \\
&= \sum_{i=1}^m \sum_{\substack{j=1 \\ j \neq i}}^m \mathbb{P}(X_i \in A) \mathbb{P}(X_j \in B)
\end{aligned}$$

$$= \sum_{i=1}^m \sum_{\substack{j=1 \\ j \neq i}}^m Q(A)Q(B) = m(m-1)Q(A)Q(B).$$

(ii) We verify that

$$\begin{aligned} & \mathbb{P}(\eta(A) = k_A) \mathbb{P}(\eta(B) = k_B) \\ &= \binom{m}{k_A} (Q(A))^{k_A} (1 - Q(A))^{m-k_A} \binom{m}{k_B} (Q(B))^{k_B} (1 - Q(B))^{m-k_B} \\ &= \frac{(m!)^2}{k_A! k_B! (m - k_A)! (m - k_B)!} (Q(A))^{k_A} (Q(B))^{k_B} (1 - Q(A))^{m-k_A} (1 - Q(B))^{m-k_B} \\ &\neq \frac{m!}{k_A! k_B! (m - k_A - k_B)!} (Q(A))^{k_A} (Q(B))^{k_B} (1 - Q(A) - Q(B))^{m-k_A-k_B} \\ &= \mathbb{P}(\eta(A) = k_A, \eta(B) = k_B). \end{aligned}$$

Hence, in general $\eta(A)$ and $\eta(B)$ are not independent.

(iii) Given that X_i are uniform with Q ,

$$Q_m(A) = \mathbb{P}(X_i \in A) = \frac{|A \cap W_m|}{|W_m|} \approx \frac{|A|}{L_m^d},$$

where $A \subseteq W_m$ when $L_m \rightarrow \infty$ as $m \rightarrow \infty$. Since $\eta(A) \sim \text{Binomial}(m, Q_m(A))$,

$$\mathbb{P}(\eta_m(A) = k) = \binom{m}{k} (Q_m(A))^k (1 - Q_m(A))^{m-k}.$$

From $\lim_{m \rightarrow \infty} \frac{m}{L_m^d} = \lambda > 0$,

$$m Q_m(A) = m \frac{|A|}{L_m^d} \rightarrow \lambda |A|, \quad 1 - Q_m(A) = 1 - \frac{|A|}{L_m^d} \rightarrow 1$$

as $m \rightarrow \infty$. By *Poisson limit theorem*,

$$\lim_{m \rightarrow \infty} \mathbb{P}(\eta_m(A) = k) = e^{-\lambda |A|} \frac{(\lambda |A|)^k}{k!}.$$

□

References

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